CalmConnect AI

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By

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CERTIFICATE

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING (BSSE).

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Project Overview

CalmConnect AI is a web-based application that has been designed to overcome the communication barrier between the patient and the psychiatrist in an intelligent, safe, and user-friendly interface. The primary objective of this project is to offer people the chance to have a secure environment to share their feelings, consult experienced healthcare workers, and monitor mental health using the help of AI-powered dialogues and instant messengers.

The issue that was considered in this project is the lack of access to and affordability of mental health care in Pakistan where a significant population of people is reluctant to go to psychiatrists because of social stigma, costs, or the absence of that specific clinic. Conventional systems fail to offer instant emotional care or even not confidences and continuity in communication between patient and psychiatrist.

CalmConnect AI is designed as a solution to these issues by providing an AI chatbot useable on top of the Gemini API of Google that provides limited emotional support, preliminary mental health screening, and self-help exercise guidance. It is also possible to perform secure messaging and an appointment booking between a patient and a validated psychiatrist on the platform. It is built on the backend based on Node.js with PostgreSQL to organize the data and a clean layout based on React and TypeScript with Tailwind CSS on the front end.

In comparison to the current system such as generic chatbots or traditional telemedicine applications, CalmConnect AI is specifically dedicated to the communication in the field of mental health, so the control of privacy, empathy, and the interaction with the context is guaranteed. It is evaluated by testing the usability, accuracy of chatbots, and the reviews made by participants of the test. The project scope is divided into three key modules, including the AI-based chat support, psychiatrist-patient communication, and mental health journaling. Several voice-enabled chat functions, sentiment analysis-based emotion recognition and mobile application development can also be incorporated into the future work to support access and userinteraction.

To sum up, CalmConnect AI suggests a feasible and culturally sensitive source of digital mental health in Pakistan. It uses AI technology and professional care to form a more friendly, responsive, and helpful system to the patients and psychiatrists.

Dedication

Firstly, we dedicate our project to the creator Allah Almighty and dedicate to whom the world owes its existence our beloved Prophet Muhammad (Peace Be Upon Him) and dedicate this to our beloved parents, our extremely dedicated and generous teachers especially our supervisor and supportive friends, their prayers always pave the way to success for us.

Acknowledgement

The requirements for the degree of Bachelor of Software Engineering.

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Chapter 1

Introduction

Mental health has emerged as one of the most glossed over and important in Pakistan. Several individuals with anxiety, depression or emotional disturbances do not see psychiatrists because of social stigma or expensive cost of consultations or ignorance about mental well-being. Subsequently, people tend to be left untreated or check up unreliable information found on the internet. The existing system lacks readily available and confidential avenues to mental health care where many individuals in the population are not attained in time.

CalmConnect AI will combat this problem by offering a safe and smart web-based platform that will match patients and psychiatrists via the use of modern technology. The system combines an AI-based chatbot based on the Gemini API provided by Google that is capable of conducting empathetic human-like interactions with interested users, providing emotional help, and conducting a preliminary analysis with subsequent guidelines on connecting them with specialists. The platform has also introduced personal journaling, booking appointments and secure messaging functions to enable the user to take care of his or her mental health at the comfort of their homes.

The frontend in this project is created in React with TypeScript and the backend is created in Node. js with Express and the database is created in PostgreSQL. Tailwind CSS guarantees smooth interaction with the site with the help of a modern and responsive design. By implementing the CalmConnect AI, the idea is to simplify, privatize, and enhance the mental health care approach based on the effectiveness of AI and the compassion of a professional psychiatrist. The project, besides supporting the application of the use of technology in the healthcare sector, intends to facilitate a positive change in the way mental health is tackled in Pakistan.

1.1 Product Purpose

CalmConnect AI is a web-based Pakistani massage application aimed at students and young adults. We are here to offer a secure, non-disclosure, and orderly environment to discuss frustrations, monitor mood, record entries, and perform exercises of the CBT genre. The platform will be connecting two key groups of users to each other, including patients and psychiatrists with a layer of verification and moderation of the the user base as an administrator. We do not establish a diagnostic/treatment alternative. One of them is the support system that helps with early help-seeking, reflection in the day-to-day and the structured follow-ups.

Most students in universities are inhibited to visit counsellors directly. They tend to consult friends or the internet. This contributes to slow assistance and discrepancy of records. This gap is filled by CalmConnect AI which provides a neutral entry point. Students are able to go through brainstorms with an AI chat and note down their own mood and pre-session preparation. Psychiatrists receive organized highlights and schedules which minimize the time on simple history taking in sessions. Administrators maintain the system in good health and in line.

- Accessible through desktop and mobile browsers.
- Simple onboarding with verified clinician accounts.
- Role-based access with logging for accountability.
- Clear boundaries for AI assistance with escalation to a human.

The project is also a learning platform to us. We use software engineering principles including; requirements, process, quality, security and testing as well as write decisions in a clear way

1.2 Product Scope

Document conventions involve various typefaces and type style that is applied in the entire document to identify a particular information. In this part, a table will include all conventions are abbreviation used in the document with its description.

Table 1.1 enumerates general abbreviations of the abbreviations applied in this Final Year Project report. It makes the reader perceive the technical words and acronyms concerning software development, artificial intelligence, databases, and system design. With such definitions in a single location, report will be easy to read, and in subsequent chapters there will be no confusion.

Table 1.1: Abbreviations

Serial #	**Abbreviation**	**Definition**
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	DB	Database
4	UI	User Interface
5	UX	User Experience
6	FYP	Final Year Project
7	SQL	Structured Query Language
8	JWT	JSON Web Token
9	CBT	Cognitive Behavioral Therapy
10	RBAC	Role-Based Access Control
11	CRUD	Create, Read, Update, Delete
12	CSS	Cascading Style Sheets
13	IDE	Integrated Development Environment

Continued on next page

Serial #	Abbreviation	Definition
14	SPA	Single Page Application
15	PGSQL	PostgreSQL Database Management System
16	API Key	Authentication Key for External Services
17	HTTP	Hypertext Transfer Protocol
18	VCS	Version Control System
19	MVP	Minimum Viable Product
20	JSON	JavaScript Object Notation
21	TSX	TypeScript XML
22	TS	TypeScript
23	FR	Functional Requirements
24	NFR	Non-Functional Requirements
25	L	Likelyhood
26	I	Impact
27	R.No	Requirement Number

1.2.1 Existing System Description

The current mental health support systems in Pakistan and most other nations are predominately traditional in nature and scope. The majority of them use the face-to-face form of therapy which is not readily available because of its high prices, tight schedules and lack of qualified psychiatrists. Several mental health facilities continue to use manual means of making appointments and patient records are usually recorded in paper or simple spreadsheets with negligible data protection. Consequently, care continuity is minimal, and it is not efficient to monitor a patient and his or her progress over the time.

There are also certain online mental health services and general-purpose chatbots, still, they have strong limitations. The generic chatbots such as Replika or ChatGPT do not focus on the field of mental health and do not provide contextual insights into the state of psychological disorders. There are local telemedicine platforms like Oladoc and Marham that establish the doctorpatient contacts but not the AI based preliminary assistances and emotional check-ins or guided mental activities. Such systems are more concerned with consultation schedule than a regular emotional support or mental health tracking. [3]

Owing to these loopholes, the current solutions cannot offer 24/7, compassionate, and affordable, care to individuals with emotional or mental health issues. No unified system that implements professional psychiatric care binding to AI assisted application and secure way of communication with patients. Such a shortcoming shows the necessity of a holistic digital solution such as the CalmConnect AI that does not only connect patients with psychiatrists, but also offers smart and persistent mental health assistance through cognitive AI.

1. **Wysa:** is the closest comparison on the AI companion + self-help side. It offers an AI-powered emotional support space and a large self-care library, so it overlaps

with our AI chat, guided reflection, and wellness support features. The difference is that CalmConnect AI also includes direct linkage to a psychiatrist, appointment management, and role-based admin control, which makes your system more like a connected care platform than a standalone self-help assistant.

2. **Youper:** is similar because it uses conversational AI for mood tracking, reflection, and journaling. That makes it close to our AI chat, mood, and journaling modules. CalmConnect AI goes further by adding appointments, communication with psychiatrist, and admin-side governance, so our project covers both self-management and supervised follow-up, while Youper is more focused on personal emotional support and self-guided reflection.

3. **BetterHelp:** overlaps strongly with our project on the professional care side. It supports therapist messaging and scheduled live sessions, which is similar to our psychiatrist messaging and appointment flow. The key difference is that BetterHelp is mainly a therapist marketplace and care delivery service, while CalmConnect AI adds an AI emotional support layer, mood logging, journaling, and CBT-oriented self-reflection inside the same system.

4. **Talkspace:** is also close in the sense that it combines therapist access with messaging and live sessions, and it also offers psychiatry on its platform. Compared with it, CalmConnect AI is smaller in scope and not a commercial nationwide provider, but our platform is more tightly structured around student-style wellness flow, including AI emotional support, mood trends, journals, CBT records, and admin verification of clinicians. That gives our system a stronger "campus or institution-ready wellness platform" shape.

5. **Headspace:** is similar mainly on the self-help and mental wellness side. It offers mindfulness tools, CBT-based exercises, therapy access, and an AI companion called Ebb. CalmConnect AI overlaps in self-help support and guided wellness direction, but our project is more centered on patient record continuity, psychiatrist interaction, appointments, and secure messaging, whereas Headspace remains broader as a commercial mental-wellness ecosystem focused heavily on meditation, habits, and coaching.

6. **Calm:** is the least similar among the seven in terms of care workflow, but it is still comparable as a mental-health support app. It focuses on stress reduction, sleep, meditation, and guided content. CalmConnect AI is much more clinical-workflow oriented because it supports registered users, communication with a psychiatrist, appointment handling, logs, and structured records, which Calm does not center in its public product positioning.

7. **7 Cups:** is similar because it provides online emotional support, one-to-one conversations, group support, and growth paths, and it also offers messaging therapy. CalmConnect AI differs by being more formally role-based and data-structured, with appointments, patient records, direct linkage to a psychiatrist, and admin control. In other words, 7 Cups leans toward community support and listener-based help, while CalmConnect AI is designed more like a managed mental-health interaction system.

1.2.2 Future System Usage Analysis

CalmConnect AI has more than a usage in the future where it will be applied as a mere communication platform between psychiatrists and patients. Upon full development and implementation, the system can become a national mental health support system where it becomes constantly supportive due to savvy automation and intelligent data analysis. In the most recent developments, an AI chatbot can be improved to detect patterns of emotion utilizing the text sentiment analysis, which can help psychiatrists to gain a better insight into the emotional condition of their patients even prior to the visit. It will turning the sessions more or less focused and productive and decrease the time of preliminary diagnosis.

CalmConnect AI can also be co-integrated with mobile applications in future iterations to make it more accessible, particularly to the users in rural or remote locations where there is a lack of access to mental health professionals. The system can be made more inclusive to a broader population through voice-based conversation and support in the regional languages in Urdu and other languages. Also, psychiatrists can be provided with advanced analytics dashboards where they could monitor the changes in patients over the period of time, in graphs, reports, and automatically created summaries based on AI-based conversations and journal entries.

There is also the potential of institutional adoption of the platform. CalmConnect AI is an internal support tool that may be used by CalmConnect in universities, hospitals, and corporate wellness programs through students, patients, and employees. The system will be capable of ensuring mental health awareness and early intervention by integrating the use of AI-based emotional support with professional care. Continuous updates, increased privacy, and enhanced performance through a combination of advanced artificial intelligence models can make CalmConnect AI a credible digital mental health system that will bring mental health culture in Pakistan to a more open and supportive direction.

1.3 Objectives

1. **Deliver a private and low-friction entry point to mental health support**
   CalmConnect AI will eliminate the psychological and technical obstacles that prevent users to request assistance. It is easy to register into the system, the interface is userfriendly, and the system offers anonymous journaling facilities to make the initial contact to a less daunting experience. Security is enhanced by means of safe authentication and limited data gathering. The site facilitates early intervention prior to mental health issues taking a spiraling turn, and thus is palatable as the first step that the student can take before they are willing to go to counsellors.

2. **Give journaling and CBT resources to facilitate session interim self-reflection.**
   The technology incorporates the cognitives techniques of Cognitive Behavioral Therapy (CBT) that include records of thoughts, moods, and automatic thoughts analysis. The tools aid the users to identify the negative thinking patterns and substitute them with rational reactions. Everyday journaling enables the user to record their feelings and progress that can be discussed with the psychiatrists during their visits.

Continuous self-reflection enhances awareness of emotions and contributes to the strengthening of positive coping processes on the sessions between the therapy.

3. **Facilitate safe messaging and appointments ensuring all history is in one location**
   CalmConnect AI is a communication platform that enables patients and psychiatrists to communicate, create appointments, and exchanges in a single location. Every one of the records, such as chat transcripts, appointment notes, or feedback, will be stored safely and in chronological order. This will provide continuity of care whereby a clinician will make informed decisions towards the reduction of redundant data entry. Cross-boundary and intra-disciplinary cooperation and adhesion become stronger as patients have access to their entire history easily but give an AI assistant guardrails and clear rules on escalation The chatbot built into the WebApp is an integrated AI assistant with preliminary guidance based on the evidence-based mental health support framework using empathetic and conversational responses.

4. **It is however not an alternative to human counselling.**
   The system applies safety guardrails, which include the identification of crisis relevant expressions and sending users to emergency resources or psychiatrists. The AI will help in mental training, setting goals, and post-processes questions and keep ethical and safety boundaries during interactions.

5. **Secure user information with RBAC, hash passwords and transport**
   The CalmConnect AI architecture is built upon security. Role-Based Access Control (RBAC) makes sure that there are only those users that have access to particular modules. The hash of the passwords is done using the bcrypt, and all data is being sent via HTTPS with TLS encryption. Access to database is heavily scoped and audit logs are used to track important operations like in attempts to log in, make changes, or export data. This multi-layered security system will ensure the confidence of users and provide international privacy standards of digital health solutions.

6. **Low-bandwidth and budget devices design is common in Pakistan.**
   The application has been customised to run on slow internet connections and lowend android phones to ensure the application is also used by a large number of users subject to the technical limitations. Small size front end elements, image compression and caching allow lower loading times. The responsive web design modifies to different screen sizes and allows some offline reading of the modules. This makes it inclusive and accessible to all students with different socioeconomic backgrounds.

7. **Ensure a tracking procedure of sensitive actions is followed.**
   All the crucial functions of calm connect AI are recorded to provide an auditory track. These involve user logging in and out, resetting passwords, appointment wills, deleting messages and administrative privileges. Logs are dated and kept safely as a matter of accountability and investigation of incident. This type of auditing is not only beneficial in terms of the transparency provided, but also in terms of the data protection compliance. It assists in the identification of suspicious practices at an early stage and gives the user more confidence in the integrity of the system.

1.4 Problem Statement / Limitations

The mental health burden is also quite hectic and the professionals who can provide the required qualifications are not excessive. The students are likely to seek help during the late hours and lose information as time goes by during the sessions. The available tools are either too general or too informal to suit the privacy and traceability requirements. The key weaknesses that we identify: the output of AI can be biased in certain scenarios, substitution of crisis support with remote follow up is impossible, and network instability may affect experience. A theistic design would be very vulnerable to such risks because it offers us no boundaries and no fallbacks in designing it. [1]

1. **No medical claims. The app provides guidance and journaling, not prescriptions**
   CalmConnect AI is strategically meant to be a supportive digital EHR, and not a clinical diagnosis or therapeutic insertion. This platform provides general mental wellness websites, journaling challenges, and CBT-based exercises based on the literature in the field, but no prescriptions, diagnosis, or medical treatment regimens. Spa has ethical issues by preventing medical claims, and legal and professional risks have also been reduced. The AI model is trained to remain within conversational safe rules, which offer educational and motivational assistance and urge the users to consult professional assistance on any medical or psychiatric issues. [2]

2. **The AI conversations are disclaimed with a notice of emergency.**
   All AI engagements are pre-empted with precise disclaimers informing people that the answers would be in informational purposes only and not professional advice. The chat interface has an emergency notice displayed that gives the contact details of local helplines and emergency services. When the user types in phrases warranting a red flag of selfharm or suicide, the system will automatically activate an escalation response showing immediate instructions and requesting the user to call a psychiatrist or hotline. This will be a moderate measure of accessibility and safety so that the users know the boundaries of AI guidance, yet they will obtain instant emotional help.

3. **We do record sensitive activities and limit access on the need to know basis.**
   The privacy and accountability of data is incorporated in the architecture of the system. Any sensitive operations such as logins and resetting passwords, updating of profiles and deletion of messages are automatically logged with time stamps and user identities. The availability of this information is done in accordance with a tight RBAC scheme meaning that only specific users (administrators, psychiatrists) are allowed to access specific information based on the duties they own. Patients do not see the information of other users, and the psychiatrists only get data of the patients provided to them. Such controlled access will support the best practices in privacy and bring traceability in the event of a security breach or any reporting conflict. [5] [6]

4. **Offline-first does not fall under MVP yet assets in the form of cache-UI features will enhance resiliency.**
   Even though the MVP lacks an offline feature, some design optimizations make it

more usable when the network is poor. Service workers localize frontend resources like CSS, scripts, and fixed-size images, which are stored in cache, which are used to make requests to the server by the browser. This enables the pages to be loaded quicker and partially work even when the internet is off. Previously loaded information such as past moods record or journals can still be viewed by the user. Nonetheless, the features based on the real-time communication, like AI chat or communicating with the help of the message, are based on the active connection. This compromises guarantee the performance efficiency but does not overcomplicate the architecture to deploy it at an academic level.

1.5 Proposed Solution

The CalmConnect AI proposed system is a web based AI-based system of mental health support and communication. It is created to close the divide between patients and psychiatrists by providing a combination of realtime messaging, AI-based emotional support as well as digital health management. It is a system that contains a number of key modules that are as follows:

1.5.1 Administrator Module:

- Manage and verify all user profiles, including:
  - Administrators
  - Psychiatrists
  - Patients
- Authorize and approve registration requests of psychiatrist using credentials.
- Track appointments, Chat logs, and messaging activity of the AI, and monitor.
- Monitor and control user feedback, reported problems and flagged discussions.
- Monitor data backups, metrics and system maintenance.
- administer platform-wide security controls, logs and policy enforcement.

1.5.2 Psychiatrist Module:

- Create and manage personal profiles with professional credentials and availability.
- View and manage assigned patients and their records.
- Schedule, approve, and update appointments directly through the dashboard.
- Chat securely with patients via encrypted direct messaging.
- Access patient journals, mood trends, CBT thought records, and AI chat summaries.
- Record post-session notes and monitor patient progress over time.
- Receive alerts for critical or high risk AI detected language during patient interactions.

1.5.3 Patient Module:

- Create, verify and own an individual wellness account.
- Communicate with the chat bot AI emotional support feature.

- Keep the daily mood records and keep personal journals and the CBT reflections.
- Book, reschedule or cancel visits with certified psychiatrists.
- Securely chat with psychiatrists using an encrypted chat.
- Get self-improvement recommendations and insights generated by the AI-based system.
- View progress dashboards, including personal levels of mood and journaling.

1.5.4 AI Chatbot Module:

- Offer 24/7 conversational support using Google's Gemini API.
- Detect sentiment and respond with empathy and evidence-based mental wellness advice.
- Suggest CBT activities, breathing exercises, and journaling prompts.
- Escalate to psychiatrist consultation when high risk content is detected.
- Maintain user privacy by anonymizing logs and limiting data storage.
- Never stop to correct the dialog quality using safe reinforcement learning.

1.5.5 Database and Security Module:

- Procedural control of the PostgreSQL in the `calmconnect` framework.
- authentication is installed with the assistance of bcrypt and JWT by authorization and password encryption.
- Enforce RBAC among the administrators, psychiatrists and the patients.
- You are recommended to maintain records of accountability.
- Permission to back up data periodically, recovery/version schema integrity.
- All API communications ought to be encrypted with the help of HTTPS and TLS.

1.5.6 System Benefits:

- Provides emotional assistance to students and patients, which is 24-hour assistance and AI assistance.
- Increases collaboration in patients and psychiatrists through consolidated data.
- Encourages dynamic mind and self observation.
- Reduces the distance in communication and administrative follow up delays.
- Offers a friendly and online system of psychology well being in Pakistan which are centralized and culturally competent.

1.6 Intended Market of Product

The CalmConnect AI will focus on patients who aspire to receive accessible and affordable mental health services, psychiatrists who wish to use a digital consultation service, and organizations that have potential to advertise the mental health. The main market includes students, professionals, and the whole population in the areas, which do not have a sufficient number of psychiatric services. Accessibility is the point of the value of the product, it is cheap in terms of operation and social impact. It offers an institution the ability to subscribe or license out to an institution as a scalable solution, to the clinical institutions and universities. CalmConnect AI could be a powerful and sustainable platform of mental health support in Pakistan.

1.7 Intended Users of Product

CalmConnect AI is targeted to accommodate three groups of users.

1. **Patients:** Stressed, anxious and depressed people who need emotional support or provision of counseling where privacy and convenience are needed. [1]
2. **Psychiatrists:** Authorized personnel who will be able to handle patient profiles, hold online appointments, and monitor the development of the patients with the help of digital history.
3. **Administrators:** System managers who ensure that users are verified, data is secured and that the platform as a whole is performing.

Universities, hospitals, and wellness organizations can be considered as the indirect users, who can implement CalmConnect AI to raise mental health awareness and easy access to support services.

1.8 Software Process Model

It will use a hybrid agile model which will be an amalgamation of Scrum and time boxed prototyping spikes. Scrum can be delivered in a short and repetitive manner and contains well-defined milestones, but spikes are associated with uncertainty regarding AI functions, such as Gemini integration, rapid design, latency, and reliability.
Scrum rythm: 2 week sprints which include planning, daily stand-up, review and retrospective. These include product owner, scrum master, developers and QA. These artifacts include: product backlog, sprint backlog and definition of done.
Definition of Done: Working code, passing tests, PII security check, linting, staging verification and updated documentation.
1-3 days Prototyping spikes: Experimental goal targeting of AI components generating short-range results and maintenance/abandonment of decisionmaking. Spike code does not require shipping.
Milestones:

1. Authentication, user roles, and chat MVP
2. Appointments, journals, and basic AI chat
3. Usability and AI quality evaluation with security hardening
4. Final release and project report

This model facilitates the development of changing needs, prompt user feedback, and responsible application of the AI, and it suits academic deadlines.

Sprint 1 - Auth/Core:
Table 1.2 describes the planning and implementation of Sprint 1. This sprint is devoted to the development of the basic authentication and access control of the system. It also determines the length of the sprint, major activities, and task breakdowns, the tasks of

the team, and the capacity planning to provide the clear and organized beginning of the development.

Table 1.2: Sprint 1

START	DUR.	ACTIVITY	DESCRIPTION	WHO
Weeks 7–10	4 weeks	Goal of the Sprint	Deliver core authentication and access control foundations.	Product Owner
		Tasks Breakdown + estimations	• 1.1 Register • 1.2 Email verification • 1.3 Login • 1.4 Password reset • 1.5 Account block • Lay groundwork for 14.1 route protection (role + ownership)	Team
		Sprint Capacity calculation	Confirm team capacity and update scope based on capacity.	Team
		Commitment	Commit to the Sprint 1 plan.	Team
After planning		Board preparation	Enter tasks/subtasks in the tool and link Definition of Done/Ready.	Team

Sprint 2 - Mood Tracking and Journals:
Table 1.3 outlines the framework and goals of Sprint 2. The sprint involves the implementation of mood tracking, journaling and patient dashboard features. It indicates the scheduled operations, the allocation of tasks, team work, and the capacity planning required to provide a prospective wellness functionality to patients in an organized way.

Table 1.3: Sprint 2

START	DUR.	ACTIVITY	DESCRIPTION	WHO
Weeks 11–14	4 weeks	Goal of the Sprint	Ship mood tracking and journaling with patient dashboard slices.	Product Owner
		Tasks Breakdown + estimations	• 6.1–6.3 mood features • 7.1–7.3 journals & consented sharing • Patient dashboard slices per 13.1	Team
		Sprint Capacity calculation	Confirm capacity; update scope.	Team
		Commitment	Commit to the Sprint 2 plan.	Team
After planning		Board preparation	Create tasks; attach acceptance criteria and test cases.	Team

Sprint 3 - Appointments and DMs:
Table 1.4 is the Sprint 3 plan and objectives. This sprint focuses on the addition of appointment scheduling and safe direct messaging between patients and psychiatrists. It provides the overview of the key activities, feature breakdown, team tasks and preparations needed to sustain professional communications in the system.

Table 1.4: Sprint 3

START	DUR.	ACTIVITY	DESCRIPTION	WHO
Weeks 15–18	4 weeks	Goal of the Sprint	Enable appointments and secure direct messaging.	Product Owner
		Tasks Breakdown + estimations	• 4.1–4.3 direct messaging • 5.1–5.4 appointments • Psychiatrist dashboard slices per 13.2	Team

Continued on next page

START	DUR.	ACTIVITY	DESCRIPTION	WHO
		Sprint Capacity calculation	Confirm capacity; update scope.	Team
		Commitment	Commit to the Sprint 3 plan.	Team
After planning		Board preparation	Create tasks; link data models and API endpoints.	Team

Sprint 4 - AI Chat and Safety:
Table 1.5 provides the plan of Sprint 4 that is concerned with the implementation of safe AI-based chat sessions. This sprint includes in the integration of the safety measures, crisis management, and chat summary to provide responsible and controlled use of the AI features in the system.

Table 1.5: Sprint 4

START	DUR.	ACTIVITY	DESCRIPTION	WHO
Weeks 19–22	4 weeks	Goal of the Sprint	Deliver safe AI chat sessions with safeguards and summaries.	Product Owner
		Tasks Breakdown + estimations	• 3.1–3.5 AI chat, disclaimers, non-clinical responses, crisis detection, transcript/summary • Align UI copy with safety acceptance 3.2–3.4	Team
		Sprint Capacity calculation	Confirm capacity; update scope.	Team
		Commitment	Commit to the Sprint 4 plan.	Team
After planning		Board preparation	Create tasks; add safety test flows and red-team checks.	Team

Sprint 5 - Dashboard, Admin, Audit, Notifications and Data management:
Table 1.6 summarizes the activities that will be carried out during Sprint 5 that will be aimed at making final arrangements of the administrative features, dashboards, logging and notification mechanisms. Another data management activity undertaken during this

sprint is report preparation, audit preparation, and controlled data operations, to make the system ready to be reviewed and released.

Table 1.6: Sprint 5

START	**DUR.**	**ACTIVITY**	**DESCRIPTION**	**WHO**
Weeks 23–26	4 weeks	Goal of the Sprint	Finalize admin, dashboards, audit/logging, notifications, and data operations.	Product Owner
		Tasks Breakdown + estimations	• 9.1 verify psychiatrists; 9.2 manage reports • 10.1–10.2 audit & query logs • 11.1–11.2 notifications • 13.1–13.3 dashboards • 12.1–12.2 data download; soft-delete/anonymise	Team
		Sprint Capacity calculation	Confirm capacity; update scope.	Team
		Commitment	Commit to the Sprint 5 plan.	Team
After planning		Board preparation	Create tasks; tag release milestones and QA sign-offs.	Team

1.8.1 Process Model Introduction

We practice hybrid agile due to the fact that scrum provides brief sprints, ceremonies, and shared definition of done. Also, prototyping removes risk of designs spikes such as prompt design, latency testing and safety controls. Our review of spikes and translating them into successful experiments on the backlogs.

1.8.2 Justification of Proposing the Process Model

1. Spikes minimize waste, AI features must be experimented on before being committed to.

2. It is a concern of mental health UX to have user feedback cycles. Scrum reviews make us listen to other people and allow us to make changes.

3. Small steps allow checking security early and migrating to a DB without massive migrations.

4. Team meetings vary throughout the semester; the beat allows us to maintain the pace.

1.8.3 Steps of Process Model

CalmConnect AI uses the hybrid agile process model, and it is carried out in the following steps:

1. **Requirement Analysis:** Collect requirements including user and system requirements together with psychiatrists, patients and supervisors to establish project scope and functionality.

2. **Planning and Sprint Establishment:** Subdivide the project into the small sprints, and establish targets, which should be accomplished by the team members in the particular sprint.

3. **Prototyping:** The design and usability are tested by making mini-prototypes of every big feature to ensure the prototyping i.e. AI chatbot conversation, book appointment, messaging.

4. **Design and Development:** Implement React and TypeScript on the frontend and Node.js on the backend and design the database using PostgreSQL. Make use of Tailwind CSS to enjoy responsive user interface.

5. **Unit, integration and user acceptance testing:** Diffuse evaluation of chatbot accuracy, security or usability each sprint completed.

6. **Deployment:** Put the application to a controlled environment and then seek feedback about the application by the users and supervisors.

7. **Maintenance and Improvements:** Add user feedback, refreeze code and upgrade the system to deliver more AI performance, performance and reliability.

It is a structured yet flexible process that will ensure effective development of CalmConnect AI in terms of collaboration, continuous experimentation and changes to the demands of the users.

Chapter 2

SOFTWARE REQUIREMENTS SPECIFICATION

CalmConnect AI has a requirement as a particular assertion of what the system is required to do or what quality is required to be attained (i.e. patients can record their daily mood on 1 5 scale, JWT tokens are not active after 60 minutes). All have been compiled in the Software Requirement Specification (SRS) on the modules, namely Patient, Psychiatrist, Admin, AI Chat, Messaging, Appointments, Journaling, Mood Tracking, and Security.

Having them clearly defined and specified, we can design the CalmConnect AI in a relatively natural manner: the database schema, the API routes and React screens can be easily traced to each of the requirements. In case the requirements are not exact or not similarity then the design becomes bulky and harder to code hence the rewriting is accomplished and the security is lost. In this project, good requirements are well discussed and clearly discuss the following:

- **Functional:** Initiate/resumptive AI chat; add/browse mood records; add/read journals; add/maintain/ hacia cancel appointments; patient role; psychiatrist role; administrator role; administrator role-controlled clinician verification.
- **Security & Privacy:** RBAC on all endpoints; password hash, bcrypt; JWTsessions; auditing other so-called sensitive actions; transporting using HTTPS.
- **Safety:** AI alerts were shown before the chat; inflation when words of risk were typed; no drug orders.
- **Usability & Performance:** Mobile first Interface; loads fast using minimal bandwidth; loads partially static data; provides Urdu friendly text where needed.

2.1 Introduction

It is at the software requirements phase that we get to study the current mode of operation and what the new system should include in its functionality. In the instance of CalmConnect AI, it will involve considering the existing process of student and psychiatrist support management: disordered WhatsApp messages, phone calls, paper notes, and haphazard spreadsheets. We break down this current workflow in several different ways, user, data, tasks, security, and pain points, and then extrapolate our results into unambiguous, testable specifications of the new platform.

Purpose: It is aimed at migrating the informal practices to a formal and secure web system. We are concerned with patient privacy protection and reliable digital records. Real needs we learn with the help of the first acquaintance with the way at which doctors and clinics treat patients today. This assists us in eliminating the errors and minimizing the rewrites in future development process. The ultimate system will assist healthcare facilities in Pakistan in an effective and viable manner.

Scope of Analysis

- **Processes:** the procedures and means of making requests, confirming and following appointments; the manner the journals and mood notes are kept; the manner of exchanging advice.
- **Information:** what do they note now (names, contact, symptoms, dates, notes), where do they keep (and who can access them).
- **Users:** patient, psychiatrist and administrator role and responsibilities, on-boarding and validation requirements.
- **Restrictions:** bandwidth, variety of devices, privacy expectations, organizational policies.
- **Potential risks:** no history involved, information leakage, inability to seek fast enough and inconsistency in following up.

Inputs

- Students and counselling staff interviews and casual conversations.
- Existing notes, messages and appointment logs (anonymised).
- Organizational regulations regarding checking, confidentiality and messages.

Activities

- Follow the current processes and identify the bottlenecks (e.g. any context lost between chats).
- Manual normalisation of data to entities and relations in PostgreSQL.
- Elicit and priority Requirements with MoSCoW (Must, Should, Could, Won't).
- Select major screens and flows prototype in order to fix assumptions.

Outputs

- **SRS Document:** Patient, Psychiatrist, Admin, AI Chat, Messaging, Appointments, Journaling, Mood Tracking, Reporting, functional, non-functional, safety and security requirements.

- **Data Model Draft:** user, profile, session, message, appointment, journal and mood entry, thought record, activity logs (the keyed and constrained tables).
- **Acceptance Criteria:** Should statements that can be tested like: patient records a mood on a 1-5 scale; VU is shown in the chart every 1 second.

Why it matters: When the requirements are precise and aligned with the actual use, it should be smooth sailing in the design and coding phases: API routes represent user actions, database tables represent real data, web interface is adjusted to the daily operations. Badly-defined requirements, conversely, result in clumsy design, security holes and expensive reworking.

Definition for This Project: In brief, this phase of software requirements at CalmConnect AI consists of a systematic study of the existing manual system and pain points, then specifications, which are testable, secure, and low-friction web platform to support students and clinicians and respect privacy and local requirements.

2.1.1 Document Scope

General

- Whatever it will not or will do is decided.
- Establishes users, roles, goals and use context.
- The functional requirements are formulated as lists.
- Meets the non-functional requirements: usability, reliability, performance, security, privacy and scalability.
- Temporalizes the external-interfaces: UI Groundwork, APIs, data-structures, integrations.
- States constraint and norms: technical, regulatory and platform constraints.
- Determines assumptions, dependencies and risk.
- Provides acceptance criteria of validation and verification.
- It has a background in design, development and testing.

CalmConnect AI

- **In scope:** Auth using JWT, RBAC, mood tracking, journals, CBT records, AI chat with disclaimers, direct messaging, appointments, validation of the administration, audit logs, and PostgreSQL schema.

- **Non-functional:** mobile-first interface, low bandwidth performance requirements, TLS, bcrypt hashing, data logs, and data retention policies.
- **Interfaces:** React frontend, REST API, Gemini API, email service (verification/reset).
- **Out of scope:** medical diagnosis or prescriptions, full offline mode, languages beyond English/Urdu.

2.1.2 Audience

General

- Project team (developers, testers) - lodge (eldest and try) features.
- Team lead / Product owner- put change first.
- Academic Evaluators / Supervisor- create completeness and rigor.
- Psychiatrists (experts in the domain) scale clinical forced damned processes and clinical forced damage.
- Security -/ Devops- execute privacy, RBAC tracking and backups and deployment boundaries.
- UI/UX designers- relates flows, content accessibility and tone.
- Institutional stakeholders (e.g. SZABIST, partner clinic) - state needs and conformity.

CalmConnect AI

- Team: Zain Ul Abideen Malik (Team Lead), Abdul Rehman Shabir (Frontend Dev), and Azhar Ali (Backend Dev).
- Supervisor: Mr. Amjad Saleem Rana.
- Clinical reviewers: verified psychiatrists using the portal for patient care.
- Operations/Security: personnel managing environment variables, SSL/TLS, backups, and monitoring.

2.2 Functional Requirements

2.2.1 Functional Requirements

FR-01 : Authentication & Authorisation:
The fundamental authentication and authorisation criteria of the system are defined in Table 2.1 These requirements explain the registration requirements, account verification, secure login and account recovery among users. Access control that guarantees the use

of protected system features by authorised and verified users is also an aspect that has been noted in the table.

Table 2.1: Authentication & Authorisation

R.No	Description
1.1	Users shall register with email, password, and role (patient by default).
1.2	The system shall send a verification link; unverified accounts cannot access protected routes.
1.3	Users shall log in using email and password to receive a JWT access token.
1.4	The system shall allow password reset via time-bound, single-use token.
1.5	The system shall revoke access if the account is blocked by admin.

FR-02 : User Profiles:
Table 2.2 explains the functional requirements associated with user profile management. It details the manner in which the patients, psychiatrists and administrators uphold and manage profile information as per their mandates. These are the requirements which aid in proper information of users, access of users with their roles, and verification of the professional accounts.

Table 2.2: User Profiles

R.No	Description
2.1	Patients shall manage a personal profile (name, contact, optional demographics).
2.2	Psychiatrists shall manage professional profiles (qualifications, license/registration, availability).
2.3	Admins shall view and edit user status (active/blocked) and trigger verification workflows for psychiatrists.

FR-03 : AI Chat:
Table 2.3 explains the functional requirements of AI chat feature. It includes the interaction of patients with the AI, utilizing safety disclaimers, and non-clinical response limitations. Other aspects to make users safe and privacy aware the table also emphasizes on include risk detection, data storage, and controlled access to chat summaries.

Table 2.3: AI Chat

R.No	Description
3.1	Patients shall start a new AI chat session and continue previous sessions.
3.2	The system shall display a disclaimer and emergency notice before the first AI response in each session.

Continued on next page

R.No	Description
3.3	The AI shall provide supportive, non-clinical responses and avoid medical prescriptions.
3.4	The system shall detect high-risk phrases and show escalation guidance; admins can configure keywords.
3.5	Chat transcripts shall be stored per patient; summaries shall be viewable by assigned psychiatrists with consent.

FR-04 : Direct Messaging:
Table 2.4 gives the functional requirements of direct messaging feature. It describes the method of interaction between patients and psychiatrists with the help of secure message threads, message status tracking, as well as user management and reporting of messages. These needs assist in providing privacy, accountability, and restricted communication in the system.

Table 2.4: Direct Messaging

R.No	Description
4.1	Patients and assigned psychiatrists shall exchange direct messages in secure threads.
4.2	The system shall record message status (sent, delivered, read) and timestamps.
4.3	Users shall be able to report or delete their own messages; deletions are logged.

FR-05 : Appointments:
Table 2.5 will specify the functional requirement of appointment management. It outlines the process of patients requesting sessions, how psychiatrists make scheduling choices and the process of making the system prevent booking conflicts. Post-session summaries are also provided in the table to facilitate continuity and transparency of care.

Table 2.5: Appointments

R.No	Description
5.1	Patients shall request appointments with verified psychiatrists by date/time.
5.2	Psychiatrists shall accept, reschedule, or cancel requests; status: scheduled, completed, cancelled.
5.3	The system shall prevent double-booking for the same psychiatrist and time slot.
5.4	After completion, the psychiatrist shall record a brief session summary; the patient shall view it.

FR-06 : Mood Tracking:
Table 2.6 explains the mood tracking functional requirements. It identifies the way that patients record their mood daily and the illustration of trends with time. The records are also able to enable the allocated psychiatrists to access mood trends and have a better grasp of patient progress.

Table 2.6: Mood Tracking

R.No	Description
6.1	Patients shall record daily mood on a 1–5 scale with optional notes and tags.
6.2	The system shall present weekly and monthly trend charts for the patient.
6.3	Psychiatrists shall view mood timelines of assigned patients.

FR-07 : Journaling:
Table 2.7 gives details of the functional requirements of journaling feature. It describes the way of keeping journal entries and arranging them with tags and search options by patients. Controlled sharing is one more feature demonstrated in the table, according to which consented entries are realized as per the assigned psychiatrists.

Table 2.7: Journaling

R.No	Description
7.1	Patients shall create, edit, and delete private journal entries.
7.2	Patients shall tag entries and search journals by date or tag.
7.3	With explicit patient consent, psychiatrists shall view selected entries.

FR-08 : CBT Toolkit:
Table 2.8 is the functional requirements of the CBT toolkit. It describes how patients make use of organised records of thoughts to comment on occurrences and feelings. Other important aspects that the table brings out include summary generation, and consent-based access where psychiatrists can access CBT records during treatment sessions.

Table 2.8: CBT Toolkit

R.No	Description
8.1	Patients shall complete thought records (situation, emotions, automatic thoughts, evidence for/against, rational response).
8.2	The system shall export a CBT summary (PDF or screen view) for session discussion.
8.3	Psychiatrists shall review a patient's CBT records with consent.

FR-09 : Admin & Verification:
These were the administrative and verification-related requirements of the system as

captured in Table 2.9 It outlines the procedure of going through the applications of psychiatrists and dealing with user reports. The requirements provide the appropriate checks and balances on the platform.

Table 2.9: Admin & Verification

R.No	Description
9.1	Admins shall review psychiatrist applications and approve/reject them.
9.2	Admins shall manage reports from users and resolve them with status tracking.

FR-10 : Activity Logs & Audit:
Table 2.10 explains the functional requirements in the activity logging and auditing. It outlines how crucial system actions are logged and how administrators go over those logs where necessary. These are necessities that promote transparency, accountability, and security monitoring in the system.

Table 2.10: Activity Logs & Audit

R.No	Description
10.1	The system shall log sensitive actions (login, password reset, role/status change, message deletion, data export).
10.2	Admins shall query activity logs by user and date range.

FR-11 : Notifications:
The system notification functional requirements are described in Table 2.11. It describes the process of updating the users who have different appointments or receive new messages. Those messages make users aware and enhance real-time communication in the system.

Table 2.11: Notifications

R.No	Description
11.1	The system shall notify users of appointment status changes.
11.2	The system shall notify message recipients of new messages (in-app).

FR-12 : Reporting & Dashboards:
Table 2.12 provides the reporting and dashboard functional requirements. It shows how relevant data is accessed by patients, psychiatrists and administrators in their role-specific views. These dashboards aid in monitoring, decision-making and general system view.

Table 2.12: Reporting & Dashboards

R.No	Description
12.1	Patients shall view personal dashboards (mood trends, journal count, and CBT progress).
12.2	Psychiatrists shall view dashboards for assigned patients (appointments, mood/journal overview).
12.3	Admins shall view system-level dashboards (user growth, usage patterns, and flagged content).

Table 2.13 will provide the acceptance criterion of each of the functional requirement outlined in the system. It serves as a strong controlpoint which would be used to verify the functionality of each feature. Given -When-Then statements are used to express requirements in a structured manner in order to eliminate ambiguity in outcomes to be expected. Such format aids developers as well as testers in knowing what needs to be done to be successful. All the criteria are connected to a certain requirement and help trace the whole process of design and testing. The validation is also made simple by the table and requirements are converted into verifiable and real life situations. In the development process, it helps to make decisions regarding implementation and minimize misunderstanding. In the case of evaluation, it offers sound grounds of verifying completeness and accuracy of the system.

Table 2.13: Acceptance Criteria:

R.No	Requirement Name	Acceptance Statement (Given—When—Then)
1.1	Register	**Given** a new user with valid email and strong password, **When** the registration form is submitted, **Then** an account is created with role=patient and a verification email is issued.
1.2	Email verification	**Given** an unverified account and a valid token, **When** the user opens the verification link within its validity, **Then** the account becomes verified and protected routes are enabled.
1.3	Login	**Given** correct credentials for a verified account, **When** the user logs in, **Then** a JWT access token is returned; with wrong credentials, access is denied.
1.4	Password reset	**Given** a registered email, **When** the user requests a reset and uses the time-bound token, **Then** a new password is set and old tokens are invalidated.

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R.No	Requirement Name	Acceptance Statement (Given—When—Then)
1.5	Account block	**Given** an account set to blocked by admin, **When** the user accesses any protected resource, **Then** the request is rejected and an audit entry is written.
2.1	Patient profile	**Given** a logged-in patient, **When** they update profile fields, **Then** changes persist and are visible on the next load.
2.2	Psychiatrist profile	**Given** a psychiatrist, **When** qualifications and availability are edited, **Then** updates save and surface in booking views.
2.3	Admin verification/status	**Given** a psychiatrist application, **When** admin approves or rejects it, **Then** the user status updates and the applicant is notified.
3.1	Start/continue AI chat	**Given** a patient, **When** they start a session or open a previous one, **Then** messages appear chronologically with timestamps.
3.2	AI disclaimer	**Given** a new AI chat session, **When** the first reply is about to be shown, **Then** the disclaimer and emergency note are displayed.
3.3	Non-clinical responses	**Given** any AI exchange, **When** medical diagnosis/prescription is implied, **Then** the response avoids medical claims and redirects to professional help.
3.4	High-risk detection	**Given** a message containing configured crisis keywords, **When** it is sent, **Then** an emergency banner appears and an escalation event is logged.
3.5	Transcript storage/summary	**Given** an ended chat session, **When** the patient reopens it, **Then** the transcript is retrievable; summaries are available to assigned psychiatrists with consent.
4.1	Direct messages (DM)	**Given** a patient and assigned psychiatrist, **When** they exchange messages, **Then** a secure thread retains the conversation with IDs and times.
4.2	Message status	**Given** a sent message, **When** it is delivered and read, **Then** status updates to *delivered/read*.
4.3	Report/Delete own messages	**Given** a user viewing their message, **When** they delete or report it, **Then** the action is applied and recorded in audit logs.

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R.No	**Requirement Name**	**Acceptance Statement (Given—When—Then)**
5.1	Request appointment	**Given** a verified psychiatrist and free slot, **When** a patient requests a booking, **Then** an appointment is created with status=*scheduled*.
5.2	Update appointment	**Given** a pending appointment, **When** the psychiatrist accepts/reschedules/cancels, **Then** both parties see the new status and receive notifications.
5.3	No double-booking	**Given** an existing appointment for a slot, **When** another booking is attempted for the same psychiatrist/time, **Then** the request is rejected with a conflict message.
5.4	Post-session note	**Given** a completed appointment, **When** the psychiatrist saves a summary, **Then** the note is stored and visible to the patient.
6.1	Record mood	**Given** a logged-in patient, **When** they submit mood=1–5 with optional notes, **Then** the entry persists and is date-stamped.
6.2	Mood trends	**Given** stored mood entries, **When** the patient opens the dashboard, **Then** weekly/monthly charts render from current data.
6.3	Clinician mood view	**Given** a psychiatrist assigned to a patient, **When** they open the patient timeline, **Then** mood trends display for the chosen range.
7.1	Journal CRUD	**Given** a patient, **When** they create/edit/delete an entry, **Then** the journal reflects the change immediately.
7.2	Journal search/tags	**Given** tagged entries, **When** the patient searches by tag or date, **Then** matching entries are listed.
7.3	Share journals (consent)	**Given** explicit consent on selected entries, **When** the psychiatrist views the record, **Then** only consented entries are accessible.
8.1	CBT thought record	**Given** a CBT form, **When** the patient completes required fields, **Then** the record is saved and versioned.
8.2	CBT export/summary	**Given** stored CBT records, **When** export/summary is requested, **Then** a readable view or PDF is produced.
8.3	Clinician review CBT	**Given** consented CBT records, **When** the psychiatrist opens the CBT tab, **Then** records are listed with dates and fields.

Continued on next page

R.No	Requirement Name	Acceptance Statement (Given—When—Then)
9.1	Verify psychiatrists	**Given** a pending application, **When** admin approves, **Then** the account role changes to psychiatrist and access is granted.
9.2	Manage reports	**Given** a user report, **When** admin sets a resolution status, **Then** the report is updated and audit-logged.
10.1	Audit sensitive actions	**Given** a sensitive operation, **When** it occurs (e.g., role change, deletion), **Then** an audit record with actor, time, and context is stored.
10.2	Query logs	**Given** admin filters (user/date/range), **When** a query is executed, **Then** matching audit entries are returned.
11.1	Notify appointment changes	**Given** an appointment status update, **When** it is committed, **Then** the patient and psychiatrist receive in-app notifications.
11.2	Notify new messages	**Given** a new DM, **When** the recipient is online, **Then** an in-app alert shows; otherwise it appears on next login.
12.1	Patient dashboard	**Given** authenticated patient data, **When** the dashboard opens, **Then** mood trends, journal count, and CBT progress appear.
12.2	Psychiatrist dashboard	**Given** assigned patients, **When** the psychiatrist opens their dashboard, **Then** appointments and key patient summaries render.
12.3	Admin dashboard	**Given** admin access, **When** the system dashboard loads, **Then** user growth, usage patterns, and flagged content are displayed.

2.3 Non-Functional Requirements

2.3.1 Software Quality Attributes

The quality targets that are measurable of the CalmConnect AI are as follows and determine the proper functioning of the system. These targets are service-level reliability, security and usability service-level objectives (SLOs). They offer objective guidelines regarding measuring the expectations of whether the platform is working as expected. Each target will be checked with the help of testing, monitoring and deployments reviews.

Performance

- **API latency:** p95 ≤ 350 ms for core endpoints (/auth/login, /mood, /messages,

/appointments) on a campus-grade network (10–20Mbps).

- **Throughput:**Sustain 100 concurrent users with no timeouts and $< 1\%$ error rate.
- **Front-end load:** First Contentful Paint $\leq 2.5\,\text{s}$ on low-end Android (2 GB RAM) over 3G; JS bundle $< 300\,\text{KB}$ (compressed), images optimized and lazy-loaded.
- **AI response time:** 95% of chatbot replies delivered within 4 s; clear typing indicator if slower.

Reliability & Availability

- **Uptime:** $\geq 99.5\%$ during semester demo window; health-check endpoint monitored.
- **Data integrity:** PostgreSQL constraints (FK, NOT NULL, UNIQUE) enforced; transactions wrap multi-row writes.
- **Retry policy:** Transient failures (HTTP 5xx, network) retried with exponential backoff (max 3).

Security

- **Transport:** All traffic over HTTPS (TLS 1.2+); HSTS enabled.
- **AuthN/AuthZ:** JWT access tokens (60 min) with refresh flow; RBAC for {patient, psychiatrist, admin}.
- **Passwords:** Bcrypt hashing ($\geq$10 rounds); no plaintext storage or email.
- **Input validation:** Centralized validation (server-side) and output encoding to prevent XSS/SQLi.
- **Secrets:** Stored only in environment variables; never committed to VCS.
- **Auditing:** Log sensitive actions (login, role change, export, delete) with actor, time, IP/device.

Privacy

- **Data minimization:** Collect only fields required for features; journals private by default.
- **Consent:** Explicit patient consent before sharing journals/CBT with psychiatrists.
- **Data export:** User can download personal data (JSON/PDF); exports auto-expire in 24 h.

Usability & Accessibility

- **Mobile-first UI:** Works on 360 px wide screens; tap targets $\geq$ 44 px.
- **Language:** English with Urdu-friendly content tone; simple wording suitable for students.
- **Accessibility:** Color contrast WCAG AA; keyboard navigation for all forms; focus states visible.
- **Onboarding:** New user can reach first mood entry within 90 s from sign-up on 3G.

Scalability

- Horizontal scaling supported for stateless Node.js API; DB uses connection pooling.
- Background jobs (emails/summaries) queued to avoid blocking requests.

Maintainability & Modularity

- **Code quality:** TypeScript strict mode; ESLint + Prettier; PR checks mandatory.
- **Layering:** Controllers $\rightarrow$ services $\rightarrow$ repositories; no direct DB calls from controllers.
- **Testability:** Unit tests for services; API tests for critical flows (auth, appointments, journals).

Observability

- **Logging:** Structured JSON logs with correlation IDs.
- **Metrics:** Request rate, latency percentiles, error rates; dashboard visible to admins.

Portability & Deployment

- **Containers:** Dockerized services; one-command boot on dev.
- **Environments:** {dev, staging, prod} with separate DBs and keys.

2.3.2 Other Non-Functional Requirements

Legal & Ethical Constraints

- **No medical claims:** The system provides self-help tools and guidance; it does not diagnose or prescribe.
- **Disclaimers & emergency notice:** Always visible in AI chat; crisis keywords trigger escalation banner.
- **Record keeping:** Audit logs retained for incident review.
- **Content moderation:** Admins can review flagged conversations for safety compliance.

Compliance & Standards

- OWASP ASVS 4.0 controls for web application security. [6]
- NIST SP 800-63B guidance for authentication and password rules. [5]
- IETF RFC 7519 (JWT) for token format and validation. [4]

Platform & Technology Constraints

- **Stack:** React+Vite+Tailwind (frontend), Node.js+Express (backend), PostgreSQL (DB), Prisma ORM, Gemini API. [7] [8] [9] [10] [11] [12]
- **Browsers:** Latest Chrome/Edge/Firefox on desktop; Chrome/Android WebView on Android (target low-end devices).
- **Hosting:** Cloud PaaS (e.g., Render/Vercel) with TLS termination; S3-compatible storage for exports (if used).

Localization & Content

- English UI with clear wording; Urdu-friendly copy where appropriate (FAQ, disclaimers).
- Date/time localized to PKT; 24-hour clock in clinical views.

Data Retention & Backup

- Daily automated DB backups; restore tested at least once per sprint.
- Soft-deletion for user-initiated removals where feasible; audit records not editable.

Disaster Recovery

- **RPO:** $\leq 24\,\text{h}$ (daily backups). **RTO:** $\leq 4\,\text{h}$ for core services.

Out of Scope for MVP

- Full offline-first operation (only cached static assets).
- Medical prescriptions/diagnosis, insurance billing, or e-prescription integrations.
- Multi-language UI beyond English/Urdu basics.

2.4 Requirement Gathering Techniques Used

In CalmConnect AI, we incorporated lightweight integrated techniques including human-friendly techniques to determine the functional requirements, in addition to sensitive UX requirements on Pakistani university context. It had a focus on transparency, confidentiality, and immediate validation. [18]

2.4.1 Interviews

Purpose: Get into the details of real workflows, pain points, and expectations; particularly regarding mental-health-related privacy and usability. [16]

- **Participants:** 3 total — 1 student (patient user), 1 psychiatrist, 1 admin.
- **Format:** 20–30 minutes per interview; consent taken; notes anonymised; audio notes only with permission.
- **Sample prompts:**
  - "When you feel stressed, how do you currently seek support?"
  - "What stops you from booking a counselling session early?"
  - "What data would you be comfortable sharing with a clinician? What should stay private?"
  - "How do you prefer reminders and follow-ups (WhatsApp, email, in-app)?"
- **Outputs:** User goals, privacy constraints, appointment and messaging flows, safety expectations (disclaimers, escalation).
- **Traceability:** Mapped to 3.1–3.5 (AI chat & safety), 4.1–4.3 (direct messaging), 5.1–5.4 (appointments), 7.1–7.3 (journals), plus NFRs on usability, privacy, and security.

2.4.2 Brainstorming

Purpose: Think through ideas, stakeholders and clusters quickly to form priorities of features, find edge cases or common privacy/safety concerns in the Pakistan student context. [17]

- **Participants:** 6 total — 3 students, 1 psychiatrist/counsellor, 1 admin/staff, and 1 facilitator (scribe).
- **Format:** 60–90 minutes; silent idea dump → round-robin sharing → voting → quick user-flow sketching.
- **Activities:**
  - **Idea prompts:** "Fastest first-use flow," "What would make you drop the app," "Signals that feel safe vs intrusive."
  - **Crazy-8s:** Eight rapid UI sketches for mood, journals, AI chat disclaimers, appointments.
  - **Red-flag review:** Identify privacy pitfalls, crisis phrases, consent touch-points.
- **Outputs:** Ranked feature themes (mood, journals, AI chat, appointments, DM), draft micro-flows, copy guidelines for disclaimers and consent, initial acceptance notes for edge cases.
- **Traceability:** Supports 6.1–6.3 (mood), 7.1–7.3 (journals), 3.1–3.5 (AI chat & safety), 5.1–5.4 (appointments), 4.1–4.3 (DM), 14.1 (route protection copy surfaces), and NFRs on usability, privacy, and accessibility.

2.4.3 Prototype

Purpose: A prototype that we created during the validation sprint was a React application that was configured as a frontend with partial Google Gemini. There was no live backend and database. We simulated flows using the lightweight mocks and local storage in the location where the behaviour of a server was required. This was aimed at UX testing, wording and safety limit controversy of CalmConnect AI, before investing in a full-stack develop. [8] [12] [18]

Prototype scope

- **Auth screens:** Registration, email confirmation page (stub), log in, role-specific patent, psychiatrist and administration landing pages.
- **AI chat:** Frame frame-first reply- disclaimer; emergency notice bullet, prompt encouraging thereof, prompt template encouragement.
- **Mood Recording:** Daily 1-5 mood record (note included); graphs according to local mocked records.

- **Journaling:** Add/Delete/ Edit journal entries; psychiatrist share option/ UI only).
- **Appointments:** Request/accept/reschedule/cancel flows shown via UI state; slot conflict message simulated.
- **Direct messaging:** Patient ↔ psychiatrist thread UI with read markers (no server persistence).
- **Admin views:** Approve psychiatrist, basic counters, and sample audit log table (static dataset).

How server behaviour was simulated

- **State/persistence:** `localStorage` for sessions, journals, and mood entries during a browser session.
- **API stubs:** Mock Service Worker (MSW) / Axios interceptors to fake REST responses (*200/401/409*) and delays.
- **Auth:** "JWT-like" token stored client-side (opaque string); role switching handled by guarded routes.
- **Charts:** Recharts (or similar) for weekly/monthly mood trends using mocked arrays.

Gemini integration

- **Live calls:** User messages were sent to Gemini using a client-side API key (demo only).
- **Guardrails in UI:** Pre-prompt system message enforced *no medical claims*; high-risk phrases detected in the client and used to *block* the next send + show crisis resources.
- **Limitations:** No rate-limiting, no audit trail and none of the server-side redaction. Key local environment variables were kept and are not safe to use.

Environment & run

- **Stack:** React + Vite + Tailwind CSS.
- **Run:** `npm install` → `npm run dev` (port 3000).
- **Config:** `.env` contained the Gemini API key (development only), and feature flags to toggle mocks.
- **Devices:** Tested on Laptops and PCs.

Evaluation procedure

- T1: Register → verification → login → land on dashboard based on role.
- T2: Open AI chat, read disclaimer, send normal prompt; then type a high-risk phrase to see escalation banner.
- T3: Log mood and confirm the dashboard charts refresh from mocked data.
- T4: Create a journal entry, tag it, toggle "share with psychiatrist", and confirm visibility badges.
- T5: Request an appointment; trigger the conflict message by selecting an occupied slot.
- T6: Send/receive two direct messages; observe read tick changes and timestamps.

Measures

- **Performance:** First Contentful Paint target ≤ 2.5 s on 3G; AI first token ≤ 4 s in 95% of tries.
- **Task success:** $\geq 85\%$ of scripted tasks completed without moderator help.
- **Safety signalling:** Participants noticed disclaimer and emergency banner before/at first AI reply.
- **Feedback notes:** Less English and urdu-friendly mood labels, more obvious empty states, larger text consent.

Gaps & planned backend work

- **Security:** Move keys server-side; enforce JWT, RBAC, and rate limits on a real API. [4]
- **Persistence:** Replace localStorage/MSW with PostgreSQL + Prisma; add audit logs for sensitive actions. [7]
- **Safety:** Server-side crisis detection and escalation logging; configurable keyword lists.
- **Messaging/Appts:** Real-time or polling backend to store threads and avoid double-booking.

Traceability

- Validated at UI level: 3.1–3.5 (AI Chat & safety prompts), 5.1–5.4 (appointments), 6.1–6.3/7.1–7.3 (mood tracking/journaling), 12.1–12.3 (dashboards).
- Deferred to backend: 9.1–9.2/10.1–10.2 (admin verification/audit/notifications), 12.1–12.3 (data access, exports, enforcement).

Note: This was a prototype that was created on the front end and was used to test FYP goals of early usability and safety test within a low-risk system. Results will be used to employ our complete stack SRS and minimize rework in API/database implementation.

2.5 Time Frame

For CalmConnect AI, The requirements, design, implementation, and validation are 32 weeks. The timeline provides protection against AI risk and privacy at early stages, then AI features are provided in dedicated sprints, and to finalize it there is hardening followed by UAT and handover. All the milestones and scope are in line with acceptance criteria in Table 2.13 (1.1…12.3).

Breakdown

- **Weeks 1–4: Initiation & Requirements**
  - Finalise problem statement, stakeholders, and non-clinical, privacy-first scope.
  - Semi-structured interviews (student, psychiatrist/counsellor, admin) and short questionnaire.
  - Synthesize into FR/NFR; draft SRS v0.1 with primary user flows and domain entities.
- **Weeks 5–6: Architecture & DB Design**
  - C4 overview; ERD (3NF); API surface; spikes for auth and Gemini boundaries.
  - Agree Given–When–Then format; start traceability to FRs in Table 2.13.
- **Weeks 7–10: Sprint 1**
  - Implement 1.1 (Register), 1.2 (Email verification), 1.3 (Login), 1.4 (Password reset), 1.5 (Account block).
  - Lay foundations (Route protection: role + ownership).
- **Weeks 11–14: Sprint 2**
  - Implement 6.1–6.3 (mood) and FR-7.1–7.3 (journals & consented sharing).
  - Patient dashboard slices per 12.1.
- **Weeks 15–18: Sprint 3**
  - Implement 4.1–4.3 (DM) and 5.1–5.4 (appointments).
  - Psychiatrist dashboard slices per 12.2.
- **Weeks 19–22: Sprint 4**
  - Implement 3.1–3.5 (AI chat sessions, disclaimers, non-clinical responses, crisis detection, transcript/summary).

  - Align UI copy with safety acceptance (3.2–3.4).

- **Weeks 23–26: Sprint 5**
  - Implement 9.1 (Verify psychiatrists) and 9.2 (Manage reports).
  - Implement 10.1–10.2 (audit & query logs) and 11.1–11.2 (notifications).
  - Complete 12.1–12.3 (dashboards).
- **Weeks 27–28: Hardening**
  - Verify protected routes; threat-model passes and load checks.
- **Weeks 29–30: UAT**
  - Run Given–When–Then acceptance for all FRs in Table 2.13; triage and fix defects.
- **Weeks 31–32: Handover & Viva**
  - Final demo, docs bundle, poster/slides, tagged repository; sign-off.

2.5.1 Milestones & Deliverables

Table 2.14 shows the project schedule with the important milestones and deliverables. Each stage is directly correlated to the acceptance criteria established in Table 2.13, which guarantees that planning and development sprint phases are linked to the validation phases. This schedule gives a clear picture of the progress of the project since its start to the completion handover.

Table 2.14: Milestones & Deliverables

Weeks	**Milestone**	**Key Deliverables (mapped to Table 2.13)**
1–2	Kick-off & planning	Team finalisation; scope statement; interview guide; consent templates.
3–4	Elicitation R1	Anonymised interview notes; questionnaire results; initial FR/NFR set; SRS v0.1 outline.
5–6	Arch & DB design	C4 overview; ERD; API map; spikes for auth/Gemini; acceptance template and traceability shell.
7–10	Sprint 1: Auth/Core	1.1, 1.2, 1.3, 1.4, 1.5 implemented; CI configured.
11–14	Sprint 2: Mood & Journals	6.1–6.3 and 7.1–7.3; patient dashboard slices (12.1).
15–18	Sprint 3: Appts & DM	4.1–4.3 and 5.1–5.4; psychiatrist dashboard slices (12.2).

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Weeks	Milestone	Key Deliverables (mapped to Table 2.13)
19–22	Sprint 4: AI & Safety	3.1–3.5 (sessions, disclaimers, crisis detection, transcripts/summary).
23–26	Sprint 5: Dashboards /Admin /Audit /Notif. /Data	9.1, 9.2; 10.1–10.2; 11.1–11.2; 12.1–12.3; completed.
27–28	Perf/Sec hardening	verification across protected routes; load tests; remediation.
29–30	UAT	Full Given–When–Then run against all FRs; defect fixes and re-tests; UAT report.
31–32	Handover/Viva	SRS v1.0 bundle; traceability matrix; poster/slides; tagged repository and sign-off.

Chapter 3

SOFTWARE PROJECT PLAN

3.1 Deliverables of the Project

The following artifacts will be submitted for CalmConnect AI. Each item is version controlled in a public/private Git repository and packaged for the final FYP submission.

- **SRS (v1.0)**—Complete Software Requirements Specification covering scope, stakeholders, FRs/NFRs, acceptance criteria, and traceability matrix.
- **System Design Pack**—Overview of architecture, C4 diagrams, PostgreSQL ERD, API specification (openapi/swagger), sequence and state charts.
- **Working Prototype (Frontend)**—React + Vite + Tailwind-UI react-integrated journey journals, mood messaging, and appointments mock data Displayed by Gemini; crisis banner; Newspaper clip scandal warnings; */disclaimers.html disclaimers.marketing plans; disclaimers.html criticism plan; disclaimers.html conduct manual; and disclaimers.html simple disclaimed instruction.
- **Source Code Repository**—Formatted/nested code with commit history sources (and issues, tags), README.mod on how to set up and run.
- **Test Plan & Evidence**—FRs, usability scripts, bug list, screenshots/video pass/fail evidence Test cases, Evidence Test plans This evidence is referred to as Test Plan.
- **Security & Privacy Checklist**—OWASP ASVS coverage summary, audit-log events, password policy, security and privacy checklist -RBAC matrix, change input validation checklist, input validation checklist, and security and privacy checklist.
- **User Documentation**—Quick Start Guide to patients and psychiatrists, Frequently Asked Questions, safety disclaimers, crisis resources.
- **Admin/DevOps Guide**—Build/Run/Mocke back-end Tests Environment variables, build/ run, mock back-end, backup/ restore.
- **Datasets & Seeds**—Dat seed information just the seed data to use on the demo (mood entries, journals, appointments) as well as scripts to load/unload the seeds.
- **Ethics Note**—Non-clinical scope statement, templates of consent, which were utilised in the interview/ prototype testing.
- **Snaponics**—Android (3G) performance of page loads and AI response time logs, and optimisation comments.
- **Presentation Slides & Poster**—Final problem, technique, demonstration, analysis, future work viva deck and one-page academic poster.
- **Project Log & Reflections**—Weekly progress log, risks/mitigations and lessons learnt in future iterations.

Submission bundle layout

```
CalmConnect-AI/
  docs/      (SRS, design pack, test plan, ethics, user/admin guides)
  src/       (fullstack prototype code)
  seeds/     (demo data i.e. admin.json)
  README.md  (how to build, run, and grade)
```

3.2 Software Project Management Plan

Software project management plan outlines how CalmConnect AI is going to be implemented within the required timeframe, scope and acceptable quality. It adopts schedule (including WBS and CPM), quality assurance, milestones, planning and documentation of resources. This proposal is aimed at the full system (React/vite/tailwind, Node.js/express, PostgreSQL, Gemini API), there was a prototype of the frontend only, which was utilized during the first 32 weeks.

3.2.1 Project Planning

The analysis is friendly rhythm classical project that is comprised of short iterations and certain gates.

Approach

- **Process:** Agile/Scrum-ban (2-week sprints), daily lightweight stand-ups, weekly review with supervisor.
- **Change control:** Any scopes changes that have been identified in Git as problems, the SRS forms the foundation, and it is altered via pull requests.
- **Security/Safety risk posture:** Done safety-risks (auth, RBAC, disclaimers, escalation, etc.) before other work to avoid duplication.
- **Definition of Done (DoD):** FINAL: coding, reviewed, ran lint tests, passing test, acceptance tests, revised documentation, demonstrable.

3.2.1.1 Milestones Plan

This summary of key project milestones and exit criteria is summarized in Table 3.1. Every milestone has definite completion requirements so that one is ready to proceed to the next stage. The structure facilitates the monitoring of the progress, quality, and ensuring that the main objectives are achieved in the project lifetime.

Table 3.1: Major Milestones & Exit Criteria

Week(s)	Milestone	Exit Criteria
1–4	Requirements Baseline	SRS v1.0 signed; FR/NFR complete; acceptance criteria drafted.
5–7	Architecture & DB Design	C4 overview, ERD (3NF), API map; spikes for auth & Gemini.
8–11	Foundations (BE/FE)	Backend scaffolding (auth/RBAC, ORM) and frontend routing/state/UI.
12–15	Sprint 1 (Core Features I)	Auth wiring; shared APIs; route guards laid.
16–19	Sprint 2 (Journals, Mood, Appts/DM, AI)	CRUD journals; mood logging; appointments & messaging; AI disclaimers/escalation.
20–24	Sprint 3 (Dashboards, Admin, Audit, Notif., Data)	Dashboards; admin review; audit logs & queries; notifications; data export/removal.
25–27	Hardening	Security/perf checks; RBAC/ownership verification; defect remediation.
28–30	UAT	Given–When–Then acceptance run; fixes and re-tests; UAT report.
31–32	Handover/Viva	Final demo; docs bundle; poster/slides; repository tagged.

3.2.1.2 Documentation Plan

- **SRS & Traceability:** FR/NFR, acceptance criteria, matrix FR $\leftrightarrow$ tests.
- **Design Pack:** Architecture (C4), ERD, API (OpenAPI), sequence/state diagrams.
- **Test Plan:** Unit/API/UX test cases, data sets, evidence screenshots, UAT script.
- **Documentation:** A guide is provided in ops guide, Environment variables, build/run, backup/optimize, seeds, li-notes.
- **User Manual:** Patients/psychiatrists quick start, safety notice, Frequently asked questions.
- **Security Checklist:** OWASP ASVS mapping, password/JWT policy, audit events.

3.2.1.3 Resources Plan

Human

- **Team Lead:** Zain Ul Abideen Malik—coordination, testing, security.

- **Frontend:** Abdul Rehman Shabir—React/Vite/Tailwind.
- **Backend:** Azhar Ali—Express, PostgreSQL, integrations.
- **Supervisor:** Mr. Amjad Saleem Rana—reviews and academic compliance.

Tools & Services: GitHub (issues/PRs), Node.js, React/Vite, Tailwind, Express, PostgreSQL, Gemini API, Postman.

Table 3.2 presents a provisional budget of the project, the projected development and academic budget. It finances all the crucial expenses such as hosting, presentation and communication final materials and a buffer fund. It is a realistic budget that should be used in an academic project that is semester-based.

Table 3.2: Indicative Budget

Item	Cost (PKR)	Notes
Domain/Hosting (semester)	6,000	Low-tier PaaS + TLS.
Printing/Poster	2,500	Final viva materials.
Contingency (10%)	850	Unforeseen.
Total (approx.)	9,350	Academic scale.

3.2.1.4 Quality Plan

Standards & Policies

- **Security:** OWASP ASVS practices; TLS-only; bcrypt ($\geq$10); JWT expiry 60 min; RBAC everywhere; audit sensitive actions.
- **Code Quality:** TypeScript strict, ESLint/Prettier, PR reviews, CI build + tests.
- **Testing:** Unit (services) tests, API (auth, journals, mood, appts, DM) tests, flow websites; acceptance (FR) [Given] [When] [Then] tests.
- **Performance:** core route API (p95) not exceeding 350ms; initial AI first token (95) not exceeding 4 s; FCP (3G) not exceeding 2.5s.
- **Usability:** Urdu user-friendly copy, WCAG AA contrast; first mood entry time of less than 90s.
- **Definition of Done:** Tests run, acceptance criteria have been fulfilled, security checklist has also been fulfilled, documentation has also been updated, demonstration has also been recorded.

3.2.2 Project Scheduling

Momentum: Sprint of two weeks, with backlog refinement on Friday, planning of sprint on Monday, mid-sprint review with supervisor and end of sprint retro.

3.2.2.1 Gantt Chart

Figure 3.1 gives a distinct narration about Sep 2025 through May 2026. The SRS is locked, architecture shaped and BE/FE foundations laid in parallel. Next we deploy three targeted sprints, namely auth/core, journals+mood+appts/DM with AI holds and dashboards/admin/audit/notifications/data. We end with hardening, a sound UAT execution with GivenWhenThen and empty handover to viva. [15]

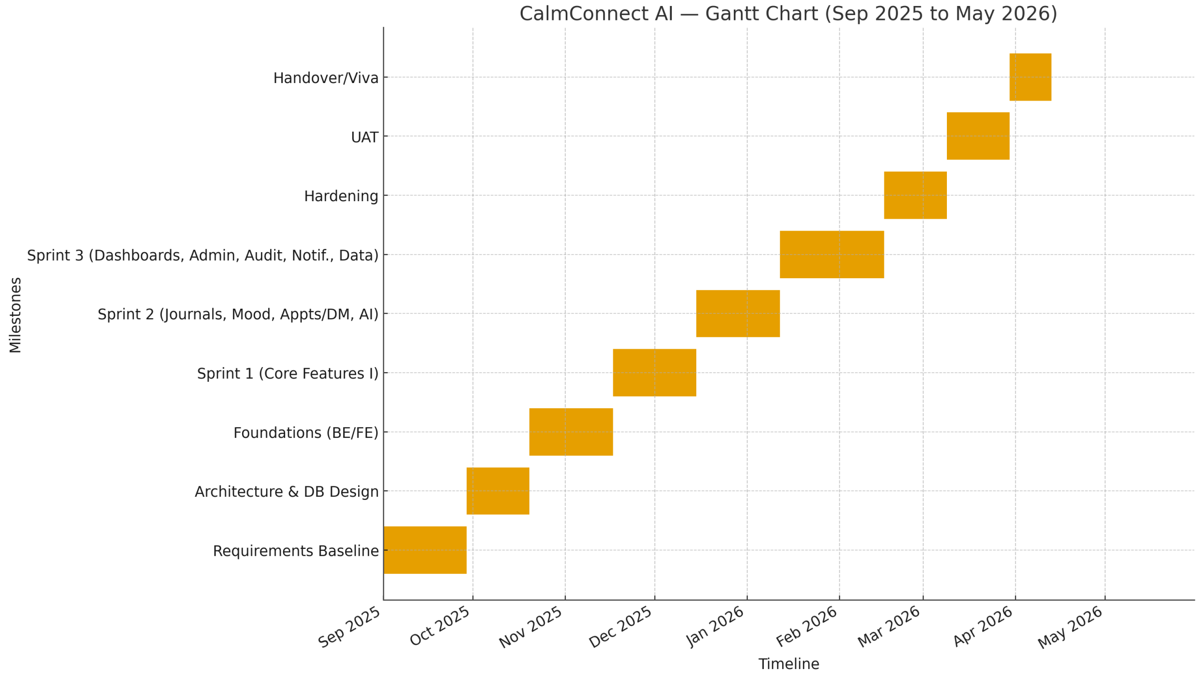


Figure 3.1: Gantt Chart

3.2.2.2 Work Breakdown Structure

- **Requirements & Analysis**
  - 1.1 Stakeholder interviews & brainstorming
  - 1.2 SRS (FR/NFR, acceptance criteria)
  - 1.3 Risk register (security, safety, privacy)
- **Design**
  - 2.1 Architecture & C4
  - 2.2 ERD & schema migration plan
  - 2.3 API design (OpenAPI)
  - 2.4 UI wireframes & style guide
- **Implementation**
  - 3.1 Backend foundation (Express, Prisma, auth/RBAC)
  - 3.2 Frontend foundation (React, routing, state)

- 3.3 Features: Journals, Mood, Appointments, Messaging
- 3.4 AI integration (Gemini + guardrails)
- 3.5 Dashboards, Admin, Audit, Notifications, Data export/removal

- **Testing & Quality**
  - 4.1 Unit/API tests, seed data
  - 4.2 UX/usability passes, accessibility checks
  - 4.3 Security checklist & fixes

- **Deployment & Handover**
  - 5.1 Hosting, TLS, environment config
  - 5.2 Backups, monitoring, runbook
  - 5.3 User/admin guides, poster, viva slides

3.2.2.3 Critical Path Method

We model activities, dependencies, and run forward/backward passes. The plan targets a 32-week timeline.

Table 3.3 presents the activities that were adopted to analyze the Critical Path Method (CPM) of the project. It displays every large activity, the estimated time of each activity and the dependence relationships among the activities. Such a breakdown assists in finding out the task sequencing and enables proper scheduling and maintaining the timeline in the project.

Table 3.3: CPM Activities

ID	Activity	Week Dur.	Predecessors
A	Requirements baseline	4	–
B	Architecture & DB design	3	A
C	Backend foundation (auth/RBAC, ORM)	4	B
D	Frontend foundation (routing/state/UI)	4	B
E	Core features I (auth wiring, shared APIs)	4	C, D
F	Core features II (journals, mood)	4	E
G	Appointments & messaging	4	E
H	AI integration & safety banners	4	E
I	Dashboards/Admin/Audit/Notif/Data	5	F, G, H
J	Hardening (security/perf) & UAT prep	3	I
K	UAT & fixes	3	J
L	Deployment & handover	2	K

The timing analysis of Critical Path Method (CPM) is given in Table 3.4. It presents the earliest and latest start and finish time of each activity and float values. The table aids in the determination of the critical tasks that have zero slack and ensures the duration of the project and the time-infrastructure.

Table 3.4: CPM Timing

ID	ES	EF	LS	LF	Float	Notes
A	0	4	0	4	0	Baseline SRS complete.
B	4	7	4	7	0	Architecture/DB locked.
C	7	11	7	11	0	Backend scaffolding.
D	7	11	7	11	0	Frontend scaffolding.
E	11	15	11	15	0	Shared contracts working.
F	15	19	15	19	0	Journals/mood done.
G	15	19	15	19	0	Appts/DM done.
H	15	19	15	19	0	AI/safety integrated.
I	19	24	19	24	0	Dashboards/admin/audit/notifications/data.
J	24	27	24	27	0	Security/perf and UAT prep.
K	27	30	27	30	0	UAT execution window.
L	30	32	30	32	0	Viva-ready.

Critical Path: A → B → C → E → F → I → J → K → L (length = **32 weeks**). Activities D, G, and H join E/F/I without float on this plan.

Risk & Mitigation Highlights

- **Security gaps** (auth/RBAC) — prioritised in Foundations and Core Features I; checklist per OWASP ASVS.
- **AI safety** — combined view of Core Features window; disclaimers/escalation verified ones.
- **Bandwidth/device restrictions** — Performance limits put in place (size of bundle, lazy loading).
- **Team availability** — Team hardening and UAT windows ensure final quality on the demos, daily commits, and tracking issues keep the momentum.

3.3 Managerial Process

3.3.1 Management Objectives and Priorities

The section below indicates how CalmConnect AI will be handled to ensure that the work is organized and predictable throughout the entire 32-week project. It is concerned with maintaining scope clarity, realistic deadlines and quality throughout. The concept of safety and reliability are not considered as the afterthoughts during development. All the objectives are established in quantifiable terms so that the progress is easy to measure and determine. These were requirements that are specific to its project milestones Table 3.1 and acceptance requirements Table 2.13.

Management Objectives

- **Scope control:** Deliver the baseline SRS features (1.1..14.1 per Table 2.13) with $\geq$ 95% acceptance tests passing by Week 30; reach 100% pass on critical FRs (1.1–1.4, 3.1–3.5, 5.1–5.4) by Week 28.
- **Schedule adherence:** Exit criteria of hitting milestones (Table 3.1) with 10 or below slippage; any lag longer than a period of 1 week raises an abort of replanning on the following review.
- **Quality:** At code-freeze There are 3 or fewer major issues per sprint; the coverage of units/API of core services is at least 60% at Week 27.
- **Security & safety:** Zero known critical OWASP results at viva; AI chat will display disruptant disclaimer and crisis banner consistently (3.2-3.4) at all times by Week 22.
- **Performance:** p95 latency of core APIs: maximum of 350ms at Week 27; first AI response: the best is 4s in 95 attempts when using campus network.
- **Usability:** SUS score 70 or above on the functioning build by Week 30 and You can first post mood in 90s since sign-up.
- **Documentation:** SRS v1.0, Design Pack, Test Plan, Ops Guide, and User Guides done by the end of Week 31 and functional with the running build.

Priorities

1. **Safety & privacy** Details of disclaimers, escalation, consent and RBAC: over addition of new features.
2. **Primary use** (author, journals, mood, appointments, messages) of cosmetic UI.
3. **Reliability & performance** Due to unneeded integrations, there is dependability and performance.
4. **Clarity of UX copy** Writing of simplicity UX vs. feature-filled writing of UX.

Table 3.5 demonstrates the RACI matrix of the key project activities. It is well defined on the duties and obligations, consultation and information on all the major work. This framework can withstand role ambiguity and work better on the relationship between the team members and administration of the project.

Table 3.5: RACI for Key Activities

Activity	Lead (R)	Approve (A)	Consult (C)	Inform (I)
SRS baseline & changes	Team Lead	Supervisor	Clinician	Team
Architecture/ERD/API	Backend	Supervisor	Frontend	Team
Security & privacy	Team Lead	Supervisor	Backend	Team
Frontend UX	Frontend	Supervisor	Clinician	Team
Testing & UAT	Backend	Supervisor	Frontend	Team
Deployment/Handover	Team Lead	Supervisor	Backend	Stakeholders

Change Control

- Every scope change created as GitHub Issues to which it has an impact (time/quality) and is assigned to FR/NFR.
- Peer-reviewed resolutions on the issue and consolidated through Pull Request; SRS refined in the PR.
- Alterations within the fourth week of weeks 28-32 must be approved by the supervisor and included on a risk note on the release log.

The communication plan in the project is as presented in the Table 3.6. It determines the large gatherings, how often the gatherings are conducted and the results. Such plan will allow tracking the progress frequently, receive timely feedback and align the team with the supervisor properly.

Table 3.6: Communication Plan

Meeting	Frequency	Purpose / Outputs
Daily stand-up (chat)	Daily (10 min)	Progress, blockers, plan for the day.
Sprint planning	Bi-weekly	Select backlog, define acceptance, estimate tasks.
Supervisor sync	Weekly	Demo increment, get feedback, confirm priorities.
Design/Code review	Per PR	Enforce standards, catch defects early.
UAT dry run	Weeks 29–30	Verify demo scripts, fix critical issues.

3.3.2 Assumptions and Constraints

The project is subject to the following assumptions, dependencies and constraints and hence determine the scope of the project, design and execution.

Assumptions

- **Academic schedule:** Taking exams/holidays; Weeks 25-27 and Weeks 29-30 buffers.
- **Internet connectivity:** Has Internet access sufficient to support Gemini API calls during demos, and the amount of free-tier quota is sufficient to test it.
- **Access to stakeholders:** Availability of supervisor and at least one clinician periodically to conduct reviews.
- **Out of clinical scope:** The point that has to be stressed is that the system is not a diagnostic and prescription service but should rather be seen as a guiding tool (advice/work journals/CBT prompts).
- **Test data:** Demo and testing data: There are no real PHI of actual patients in the development environments, Demo and testing data.
- **Devices & network:** The device will also have the low end android gadgets and variable bandwidth (3G/4G); interface will be as such.

Dependencies

- **External APIs:** Availability, rate limits, pricing of Google Gemini, and latency.
- **Hosting:** uptime and termination of TLS (Cloud PaaS e.g. Render/ Vercel)
- **Email service:** The service of verification and resetting password requires SMTP/provider reliability.
- **React/Vite/Tailwind/Libraries:** Libraries do not require any customization at the conclusion of the project.

Constraints

- **Time:** 32 weeks total number; Tables 3.3 and 3.4 constitute the critical path. Late slips will squeeze modalities and handover modalities.
- **Budget:** Free primarily, minimal sum of money on hosting, printing of the posters and contingencies.
- **Security & privacy:** Easy RBAC is mandatory, passwords will be hashed using the bcrypt hash algorithm, and transport will be carried out over TLS and sensitive operations will be audited.
- **Legal/ethical:** No medical statement; well marked-off disclaimers and emergency departments; permission of journal/CBT disclosures of records.
- **Performance:** The low-bandwidth users will have their performance requirements met with respect to bundle size and latency budgets.

- **Data retention:** Backups Backups are retained; soft-deletion/anonymity Backups are good, audit records read-only.
- **Technology:** Relational store: PostgreSQL; no changes of DB in-mid-project.

Risk Highlights & Mitigation

- **API limits or outages:** Backup had safe response, stored responds; are willing to make interaction records visible.
- **Security recession:** Checklists and PR review Weeks 28-30.
- **Bandwidth/device variance:** Trigger on devices with low end Compress and lazy-load Android early.
- **Variable availability of a team:** Visibly track activities on GitHub Projects; make smaller and independent of each other; have some buffer between a mid-sprint.

3.4 Project Risk Management

Risk is the likelihood that there will be unfortunate consequences of uncertain events or conditions on the scope, schedule, cost, quality, use of safety, or user confidence of CalmConnect AI. This section clarifies the ways of identifying, analysing, responding to, and monitoring risks in the project.

3.4.1 Risk Management Plan

The plan takes into account high impact risk and the low impact risk. It determines usable scales of likelihood and impact, roles and creates artefacts (risk register, reviews, and dashboards) to monitor the project within the 32-week schedule.

3.4.1.1 Purpose

- Proactively minimise threats to delivery (schedule slippage, quality regressions, AI-safety incidents).
- Offer a systematic way of rating risks and select answers (avoid, reduce, transfer, accept).
- Maintain traceability from risks to mitigation actions and owners.

3.4.1.2 Roles and Responsibilities

The RACI matrix of risk management activities is provided in Table 3.7. It spells out the clear roles and accountability in the identification, monitoring and mitigation of project risks. The structure will provide proactive management of risks and provides the information about the stakeholders at every stage of project life.

Table 3.7: Risk Management RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Maintain risk register	Team Lead	Team Lead	Backend, Frontend	Team
Security & privacy risks	Team Lead	Team Lead	Backend	Team
AI safety & escalation risks	Frontend	Team Lead	Clinician	Team
Schedule/capacity risks	Team Lead	Team Lead	Backend, Frontend	Stakeholders
Mitigation action owners	Backend	Team Lead	Supervisor	Team

3.4.2 Risk Management Activities

3.4.2.1 Risk Identification

- **Sources:** requirements workshops, architecture/design reviews, code reviews, security checklist (OWASP ASVS), dependency updates, prototype tests, stakeholder feedback.
- **Triggers:** missed exit criteria, rising defect trends, API quota warnings, performance regressions, policy changes.
- **Artefact: Risk Register** (Table 3.9) updated weekly.

3.4.2.2 Risk Analysis

Table 3.8 defines the scales which are used in the determination of the likelihood and impact in the risk in the project. The frequency of the risk happening and the level to how much it would affect schedule, quality or safety is the level of each score meaning. These scales are considered to provide a coherent structure of evaluation and ranking of risks.

Table 3.8: Likelihood and Impact Scales

Score	Likelihood (L)	Impact (I)
1	Rare (<5%)	Negligible (minor annoyance, no rework)
2	Unlikely (5–15%)	Low (small task slip, trivial fix)
3	Possible (15–35%)	Medium (rework in one sprint, minor scope cut)

Continued on next page

Score	Likelihood (L)	Impact (I)
4	Likely (35–60%)	High (slip $\geq$ one sprint, user-visible quality hit)
5	Almost certain (>60%)	Critical (derails milestone, safety/security risk)

Risk Exposure:

$$\text{Risk Score} = L \times I \quad \text{(range 1 to 25)}$$

Heat levels: **Low** (1–6), **Moderate** (7–12), **High** (13–19), **Critical** (20–25).

3.4.2.3 Risk Response Planning

- **Avoid:** Change plan/technology to eliminate the risk cause.
- **Reduce:** Add tests, spikes, reviews, budgets to lower L or I.
- **Transfer:** Use external service/support; schedule buffer.
- **Accept:** Document and monitor; prepare contingency.

3.4.2.4 Rating Risk Likelihood and Impact

- Each risk gets initial L/I scores; residual L/I are re-scored after mitigation.
- High/critical risks require a dated mitigation task and an *owner*.

3.4.2.5 Risk Monitoring and Control

- Weekly review in supervisor sync; dashboard shows open risks, due mitigations, and trend.
- Triggers create issues automatically (e.g., AI escalation failure → create high-priority bug).
- Closure criteria: mitigation delivered, risk re-scored to Low, or risk no longer applicable.

3.4.2.6 Risk Assessment

- Risks are linked to the **Critical Path** and **Milestones** (see CPM and Milestones tables).
- Security & privacy risks must be closed before the **Hardening/UAT** gate (Weeks 25–30).

Table 3.9 The project risk register, which contains the main technical, schedule, and operational risks, is represented. The likelihood and impact scores are used to evaluate each risk and give them the responsible owner with the response strategies and mitigation measures. This register aids the process of constantly monitoring the risks and making a sound decision during the project.

Table 3.9: Risk Register

ID	Risk	Owner	L	I	Score	Response & Key Mitigations
R1	**Schedule slip**	Team Lead	3	4	12	Reduce: decompose work; daily stand-ups; WIP limits; early integration spikes; buffer in Weeks 28–30.
R2	**AI safety failure**	Frontend	2	5	10	Reduce/Avoid: UI guardrails; unit tests on crisis keywords; UX walkthrough; supervisor sign-off before UAT (Week 30).
R3	**Security regression**	Team Lead	3	5	15	Reduce: OWASP ASVS checklist; code-review gates; secrets in env; JWT expiry 60 min; audit logs for sensitive actions.
R4	**Performance**	Frontend	4	3	12	Reduce: code-splitting; image compression; caching; API p95 $\leq$ 350 ms; FCP $\leq$ 2.5 s.
R5	**External API**	Backend	3	4	12	Transfer/Reduce: timeouts; graceful fallback; cache last safe response; pre-record demo.
R6	**DB integrity**	Backend	3	4	12	Reduce: FK/unique constraints; transactional writes; no double-booking checks; migration tests.
R7	**Privacy breach**	Backend	2	5	10	Avoid/Reduce: consent flags; server-side enforcement; audit every access; least-privilege exposure.
R8	**Team capacity**	Team Lead	4	3	12	Accept/Reduce: smaller stories; cross-pairing; move non-critical items past Week 24; keep buffer near viva.
R9	**Requirements churn**	Team Lead	3	3	9	Reduce: baseline SRS by Week 4; change control via issues/PR; time-box discovery spikes.
R10	**Hosting/infra outage**	Team Lead	2	4	8	Transfer/Accept: health checks; simple failover (local run); pre-recorded video; printed screenshots.

Continued on next page

ID	Risk	Owner	L	I	Score	Response & Key Mitigations
R11	**Data loss**	Backend	2	4	8	Reduce: nightly backups; restore test once per sprint; seed data for demo.
R12	**Accessibility**	Frontend	3	3	9	Reduce: usability tests; Urdu-friendly copy; WCAG AA contrast; refine labels/tooltips.
R13	**Legal/ethical**	Team Lead	2	5	10	Avoid: copy review with supervisor; fixed disclaimers; escalation to human help.
R14	**Audit log gaps**	Backend	2	4	8	Reduce: central audit middleware; log access/deletes/exports; restricted viewer; tamper-evident storage.
R15	**Dependency update**	Backend	3	3	9	Reduce: freeze versions by Week 20; CI smoke tests on lockfile changes; post-viva upgrades.

Table 3.10 presents the risk heat map that is qualitative and performed in the project. It brings together both likelihood (L) and impact (I) in classifying risks into Low, Medium, High and Critical. This is a matrix used to prioritize the risks and which risks need to be mitigated or kept a closer eye on.

Table 3.10: Qualitative Heat Map

L\I	1	2	3	4	5
1	L	L	L	L	M
2	L	L	M	M	H
3	L	M	M	H	H
4	M	M	H	H	C
5	M	H	H	C	C

Legend: L=Low, M=Moderate, H=High, C=Critical. Items in H/C must have an owner, a dated mitigation, and appear on the weekly review agenda.

Monitoring Artefacts

- **Risk Register** (Table 3.9) with current/residual scores.
- **Risk Burn-down** (optional): count of H/C risks over weeks.
- **Gate Checks:** Security & AI-safety risks must be *Low* before Hardening/UAT (Weeks 25–30).

Chapter 4

FUNCTIONAL ANALYSIS AND MODELING

4.1 Use Case Modeling

CalmConnect AI has a use case diagram that gives a rough overview of how various users engage with the system. The system identifies all features that make CalmConnect AI as the system boundary, and the real persons who make use of the features can be found on the left as part of the external actors. The primary participants include Patient, Psychiatrist and Admin. Every of the ovals within the border reflects a service or an activity provided by the platform, including account management, AI-assisted service, appointments, journals and administration. [14]

Most of the blue and light blue use cases are interacted with by patients. They create accounts, activate, change passwords, and check their mail in order to do any mental-health functionality. Upon authentication, a patient begins to chat using AI, perform CBT activities, use journals, and see mood patterns. View role based dashboard use case provides a quick look at the mood history, future appointments and recent activity organizational to the role of the user. The patients also request their appointments and watch the status of their appointments and send and receive direct messages with psychiatrists. Another use case encompasses account deletion requests, which is based on the privacy and data-protection concerns in the report.

Psychiatrists are interested in the use cases involving treatment. They log in, use the role-based dashboard and manage the flow of appointments: they receive appointment requests, update the status of appointments, and see scheduled appointments. Direct messaging is a follow-up question or short check-in between the psychiatrists and patients. With these use cases, psychiatrists can see the progress of the patient using mood trends, CBT work and journals whilst the system also abides by the access rules and consent model described in the requirements.

Admin actor takes care of supervision and the governance of platforms. The user management, seeing activity logs, and registering psychiatrists are all examples of user cases. Three additional specific user management actions, including deleting user, blocking user, and unblocking user, are the extensions of the user management use case. Such relations demonstrate that deletion or blocking will not exist in isolation; both processes are always founded upon the wider usermanagement process. The use of activity logs helps in security, audit trail and review of incidents as stipulated in the nonfunctional requirements.

Emotional support given by AI is presented as a stand-alone yellow use case, which is called Receives AI emotional support (non clinical). Detect high risk phrases and show escalation is an extension of this use case. The extend relationship implies that the second use case will only be caused in special situations, such as when the message has self-harm or crisis words. Such cases will also show sign signals of escalation (e.g. crisis contacts, safety notes), but regular supportive conversations will still take place through the primary AI chat use cases.

Use Case Diagram:
Figure 4.1 in a single perspective, summarises the operational capabilities of CalmConnect AI. It displays the features associated with which type of user, the administration and safety features and core wellness features are adjacent, and specialised behaviour (impeding users or escalating a crisis) are pulled out of the normal sequence of actions. This model will subsequently be used to direct elaborate scenarios, interface, and implementation of the respective modules. [14]

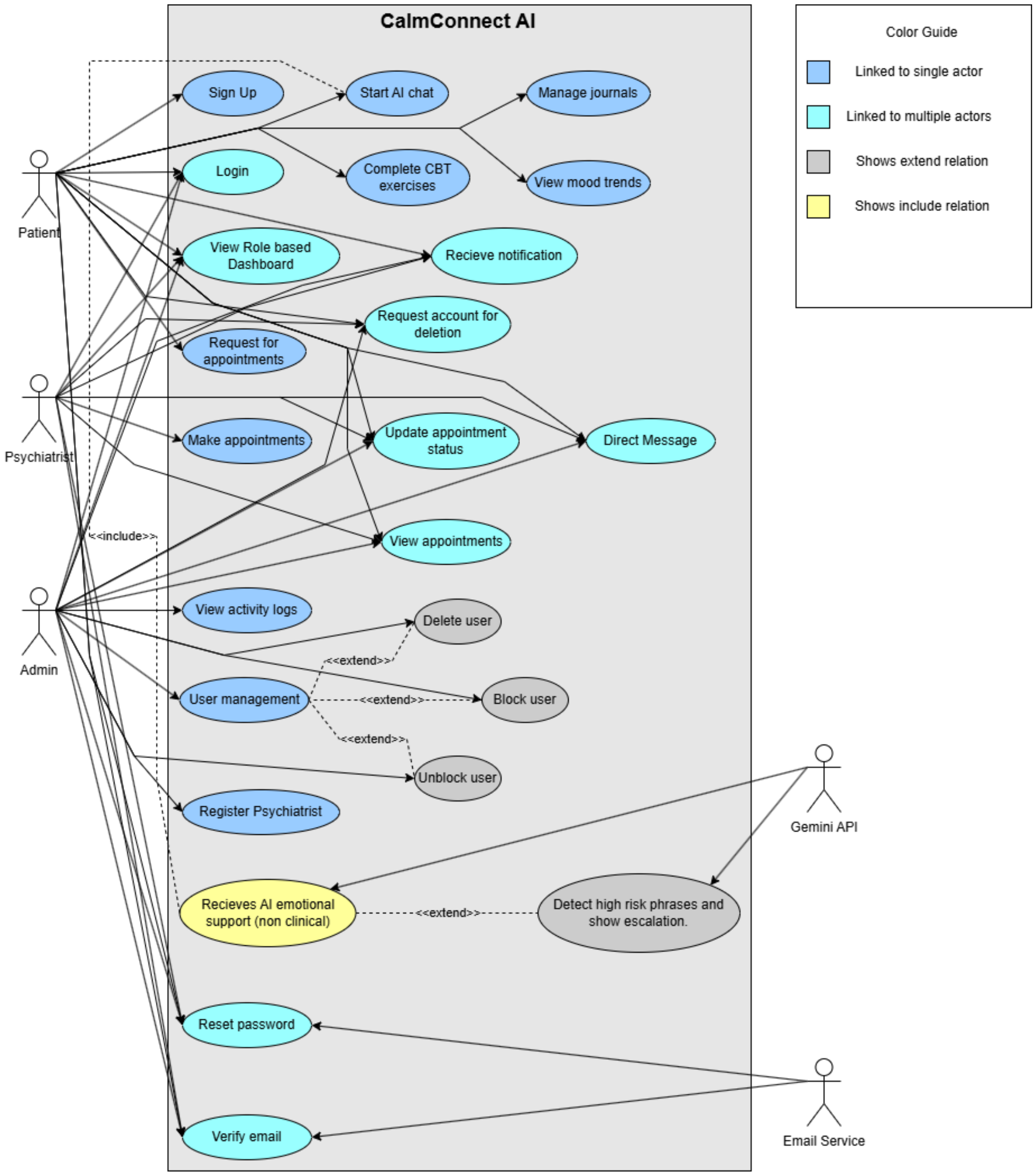


Figure 4.1: Use Case Diagram

4.1.1 User Stories

The user stories will outline the system as viewed by those who will be using CalmConnect AI. Both stories focus on what the user desires to undertake and its significance to his/her work or wellbeing. These tales are related to the characteristics depicted in the use case diagram.

1. **Register**
   - As a new user, I want to register with a unique email and password so I can create a secure CalmConnect AI account.
   - As a new user, I want the system to validate required fields and reject duplicate emails so my registration stays correct and complete.
   - As a new user, I want to receive a verification email after registration so I can confirm ownership of my email address.
2. **Login**
   - As a registered user, I want to log in using my email and password so I can access the system securely.
   - As a registered user, I want to be redirected to my role-based dashboard after login so I see the features meant for my role.
   - As a registered user, I want clear error messages when credentials are invalid or the account is blocked so I understand why access is denied.
3. **Start AI Chat Session**
   - As a patient, I want to start an AI chat session so I can share my thoughts and receive supportive responses.
   - As a patient, I want the system to save my chat messages so they can be used later for tracking and review.
   - As a patient, I want the system to handle AI service downtime gracefully so I am informed when chat is unavailable.
4. **Request Appointment**
   - As a patient, I want to request an appointment with a psychiatrist so I can seek professional support inside the platform.
   - As a patient, I want to select a psychiatrist, date, and time so my request matches my preferred availability.
   - As a patient, I want the system to prevent booking requests when no slots are available or the psychiatrist is inactive so I do not submit invalid appointments.
5. **Manage User Accounts**
   - As an admin, I want to view user accounts so I can monitor the platform users.
   - As an admin, I want to block or unblock a user so I can respond to misuse or restore access when appropriate.

- As an admin, I want to delete accounts when required so the platform data remains clean and controlled.

6. **Verify Email**
   - As a patient, I want to verify my email through a secure link so my account becomes trusted and activated.
   - As a patient, I want the system to validate the verification token so only the correct owner can verify the account.
   - As a patient, I want the system to handle expired or invalid tokens safely so my account cannot be verified incorrectly.
7. **Reset Password**
   - As a user, I want to request a password reset so I can regain access when I forget my password.
   - As a user, I want to receive a reset link by email so only I can change my password securely.
   - As a user, I want the system to invalidate old sessions after password change so my account stays protected.
8. **Crisis Language Detection and Escalation**
   - As a patient, I want the system to detect crisis language in my chat messages so I receive safety guidance during high-risk situations.
   - As a patient, I want an escalation banner with emergency contacts so I can quickly find help when needed.
   - As an admin, I want crisis events to be logged so sensitive safety actions remain auditable.
9. **Update Appointment Status**
   - As a psychiatrist, I want to approve an appointment request so I can confirm the session with the patient.
   - As a psychiatrist, I want to reschedule an appointment by selecting a new time slot so I can manage availability changes.
   - As a patient, I want to receive a notification when the appointment status changes so I stay updated about my request.
10. **Record Session Summary**
    - As a psychiatrist, I want to record session summary notes so key outcomes and follow-up actions are stored for continuity of care.
    - As a psychiatrist, I want the summary to be linked with the appointment and patient so I can retrieve it later easily.
    - As a psychiatrist, I want the system to prevent saving empty or invalid summaries so stored notes remain meaningful and correct.
11. **Direct Messaging**

- As a patient, I want to send direct messages to my psychiatrist so I can share updates between appointments.
- As a psychiatrist, I want to receive message notifications so I can respond in a timely way.
- As a user, I want the system to show delivery errors when the recipient is blocked or inactive so I know why messaging failed.

12. **View Activity Logs:**
    - As an admin, I want to view activity logs with filters so I can review sensitive actions by user, date, and action type.
    - As an admin, I want the system to display log records in a clear table so auditing becomes quick and reliable.
    - As an admin, I want the system to show a clear result when no records match my filters so I understand the outcome of my search.

These narratives address the key objectives of any given user group and are indicative of the functional needs of CalmConnect AI. They provide a basis on which tasks, screens, work flows, and acceptance criteria are designed at advanced stages of development.

4.1.2 Individual Actor Use Cases

This section gives the important use cases of each actor in CalmConnect AI. The tables conclude the aims, stimuli, and the consequences of actions carried out by the Patients, Psychiatrists, and Administrators.

Use Case UC-01 Register:
The Register use case is explained in Table 4.1. This use case arrives at a beginning of the interaction between the patient and the CalmConnect AI. The aim is to secure an account with a good and valid email and a password that will enable the user to log-in later and use mental-wellness tools. As mentioned in the table, the site and the database should be internet enabled and the email address should be one that is distinct. Normal flow is depicted by the way the patient opens the site and clicks on sign up, complete the form and submits it. After this, the new record is stored in the system and an email of verification is sent. Some of the routine problems that are covered by the exception flow are duplication of emails or lost data. Such checks safeguard against invalid/inreplaceable registrations in the platform.

Table 4.1: Use case table of Register

Field	**Details**
UCT ID	UC-01
Case Name	Register
Use–case overview	To create a new patient account on CalmConnect AI by providing personal details and secure credentials.

Continued on next page

Field	**Details**
Actors	Patient
Pre–conditions	• Website and database are online. • User has a valid email address.
Triggers	User clicks the ”Sign Up” button.
Steps	1. User opens the CalmConnect AI website. 2. User selects the sign–up option. 3. User enters required information. 4. User submits the form.
Post–conditions	• A new patient account is created. • Verification email is sent to the user.
Exception Flow	• Duplicate email entered. • Required data missing or invalid.

Use Case UC-02 Login:
Table 4.2 describes the Login use case. All the three roles share this use case. It is used to verify a registered user and redirect him/her to the appropriate role-based dashboard. The pre-conditions keep in mind that the account should exist and should not be blocked. The steps depict a straightforward log in process. One clicks on the login page and then provides their credentials and makes a post. In the event that the information is valid, the system goes ahead to establish a session, and redirects the user to his perception of the system. Exception flow contains the items where the password or email is incorrect or the account is blocked. A clear message is displayed in such a case rather than a dashboard.

Table 4.2: Use case table of Login

Field	**Details**
UCT ID	UC-02
Case Name	Login
Use–case overview	To authenticate a registered user and provide access to their role–based dashboard.
Actors	Patient, Psychiatrist, Admin

Continued on next page

Field	Details
Pre–conditions	• User is registered. • Account is active.
Triggers	User clicks the “Login” button.
Steps	1. User navigates to login page. 2. User enters email and password. 3. System verifies credentials. 4. System redirects user to their dashboard.
Post–conditions	• A valid session is created.
Exception Flow	• Invalid credentials entered. • Account is blocked.

Use Case UC-03 Start AI Chat Session:
The use case of Start AI Chat Session is discussed in Table 4.3. It is the usage case that CalmConnect AI focuses on as it manages the AI chat which assisting the patient. The pre-conditions demonstrate that the patient should be logged in and the AI engine should be available. The primary flow is the experience as it walks through. The system opens the chat screen and the patient provides the first message. The power of AI provides a response and the machine indicates this reply and at the same time logs the interaction to be analyzed later. The exception flow refers to what occurs in the event of non availability of the AI service or where crisis language is exhibited. In such instances the regular communication is disrupted by explicit indication that there is a message failure rather than the failure to communicate.

Table 4.3: Use case table of Start AI Chat Session

Field	Details
UCT ID	UC-03
Case Name	Start AI Chat Session
Use–case overview	To allow a patient to begin a supportive, non–clinical AI conversation.
Actors	Patient, AI Service
Pre–conditions	• Patient is logged in. • AI engine is available.

Continued on next page

Field	**Details**
Triggers	Patient clicks "Start AI Chat."
Steps	1. System loads chat screen. 2. Patient sends first message. 3. AI generates response. 4. System displays reply.
Post–conditions	• Chat session is created. • Messages stored for emotional tracking.
Exception Flow	• AI unavailable. • Crisis language detected.

Use Case UC-04 Request Appointment:
Request appointment use case is presented in Table 4.4. It targets facilitating the patient to make an appointment with a psychiatrist on the platform as opposed to the external medium. The pre-conditions prove that the patient must be registered, and the active psychiatrist must be present. The procedures are to be followed on how the patient will choose a psychiatrist, date and time then he/she will make the request. The request is entered into the system with the pending status and alerted to the selected psychiatrist. The exceptions indicate what happens in the conditions of unavailability of free slots or working psychiatrist. This renders the booking process realistic and the invalid appointments to be non-stored.

Table 4.4: Use case table of Request Appointment

Field	**Details**
UCT ID	UC-04
Case Name	Request Appointment
Use–case overview	To allow a patient to request a meeting with an available psychiatrist.
Actors	Patient, Psychiatrist
Pre–conditions	• Patient is authenticated. • At least one psychiatrist is active.
Triggers	Patient clicks "Request Appointment."

Continued on next page

Field	Details
Steps	1. Patient chooses psychiatrist. 2. Patient selects date and time. 3. Patient submits request. 4. System records request. 5. Psychiatrist is notified.
Post–conditions	• Appointment status = Pending.
Exception Flow	• No available slots. • Psychiatrist inactive.

Use Case UC-05 Manage User Accounts:
Table 4.5 manages the User Accounts use case of the admin. This use case regulates the platform in terms of misuse response, account-related issues and data-cleaning requests. The pre-conditions are that the user records are already present and the user should be logged in. The overall flow is that the admin initially views the list of the users, then picks one of them and then picks a can of view, block, unblock or delete. The system modifies the records and nullifies the sessions where necessary. Permission and database errors are in the exception flow. These cases are significant in security and audit perspective as they give a clue as to when an administrative change can fail.

Table 4.5: Use case table of Manage User Accounts

Field	Details
UCT ID	UC-05
Case Name	Manage User Accounts
Use–case overview	To let the admin view, block, unblock, or delete user accounts.
Actors	Admin
Pre–conditions	• Admin is logged in. • User records exist.
Triggers	Admin selects the user management module.

Continued on next page

Field	**Details**
Steps	1. Admin views list of users. 2. Admin selects a user. 3. Admin chooses an action (view, block, unblock, delete). 4. System updates user record.
Post–conditions	• Updated user status saved. • Blocked or deleted accounts lose access immediately.
Exception Flow	• Permission error. • Database update failure.

Use Case UC-06 Verify Email:
Table 4.6 describes the Verify email use case. This use case establishes the fact that email used during registration is the property of the patient. It enhances the confidence in the system and decreases dummy accounts. The pre conditions indicate that the registration form is already completed and the email service is present. The steps outlines the process by which the system generates a token, dispatches a link followed by verifying the token on clicking of the same by the patient. As soon as this is confirmed, the account level is altered to Verified. The exception flow refers to the token expiry and delivery problems. These facts demonstrate that CalmConnect AI can handle the issue of late clicks or inaccessible inboxes and still maintain the safe verification process.

Table 4.6: Use case table of Verify Email

Field	**Details**
UCT ID	UC-06
Case Name	Verify Email
Use–case overview	To confirm that the email address provided during registration belongs to the patient.
Actors	Patient, System, Email Service
Pre–conditions	• Patient has submitted the registration form. • Email service is available.
Triggers	Patient clicks the verification link received by email.

Continued on next page

Field	Details
Steps	1. System generates a unique verification token. 2. System sends an email that contains the verification link. 3. Patient opens the email and selects the link. 4. System validates the token. 5. System marks the account as verified.
Post–conditions	• Patient account status is set to "Verified".
Exception Flow	• Token expired or invalid. • Email not delivered because of network or provider issues.

Use Case UC-07 Reset Password:
Table 4.7 shows the use case of Reset Password. This application is useful when the user has forgotten his/her password or when he suspects that a secret has been leaked. The pre conditions ensure that the user exists and the email service is operational. The flow of normal states that the user gets a request to reset the account and s/he logs in with a registered email, gets a link and creates a new password. The system then updates the credentials and negates the older sessions to secure the account. The flow exception includes invalid emails and invalid or used tokens. Such regulations make it very difficult to abuse the reset process by the attackers.

Table 4.7: Use case table of Reset Password

Field	Details
UCT ID	UC-07
Case Name	Reset Password
Use–case overview	To let a user recover access when the current password is forgotten or compromised.
Actors	Patient, Psychiatrist, Admin, System, Email Service
Pre–conditions	• User exists in the system. • Email service is reachable.
Triggers	User selects "Forgot password" on the login page.

Continued on next page

Field	Details
Steps	1. User enters registered email address. 2. System checks that the email exists. 3. System generates a reset token and sends a reset link. 4. User opens the link and enters a new password. 5. System validates the token and updates the password.
Post–conditions	• User password is updated. • Previous sessions are invalidated.
Exception Flow	• Email not found in the system. • Token invalid or used already.

Use Case UC-08 Crisis Language Detection and Escalation:
The Crisis Language Detection and Escalation use case is described at Table 4.8. This application case provides an additional safety measure to the AI chat. It scans messages on the possibility of phrases indicating self-harm or intense distress. The pre-conditions tell that the chat session is active and that there are detection rules set. These steps indicate how the system examines every incoming message, analyzes it and identifies this message as normal or high risk. Once a threshold is hit an escalation banner will be displayed containing emergency contacts and other safety data. Audit of the event is also logged. The exception section is used to describe the scenario in case of the unavailability of the detection service. A failure would be noted in the system in such a case, to investigate the issue subsequently.

Table 4.8: Use case table of Crisis Language Detection and Escalation

Field	Details
UCT ID	UC-08
Case Name	Crisis Language Detection and Escalation
Use–case overview	To identify high–risk phrases in AI chat and present crisis information to the patient.
Actors	Patient, System, AI Service

Continued on next page

Field	**Details**
Pre–conditions	• AI chat session is active. • Crisis keyword rules are configured.
Triggers	Patient sends a new chat message.
Steps	1. System receives the message. 2. System or AI service scans the text for crisis indicators. 3. If threshold is reached, system marks the message as high risk. 4. System shows escalation banner with emergency contacts and safety advice. 5. System logs the event for audit.
Post–conditions	• Patient sees crisis guidance on screen.
Exception Flow	• Detection service unavailable, system skips escalation and records the failure.

Use Case UC-09 Update Appointment Status:
The Update Appointment Status is the use case that is covered in Table 4.9 and is primarily related to the psychiatrist. After a patient has ordered an appointment, there should be an option of him/her approving, scheduling or canceling the appointment with a psychiatrist. The pre -conditions serve to remind us that the request should already be there and that the psychiatrist should be signed in. The steps outline the process by which the psychiatrist accesses request, selects an action and a new time slot in case of a rescheduling. The system makes an update to the record and notifies the patient. Exception cases include the failures of databases or notifications. This information assists developers to consider how they can implement retry logic and error messages that would keep both sides in the loop.

Table 4.9: Use case table of Update Appointment Status

Field	**Details**
UCT ID	UC-09
Case Name	Update Appointment Status
Use–case overview	To let a psychiatrist accept, reschedule, or cancel a patient's appointment request.
Actors	Psychiatrist, Patient, System

Continued on next page

Field	Details
Pre–conditions	• Appointment request exists with status "Pending". • Psychiatrist is logged in.
Triggers	Psychiatrist selects an appointment from the schedule or notifications.
Steps	1. Psychiatrist opens appointment details. 2. Psychiatrist chooses an action: approve, reschedule, or cancel. 3. If reschedule is chosen, psychiatrist selects new time. 4. System updates appointment record. 5. System sends notification to patient.
Post–conditions	• Appointment holds updated status and time. • Patient sees the latest status on their dashboard.
Exception Flow	• Database update fails. • Network error while sending notification.

Use Case UC-10 Record Session Summary:
The description of the Record Session Summary use case is provided in Table 4.10. This use case provides the most crucial outcomes of a therapy session. It promotes continuity of care in the sense that the notes would be callable later by both psychiatrist and the patient based on the access regulations. The pre conditions are the fact that the appointment is arranged and the psychiatrist is authenticated. The routine procedure entails appointment-setting on the part of the psychiatrist, as well as the summary and saving. The data is recorded in the system and is associated with the appointment and patient. Validation and storage error is present between the exception flow. Such instances are to be noted as the loss or corruption of a summary can affect the clinical follow-up.

Table 4.10: Use case table of Record Session Summary

Field	Details
UCT ID	UC-10
Case Name	Record Session Summary

Continued on next page

Field	**Details**
Use–case overview	To let the psychiatrist store key points and recommendations after an appointment.
Actors	Psychiatrist, Patient, System
Pre–conditions	• Appointment has status “Completed” or is near completion. • Psychiatrist is authenticated.
Triggers	Psychiatrist selects “Add session summary” for a finished appointment.
Steps	1. Psychiatrist opens the appointment record. 2. Psychiatrist enters summary notes and follow–up actions. 3. Psychiatrist saves the summary. 4. System stores the notes and links them with the appointment and patient.
Post–conditions	• Session summary is available for later review by psychiatrist and patient (according to access rules).
Exception Flow	• Validation error on summary form. • Storage error logged by the system.

Use Case UC-11 Direct Messaging:
Table 4.11 gifts the use case of Direct Messaging. This application provides a safe communication line between a patient and the given psychiatrist during out-of-session periods. The pre-conditions demonstrate that both accounts are not closed yet and that the assignment between them is present.

The steps explain the process through which messages are written, sent, stored before being presented to the recipient with notifications. History of conversation expands as time goes by and both the sides are provided with a mutual perception of the past dealings.

Exception flow contains the cases when the receiver is blocked or the delivery is interrupted due to some network problems. Such information will inform the sender in terms of how the app will notify him that the message can no longer be sent.

Table 4.11: Use case table of Direct Messaging

Field	Details
UCT ID	UC-11
Case Name	Direct Messaging
Use–case overview	To provide a secure text channel between a patient and psychiatrist for short updates and questions.
Actors	Patient, Psychiatrist, System
Pre–conditions	• Patient and psychiatrist have active accounts. • Patient is assigned to the psychiatrist.
Triggers	User opens the direct message thread and sends a new message.
Steps	1. User types a message in the chat box. 2. User sends the message. 3. System stores the message with timestamps and sender identity. 4. System pushes a notification to the recipient. 5. Recipient opens the thread and views the message.
Post–conditions	• Conversation history is updated.
Exception Flow	• Recipient account blocked or inactive. • Network error while sending the message.

Use Case UC-12 View Activity Logs:
Table 4.12 outlines View Activity Logs use case of the admin. This usage case facilitates security, compliance and debugging. Through logs, the admin can be able to view regarding who did what and at what time. The pre-conditions are that the admin should have the permission to do this and logging should be enabled. The steps detail the process by which the admin is executing the query by selecting filters and performing the query and viewing results. Even the possibility to export or record findings is mentioned. Exception flow illustrates what occurs when there is any failure in the log storage, or when there are no records to be returned as a result of a query. Combined, these facts demonstrate how CalmConnect AI can be used to trace the sensitive processes like blocking accounts or crisis events.

Table 4.12: Use case table of View Activity Logs

Field	**Details**
UCT ID	UC-12
Case Name	View Activity Logs
Use–case overview	To let the admin review important system actions for audit and security checks.
Actors	Admin, System
Pre–conditions	• Admin is authenticated with proper rights. • Logging subsystem is active.
Triggers	Admin opens the "Activity Logs" section on the admin dashboard.
Steps	1. Admin selects filters such as date range, user, or action type. 2. System queries stored logs. 3. System displays matching records in tabular form. 4. Admin reviews entries and, if needed, exports or records findings.
Post–conditions	• Admin gains visibility into sensitive operations.
Exception Flow	• Log storage unavailable. • Query returns no records for given filters.

4.2 Functional Modeling

Functional modeling describes the flow of data/information among users and system processes in addition to data stores of CalmConnect AI. This part uses the functional requirements as developed in Chapter 2 and converts them into a graphical and logical conceptualisation of system behaviour consisting of data and processes. The models help in ensuring that every interaction between users, system activity and data flow among others have a thorough understanding before implementation. [15]

4.2.1 Entity Relationship Diagram

Entity Relationship Diagram (ERD) is a logical organization of data of CalmConnect AI and represents the connection between the principal entities of the PostgreSQL database. It is focused on unamenable data required to help with authentication, communication, mental health tracking, and administration.

The significant objects of the system are User, Patient Profile, Psychiatrist Profile, Chat Session, Chat Message, Appointment, Mood Entry, Journal Entry, CBT Thought Record, and Activity Log. The Tenets of business that can be enforced contacts between these parties are role-based access, allocation of a patient to a psychiatrist, and keeping of sensitive records.

Figure 4.2 shows the Entity Relationship Diagram (ERD) of CalmConnect AI that represents the essence of entities, attributes, and relationships of the system. The User entity is the focal point and controls the authentication, roles and account status. To facilitate mental-wellness tracking, users have the capacity to start AI chat sessions, record moods, write journal entries, as well as, complete CBT thought records. Users can also use the system to request an appointment and even give direct messages, which enables one to interact with mental health professionals. Users get notifications to inform them about important events and updates in the system. The ActivityLog entity has administrative and system-level actions that are kept in the ActivitiesLog to guarantee duty of activity and security. The ERD is based on the concept of normalization and helps to entertain a scaled and secure database design. [13]

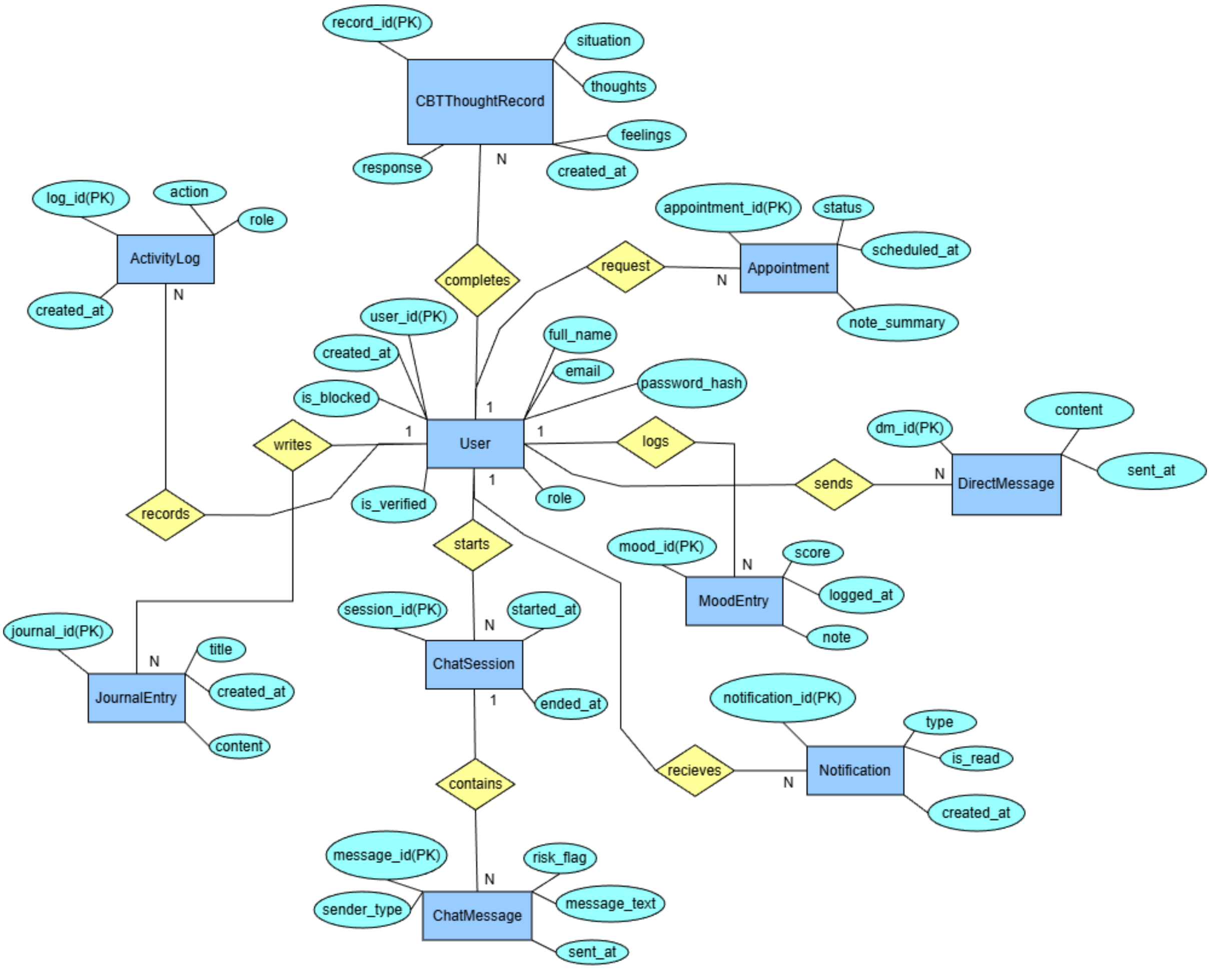


Figure 4.2: ER Diagram

4.2.2 Data Flow Diagram

Data flow diagrams (DFDs) reflect the process of information transfer across the system, communication with CalmConnect AI on the part of third-party organizations, and data processing and data storage. The DFDs are the complements of the ERD since they are interested in the system behaviour rather than in data structure.

These DFDs have three levels. Level 0 provides an overview of the system at the high level Level 1 splits out the main system into the core functions and Level 2 provides a further description of some of the major workflows that the specific AI application will entail, such as chat management, appointment management, and journaling. [19]

4.2.2.1 DFD Level 0

The Level 0 Data Flow Diagram can also be called context diagram; it presents CalmConnect AI as a unitary system. It also represents associations among the system and external systems: Patients, Psychiatrists and Administrators, Email Service and the AI Service (Gemini API).

The Level 0 Data Flow Diagram of CalmConnect AI illustrated in Figure 4.3 gives a clear overview of the responsiveness of the system to the environment. The entire platform can be depicted by the central unit process which will be connected to the outside participants who include Patient, Psychiatrist, and Admin. The system enables patients and psychiatrists to chat with AI, book an appointment, communicate, and check the activity of wellness. Gemini API offers emotional assistance by the use of artificial intelligence by responding to the input of the user. When sending products such as verification and password reset messages through the system it is accomplished via the SendGrid API through email. The central system also processes monitoring and user control administrative actions. This is the diagram that defines the system boundary and provides the basis of the further functional decomposition of the system. [12]

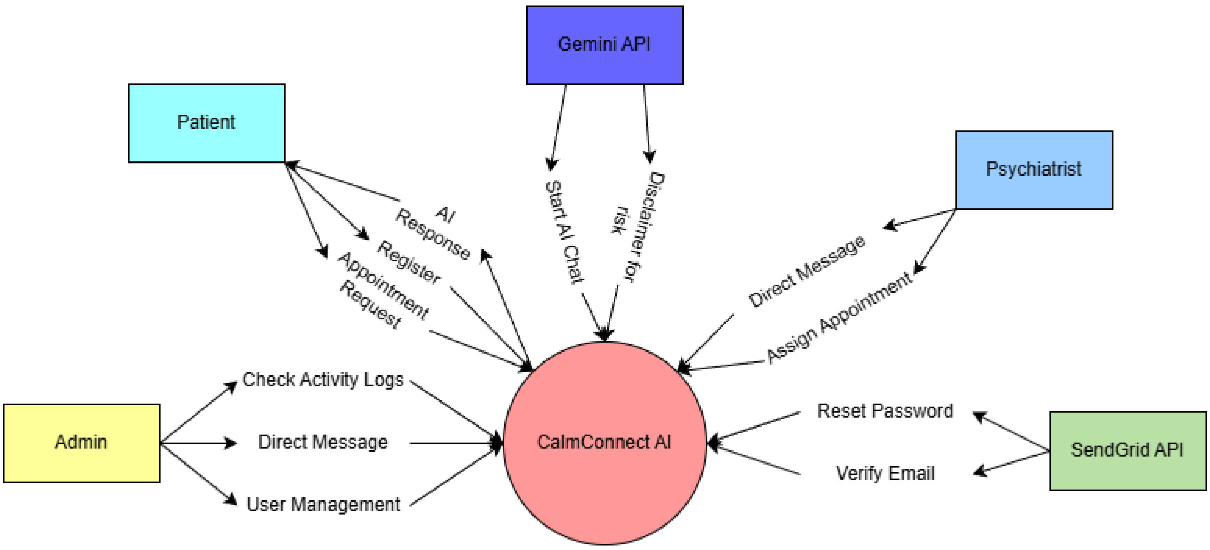


Figure 4.3: DFD Level 0

4.2.2.2 DFD Level 1

The level 1 Data Flow Diagram splits the CalmConnect AI mega processes. They are Authentication and Authorisation, Account Management, AI Chat Processing, Direct Messaging, Appointment Management, Mental Health Tracking (Mood, Journals, CBT), Administration and Notifications.

The Level 1 Data Flow diagessment of CalmConnect AI is shown in Figure 4.4 due to the fact that it divides the system into the most essential taking processes. User Authentication and Account Management (1.0) handles the process of registering as well as the process of logging in and restricting the roles of patients, psychiatrists and administrators. AI Chat process (2.0) is the process that addresses the user interaction with the Gemini API, and creates a chat session. Wellness Tracking offers mood logging, journaling, and CBT, and the user can get the information regarding his or her mentalhealth. The Appointment Management (4.0) can be used to facilitate the procedure of booking, updating and administration of the sessions between the patients and the psychiatrists. The Direct Messaging (5.0) is the secure enough tool of communication between the users, and the Notification (6.0) is the external services of delivery of the mailing messages. Such decomposition explains functionality and information flow of key modules of the system.

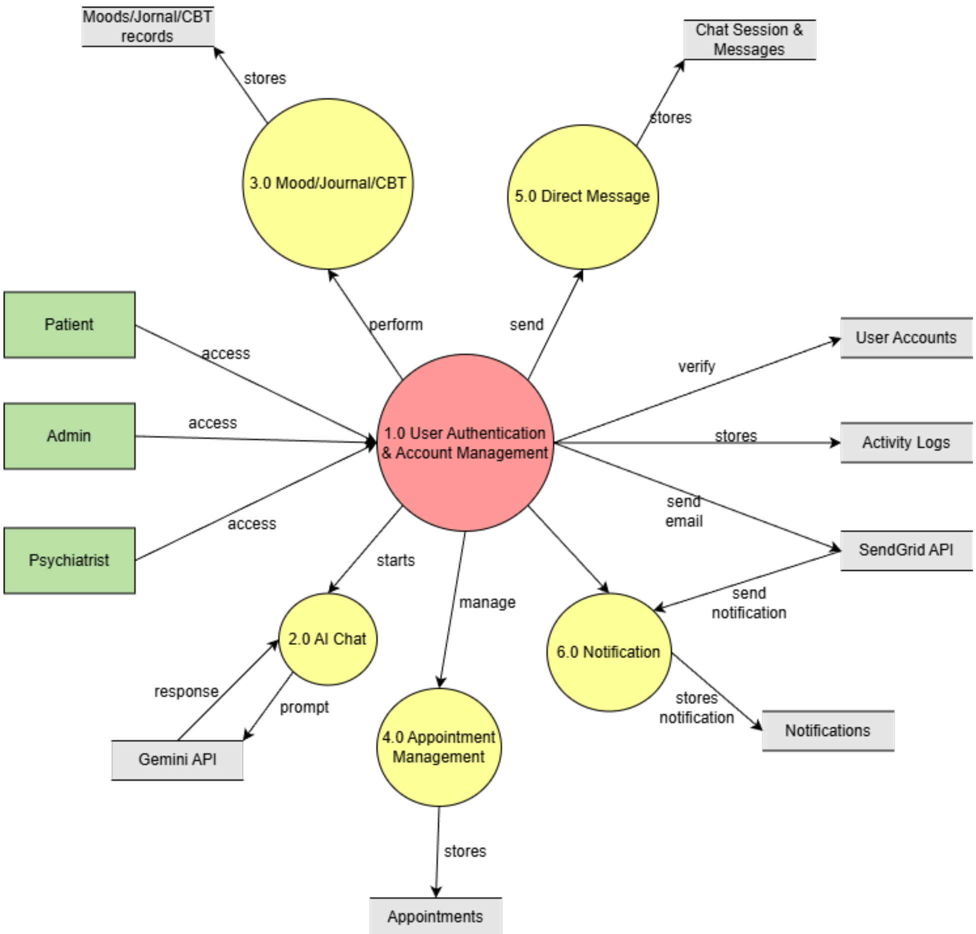


Figure 4.4: DFD Level 1

4.2.2.3 DFD Level 2

Level 2 Data flow diagrams provide a detailed view of the process chosen as being important. They can include the use of AI chat featuring safety measures, booking of appointments containing conflict detection, and the use of journals including consent sharing.

Figure 4.5 represents Level 2 Data flow diagram of User Authentication and Account management module of CalmConnect AI. This flow chart subdivides the form of authentication into subprocesses that are more specific including registration, email, lifting, password reset, accessing authorisation and account status management. These subprocesses get shared with peers; psychiatrists, administrators and patients depending on their ability to access. The user account data and user profile data is stored and accessed by the use of User Accounts and User Profiles data stores. Email verification and password-reset email is sent through the utilization of Sendgrid API, as well as all rivalrous processes are recorded in the Activity Logs. The following Level 2 DFD is the correct image of the implementation of the presence of the secure access and role-based control of the system.

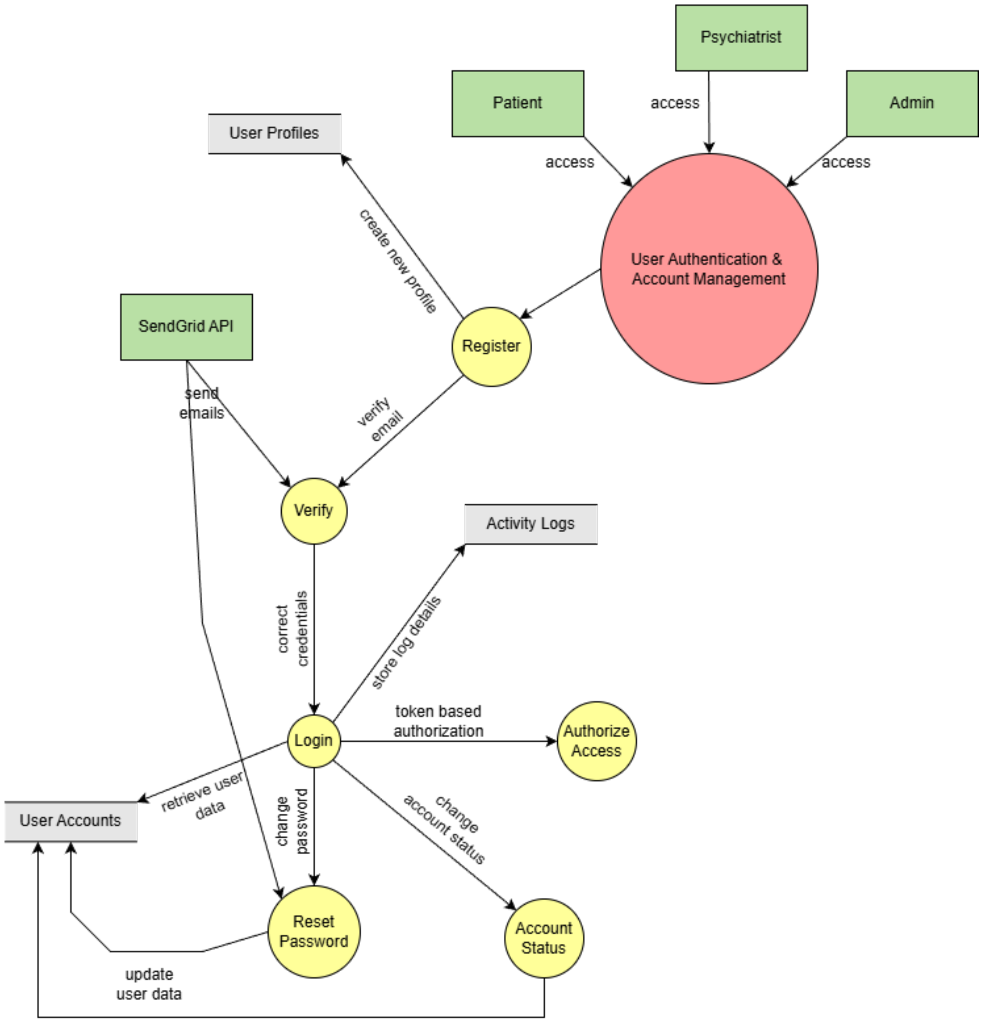


Figure 4.5: DFD Level 2

Chapter 5

SYSTEM DESIGN

The subsequent chapter explains system design of the CalmConnect AI. At this phase, the needs identified above are translated to an actual practical technical framework. The design involves the distribution of various parts of the system, the assignment of responsibilities, and flowing the actions of the user within the platform. The focus is made more on the keeping of the design simple, modular, secure and scalable without exceeding the identified project scope, as well as academic constraints.

The system design is divided into two major areas namely, structural design and behavioural design. Structural design deals with how the components of a systems are organized and how they co-relate to one another whereas behavioral design deals with the way in which these components interact and react with each other during system operation.

5.1 Structure Diagrams

The CalmConnect AI system has a diagrammatic representation in structure diagrams, which is the static view of the system. They define the key classes, system components, elements of deployment, and the connections between the classes and components. These systems can be used to simplify the understanding of system boundaries and allocate responsibilities at the beginning of development so that the developers can weigh up how the system is structured before transitioning into implementation. [23]

5.1.1 Class Diagram

The class diagram depicts the fundamental data model and basis of CalmConnect AI. It is a model of the principal system entities based on the requirements analysis and the ERD, as User, Patient, Psychiatrist, Admin, Appointment, Message, Journal, MoodEntry, CBTRecord, AIChatSession and ActivityLog. [22]

The User entity is a base and is differentiated into Patient, Psychiatrist as well as Admin using roles. Underneath every specialised class, there are the attributes and operations that the particular class performs in the system. To provide an example, a Patient manages entries of mood, journals, CBT records, and a Psychiatrist deals with appointments, session summary materials, and patient information review.

Class relations are related to real life. A Patient can place numerous orders in the Appointment section, one Psychiatrist is related to one Appointment, and all Messages are organized in a safe and secure dialogue. ActivityLog entity relates to the critical user actions to facilitate auditing and traceability.

The class diagram encourages the separation of duties and excellent encapsulation. Data storage issues, security restraints, and business regulations are grouped in clearly defined classes, thereby enhancing system preservation and the expansion in the future without a significant re-structuring.

The CalmConnect AI system class diagram presented in Figure 5.1 demonstrates the key entities and relationships of the system. The User class serves as a foundational class with account common details and authentication methods and it is further subdivided into Patient, Psychiatrist and Admin. All of the roles build upon the core user functionality in accordance with the system roles.

It has Patient class, which as a mood tracking, journaling, appointment request and messaging service, where personal data are kept in MoodEntry and JournalEntry classes. The Psychiatrist class handles appointments and is used in communicating with the patients, and the Admin class does user verification and access control. The class Appointment is used to model an appointment and two classes used to communicate are MessageThread and Message. The system overview diagram displays the key object makeup necessary in user management, mental health record, and messaging and appointment management in CalmConnect AI.

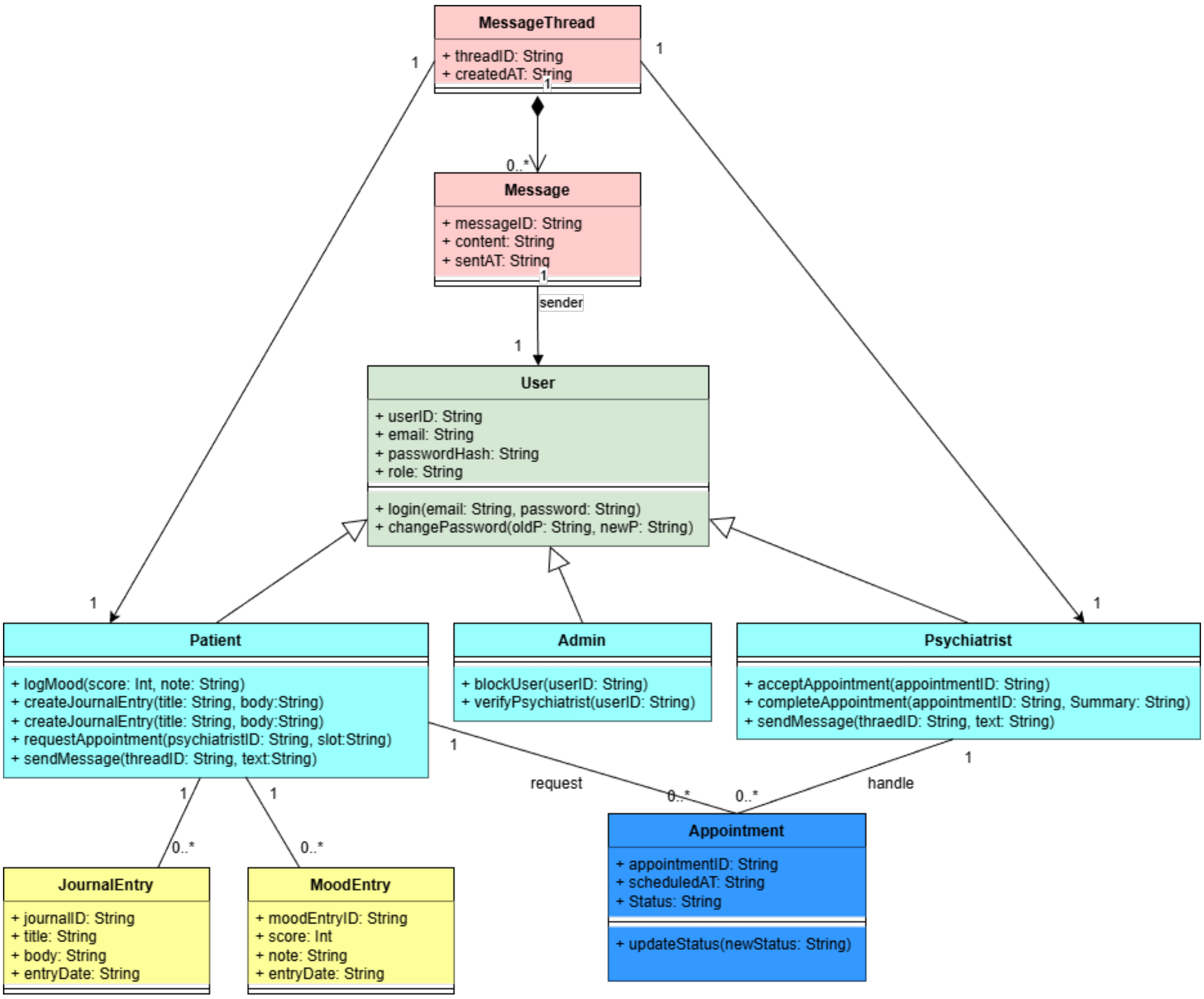


Figure 5.1: Class Diagram

5.1.2 Deployment Diagrams

The deployment diagram illustrates the deployment of CalmConnect AI in the various system environments. It employs a client server architecture, in which the front end, backend, external services and data storage services are well separated in order to provide clarity and maintainability. [21]

The platform is accessed by the user using a web browser via the desktop or mobile devices. The frontend is deployed in the form of a single-page application that uses the secure React APIs over HTTPS to make interface with the backend. This decoupling enables user interface to be responsive and business logic is left to a different code. The backend is built on the Node.js platform and performs authentication, authorization, business core, and service integrations. Structured system data, which consists of user accounts, messages, appointments, and logs on activities is stored in a dedicated PostgreSQL database server. The system also incorporates the use of external services such as the AI service that has been energized on the basis of Google Gemini API to provide conversational assistance and email service that can be used by verifying an account and receiving a new password. TLS encryption and controlled network boundaries has been secured to ensure all communication happening between components.

The deployment model of the CalmConnect AI can be seen in Figure 5.2. Users enter the site using a web browser on their computer, and the frontend application created with the React basis and developed with Vite and Tailwind CSS is loaded. The front end is served on a cloud-based hosting service, like Vercel or Netlify and serves up the front end as a static aspect of the app in terms of HTML, CSS, and JavaScript.

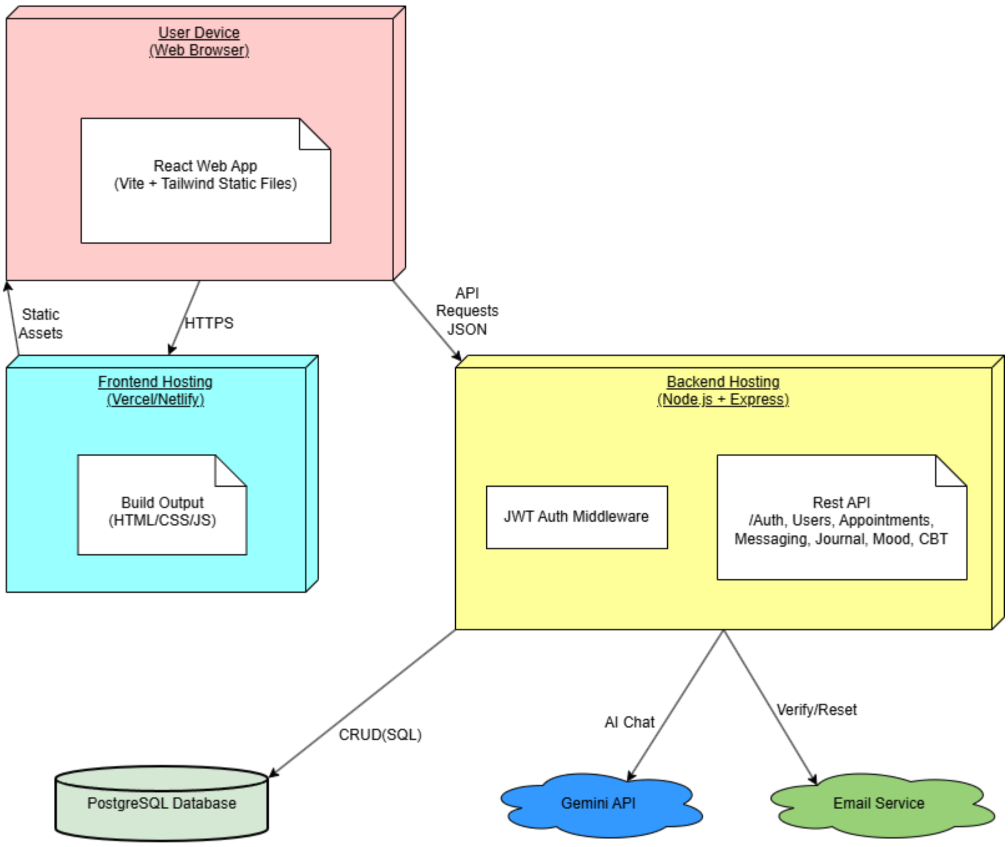


Figure 5.2: Deployment Diagram

5.2 Behavioral Diagrams

Behavioral charts define the way CalmConnect AI will work in the runtime. They pay attention to the user interactions, decision point and the communication of messages between various system parts. These charts have their basis directly on the use cases and user stories given in Chapter 4 and so system behavior can be absolutely in line with known user requirements. [24]

5.2.1 Activity Diagrams

Activity diagrams help to show the step-by-step flow of actions of key processes of the system, such as user registration, log in, chat interaction between AI and users, appointment management, and crisis management. They indicate the sequence of actions, critical decision points, and parallel actions that take place under the system in operation that makes the complex workflows easier to see. [25]

To illustrate, AI chat activity flow will start with patient authentication, submission of messages and compilation of AI responses. Each of the replies is considered in terms of crisisrelated language, and in case the risk indicators are discovered, the flow changes to the steps of escalation, including alerts or support recommendations. Likewise, the matters of appointments vividly demonstrate alternative methods of approval, rescheduling or cancelling on decisions of the psychiatrist, and system regulations.

In general, these diagrams facilitate in making sure that normal interactions, exceptions cases, and safety-critical behaviors are always realized. They are also supplementary to developers and other stakeholders because of the ability to give an overview of how the system would act in given conditions, which eliminates confusion when implementing and testing the system.

Figure 5.3 shows the flow of user registration activity of the system. This will start by a new user opening the registration screen where he or she will be requested to fill in the necessary information, which includes: name, email address, and password. The evaluation of the system will be first to verify the presence of all mandatory fields to be complete; this is to guarantee that all the required information is in place and that there are no validation errors that should be put to guide the user. After validation of the inputs, the system is able to check whether the email address given already exists to limit the occurrence of duplicate accounts. In case the email is already registered a suitable error message appears. In case the email is unique, the system will create a new user account in an unverified state, the system generates verification token and sends a verification email to the user. This is followed by a confirmation message which makes the user confirm that he or she is the owner of that email address before the system is accessed. The focus of this flow of activities is the accuracy of the data, elimination of duplicates, and the authenticated activation of an account by email validation.

Moreover, this registration flow keeps the full system access of the user until the possession of the email address issued has been verified. The system minimizes the chances of counterfeit or unauthorized accounts by ensuring unconfirmed state of account and validation of email by use of a token that is not yet verified. The method also enhances the general security of systems and facilitate valid user identity management during the onboarding phase.

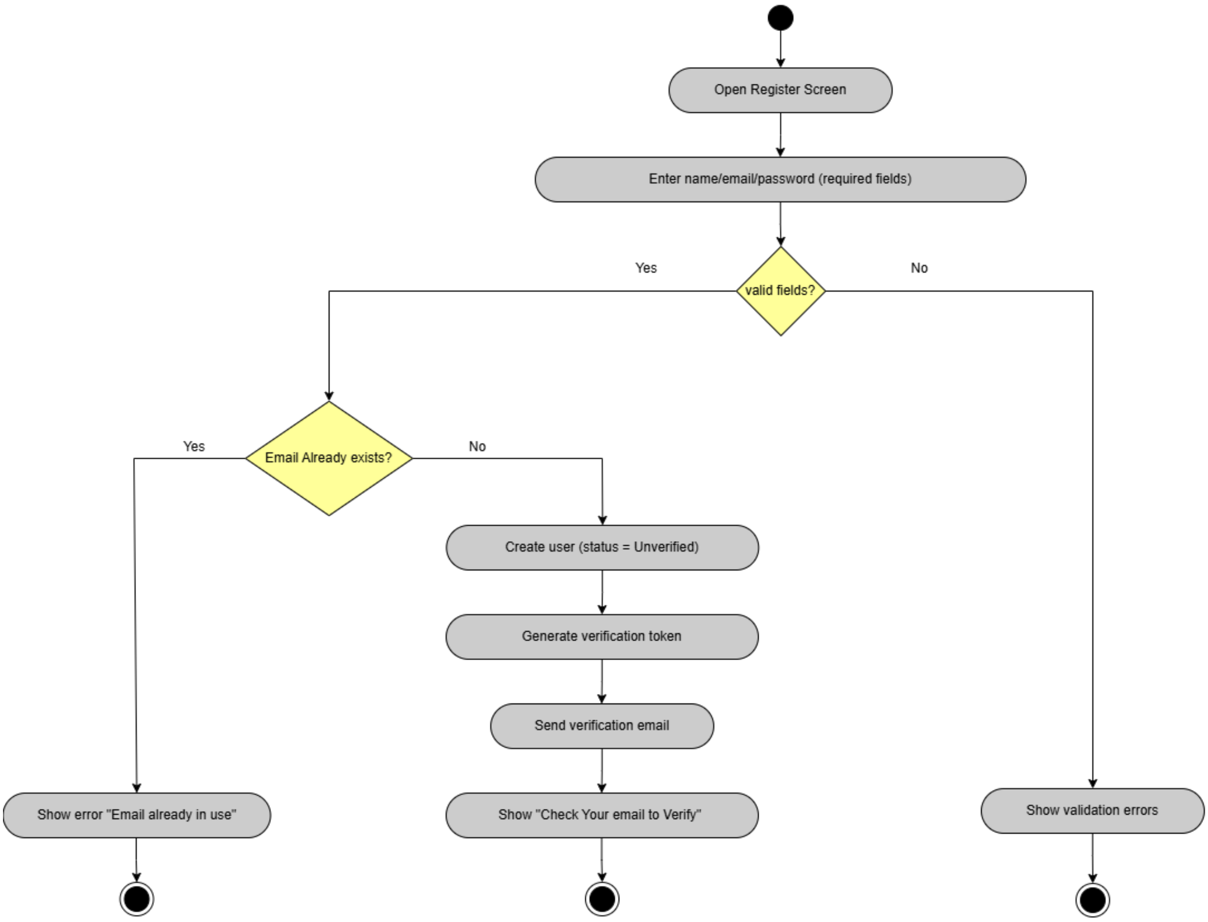


Figure 5.3: Register Activity Diagram

Figure 5.4 symbolizes the business process of the CalmConnect AI user login experience. The flow starts with a user who opens the login interface and inserts an email address that he has registered and uses the password before placing the request to log in. The system initially verifies that there is an account with the given email to eliminate unapproved attempts to enter the account, in the event that nothing is located, a message of invalid credentials is shown. In case the account is there, the system checks whether the account is blocked by administrative measure or policy action and informs the user in case it is restricted. In active accounts, the password typed in with an account is compared with the corresponding credential hash and a lack of match returns an error message stating invalid credentials. On success of the authentication, the system creates a secure JWT and a session, reads the role associated with the user and redirects the user to the corresponding role-based dashboard. The focus of this activity flow is on a secure authentication, step-by-step validation, clear user feedback, and controlled access, which is in accordance with assigned system roles.

This login process helps in the security of a system by performing various validation processes before a user is granted access such as account existence, account status and credential validation. The system should manage its sessions by ensuring that a JWT and session are issued only after successful authentication thereby makes sure that there is secure session management and that there is no unauthorized access. Role-based redirection also ensures that users are only allowed to access the features that their designated role allows them to.

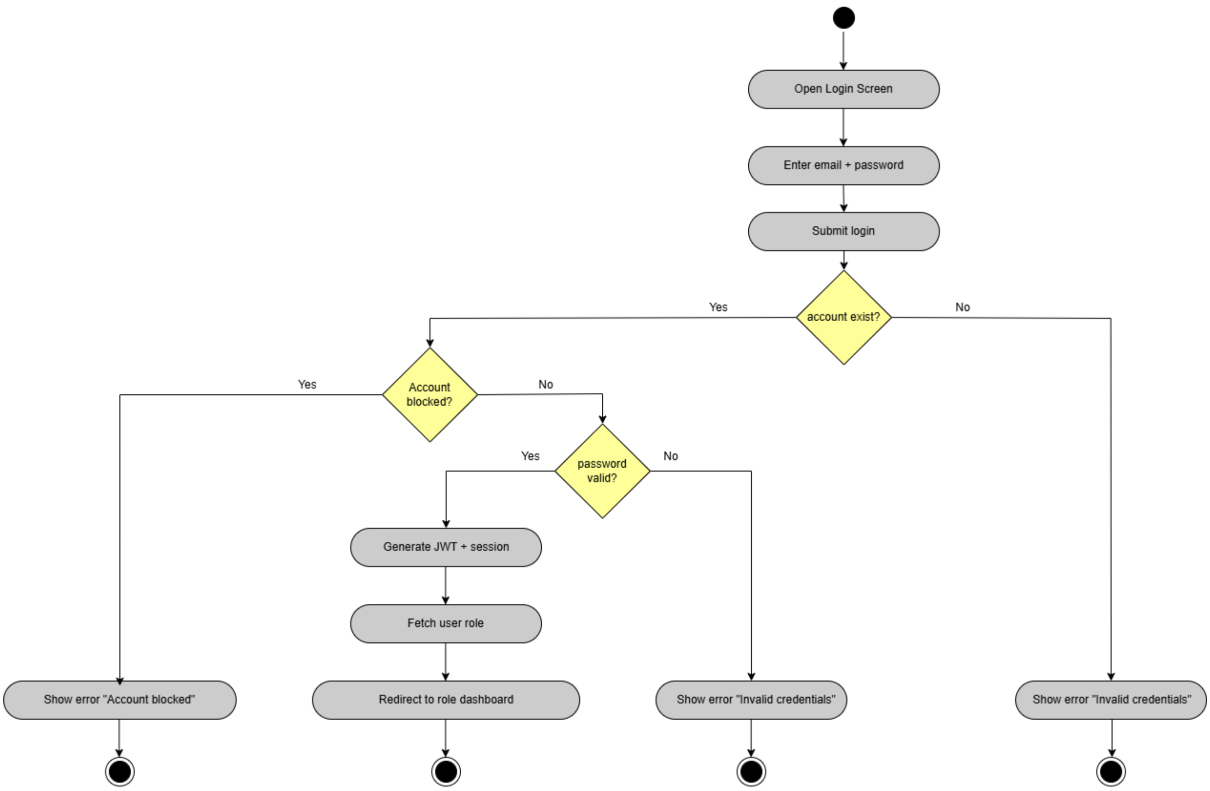


Figure 5.4: Login Activity Diagram

Figure 5.5 illustrates the flow of activities of the AI chat option in CalmConnect AI, which begins when a patient opens the chat interface, types in a message and sends it to the system. When the patient submits a message, the message is stored immediately as a history of conversation to make it traceable and to be analyzed or reviewed later. The system then queries the existence of the AI service to establish whether automated processing can be done. In case the service is on the saved message is sent to the AI API, the response generated is received and safely stored and the message can be displayed to the patient to form the interaction in real time. In case the AI service is inaccessible, the system will proceed with the next step and notify a patient that the chat is currently unavailable and the event of downtime will also be captured to be monitored, diagnostics and further measures to enhance the reliability of the system. This flowchart indicates both regular and abnormal flows, illustrating how fault is handled, data is persisted, and users receive similar feedback with the use of AI chat module.

Moreover, such a flow of activities demonstrates a resilience-oriented architecture since the message is not stored and processed by AI, which means that patient inputs are not lost when the service fails. The recording of the downtime incidents assist in monitoring operations and aiding in enhancing the availability of the systems over time. User notifications during failure are transparent and trustworthy and do not affect the overall chat experience.

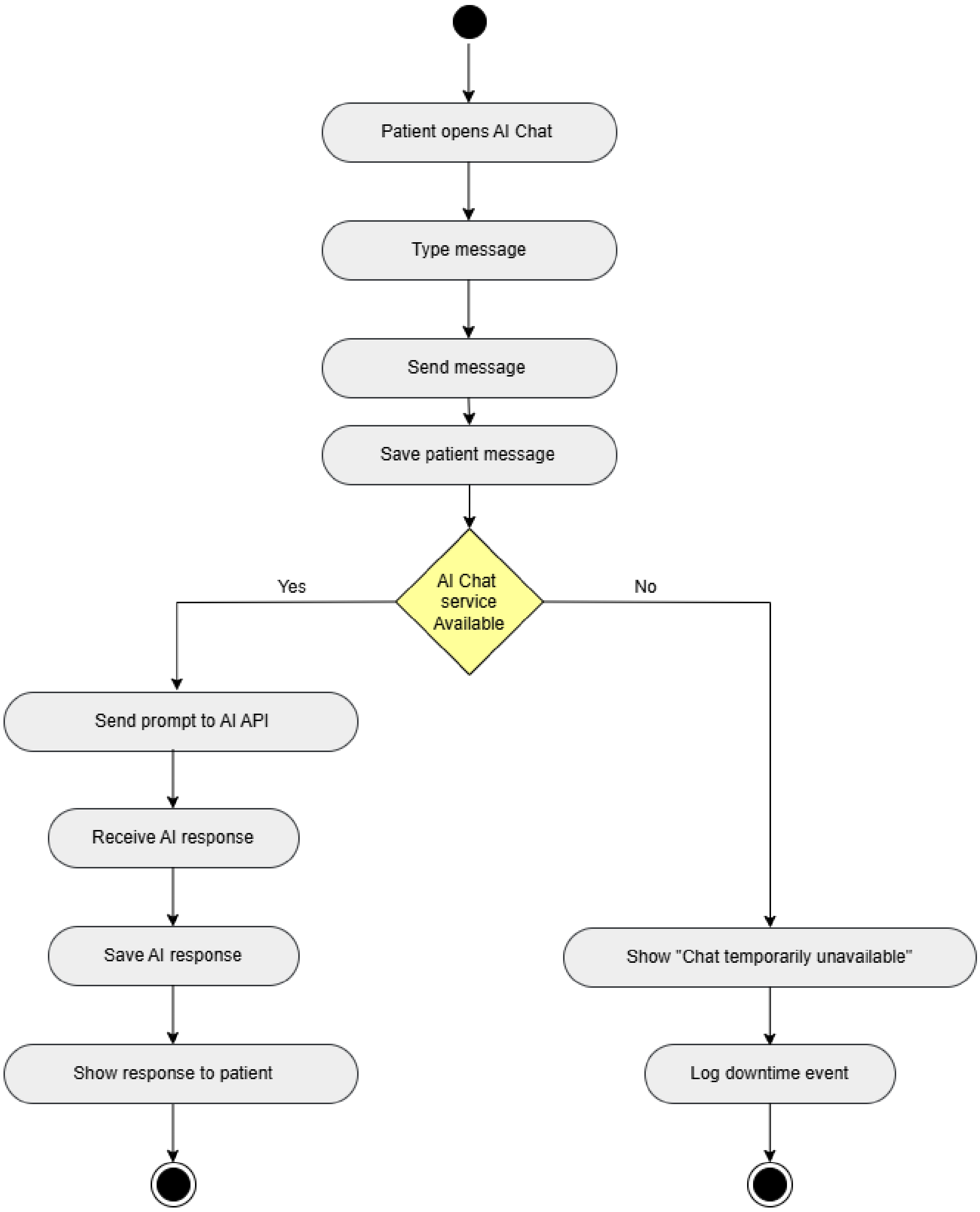


Figure 5.5: AI Chat Activity Diagram

Figure 5.6 illustrates the sequence of activities of requesting appointment in Calm-Connect AI. It starts with the patient opening the appointments section and choosing a psychiatrist, an option of a date and time he prefers, and placing the appointment request. The first step that the system performs is checking the activity of the psychiatrist chosen the psychiatrist might be a non-active one, the patient might be informed of this fact and the process stops. The next step which is to check slot availability is then taken by the system in case the psychiatrist is active. Upon event of slot availability, an appointment is made with a pending status, psychiatrist is informed about the request, and the patient can be informed and a confirmation message is sent to him. In a case where slots are not available it will inform the patient that there are no appointments that can be made on the time that has been selected. This chart simply reflects the verification checks, alternative routes and user input in such a way that there is controlled business schedules and open communication during the entire process of making an appointment request.

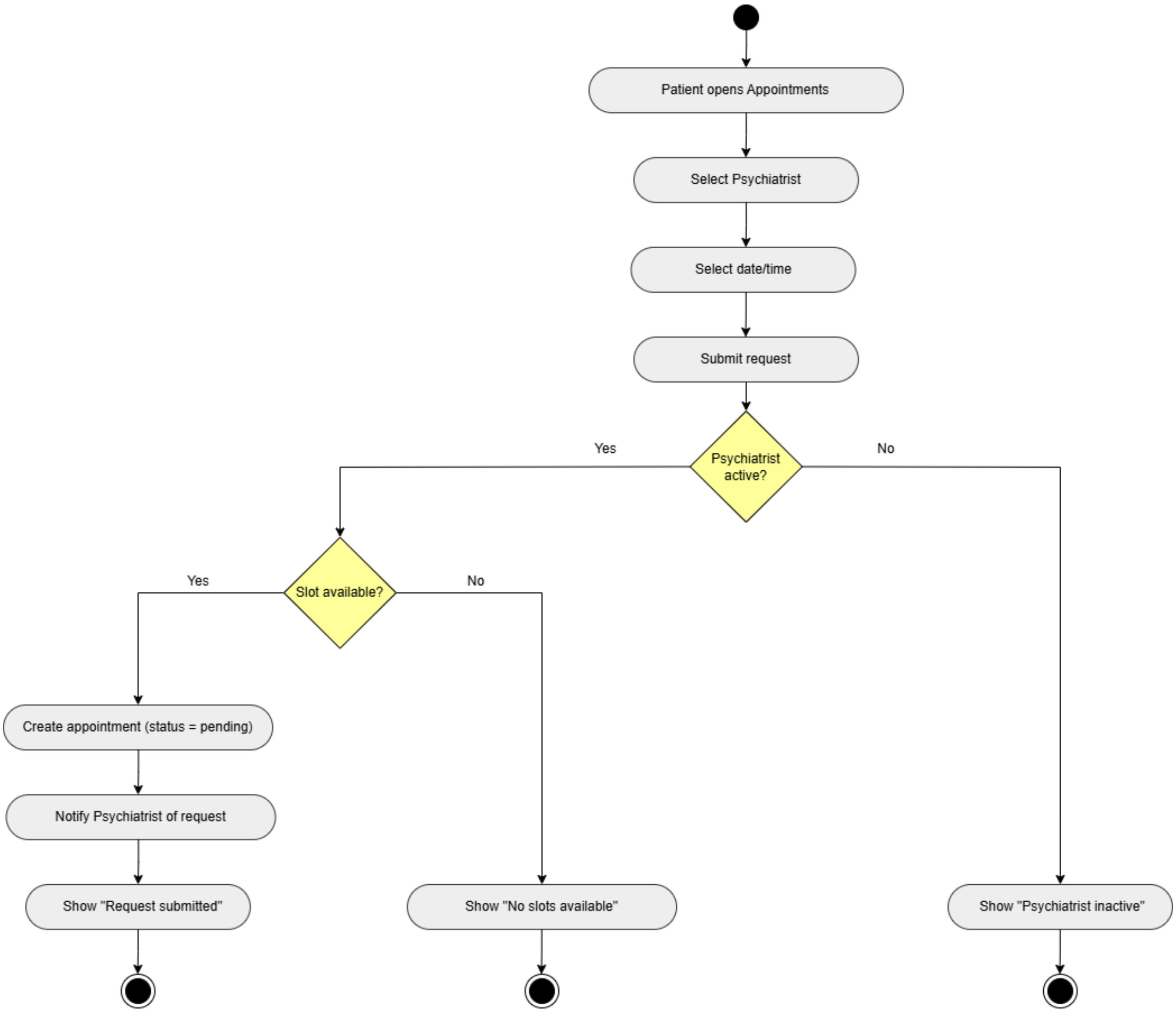


Figure 5.6: Request Appointment Activity Diagram

Figure 5.7 illustrates the flow of activities of the user management module of Calm-Connect AI as viewed by the administrator. When the user is opening the user management portion of the website, the list of registered users is displayed and a particular user account is selected, this is where the process starts. The admin then either blocks or allows the account in question in this case; nevertheless, in this case the system not only alters the user status but also logs the administrative action so that it could be audited and just shows the new status. In case the admin fails to block or unblock the account, the secondary decision is taken; whether to delete user account or not.

On confirmation, the system removes the account as per the laid policies, captures the action of the admin, and shows the result of deletion.

In case no action is chosen, the process terminates. This diagram emphasizes workflows which are controlled by administration, accountability with action logging, and policies-based user management.

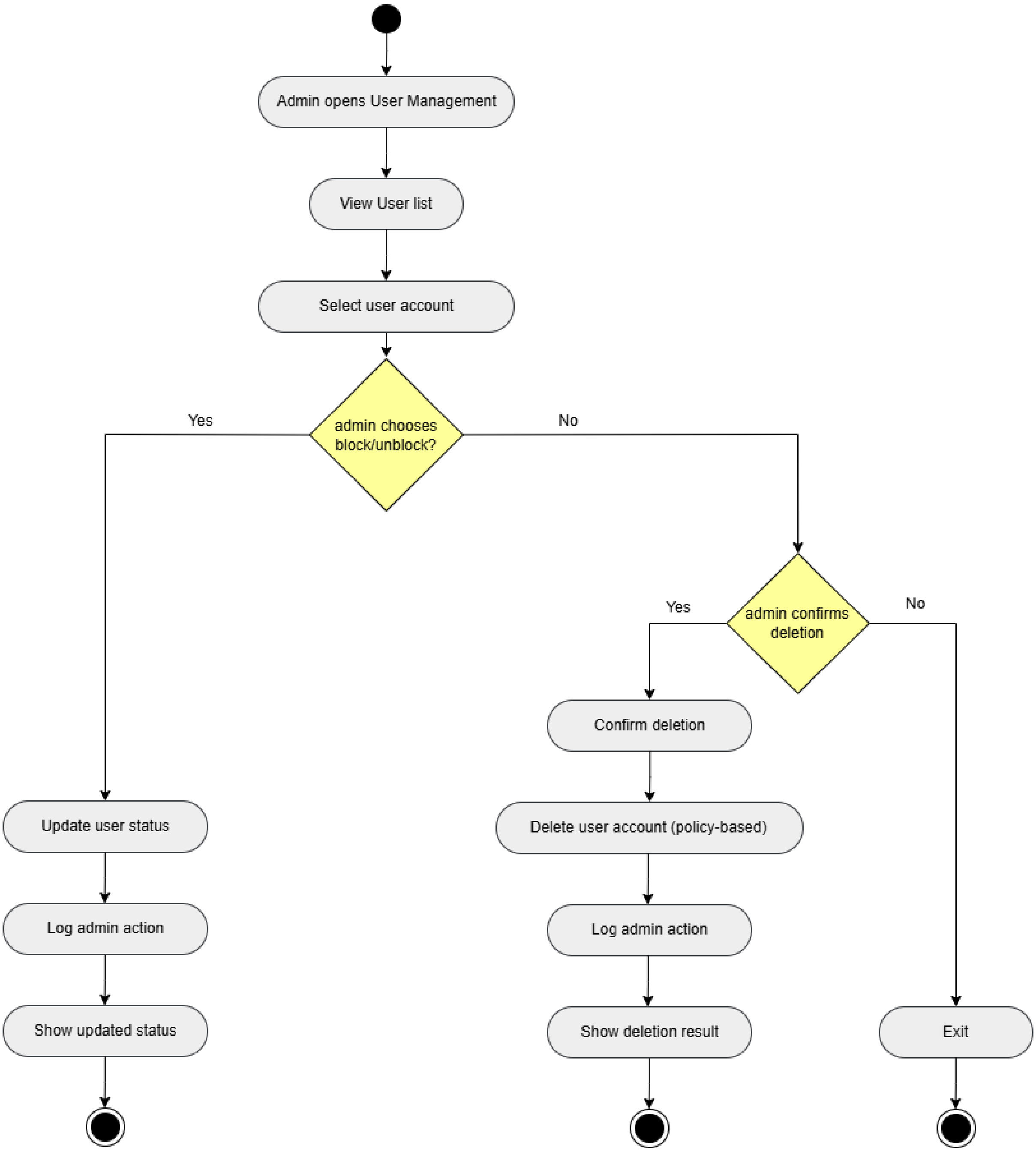


Figure 5.7: User Management Activity Diagram

Figure 5.8 document the flow of activities of the email verification process in Calm-Connect AI. It works by a user clicking the verification link that they receive by mail requesting a verification token that gets sent to the backend which is then verified. It will first authenticate a token; the second step will involve checking whether a token is correct or not; in case of an incorrect token, the user is informed and the process stops. In case of a valid token, the system will test rather its expiration date. The user is notified of the fact that the token has expired and is not verified in case of expiration. When the token is valid and it is still unexpired, the system will indicate that the user electronically ID has been confirmed, the token will be invalidated such that it cannot be used ever again and a message is displayed to inform the customer that the email account has been verified successfully. This diagram focuses on the safe activation of the accounts, the adequate verification process, and the clear feedback to the user during the procedure of verification.

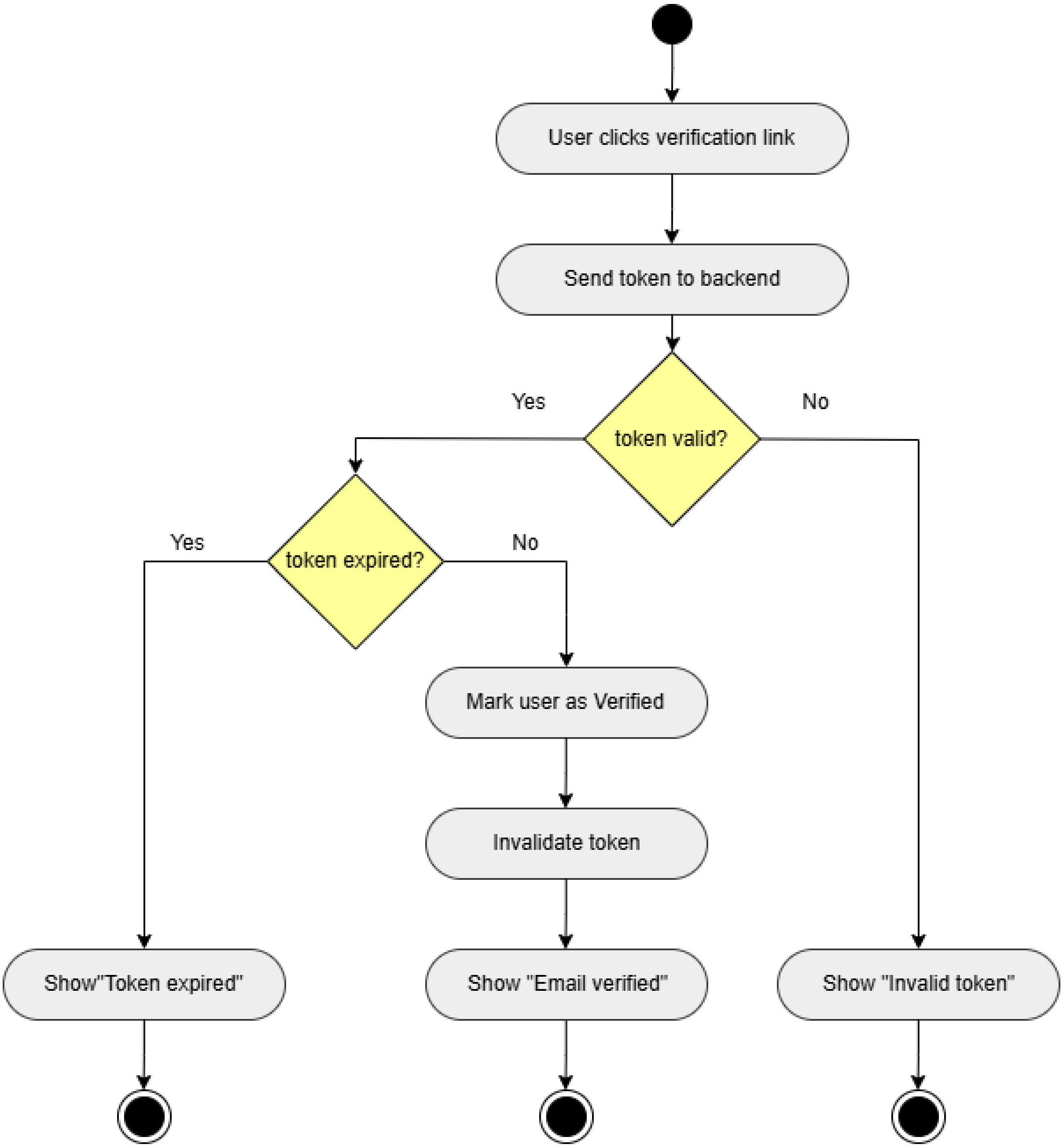


Figure 5.8: Email Verification Activity Diagram

Figure 5.9 describes the flow of activities involved in the process of resetting passwords within the CalmConnect AI. It starts with steps in which a user requests a password reset through entering their registered email address and then pressing the request. In case the user account is present, the system will generate a reset token and send a password reset link to the user through email. Once the user has opened the link he/she enters and confirms a new password which is again checked by the system.

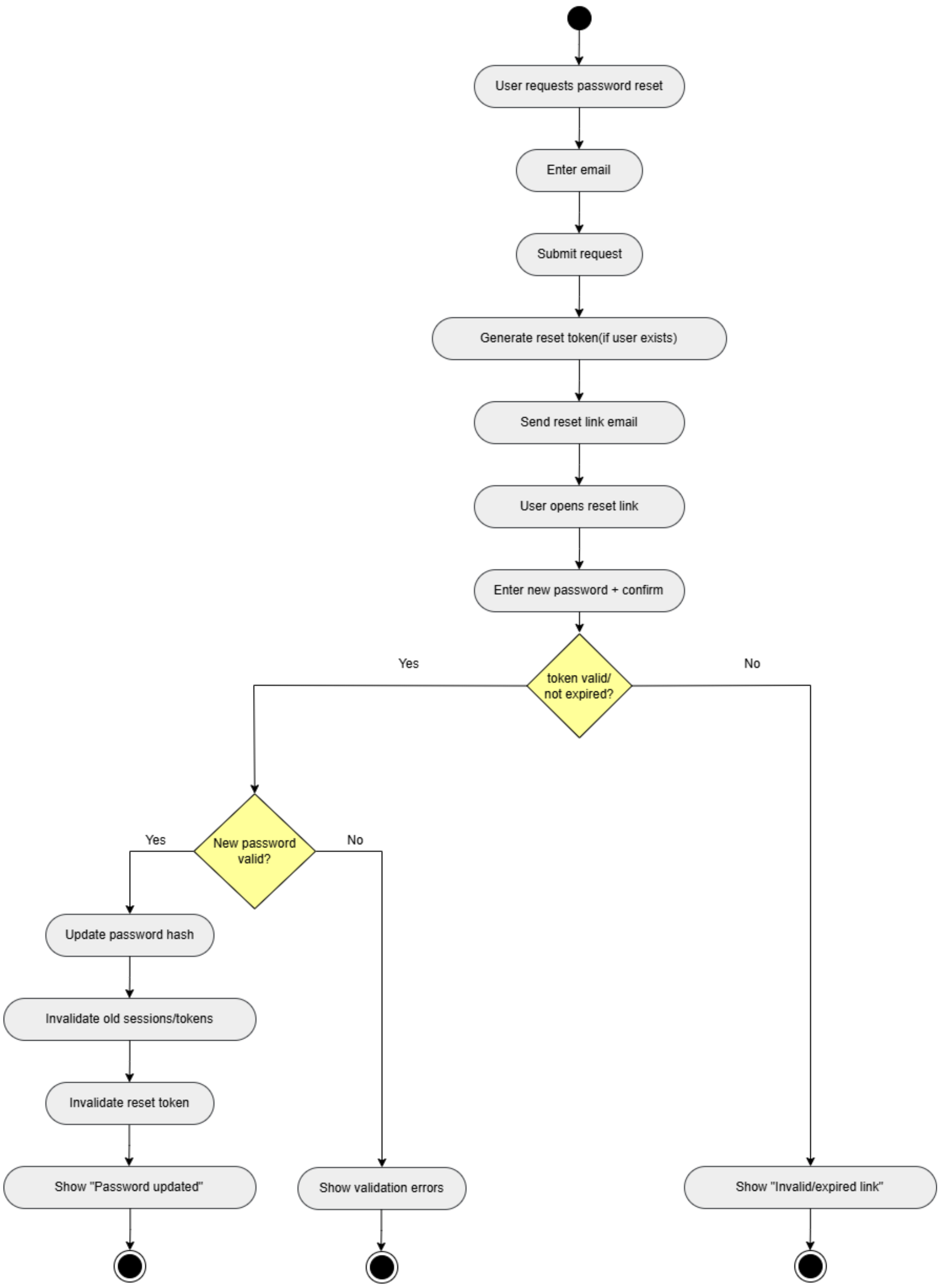


Figure 5.9: Password Reset Activity Diagram

Figure 5.10 shows the flow of activities when a patient sends a chat message in Calm-Connect AI. It begins when the patient sends the message, after which the system analyzes it to identify any implication of a crisis. In the case of no crisis being detected, the system will continue with the normal AI response workflow and the interaction stops. Upon identification of a crisis, an escalation banner and emergency contact information is shown instantly by the system to direct the patient towards immediate assistance. An audit and monitor record of the crisis event is registered and the system can also recommend calling of relevant support services. In this diagram, proactive risk identification, the responsible escalation management, and patient safety are highlighted in the chat communication process.

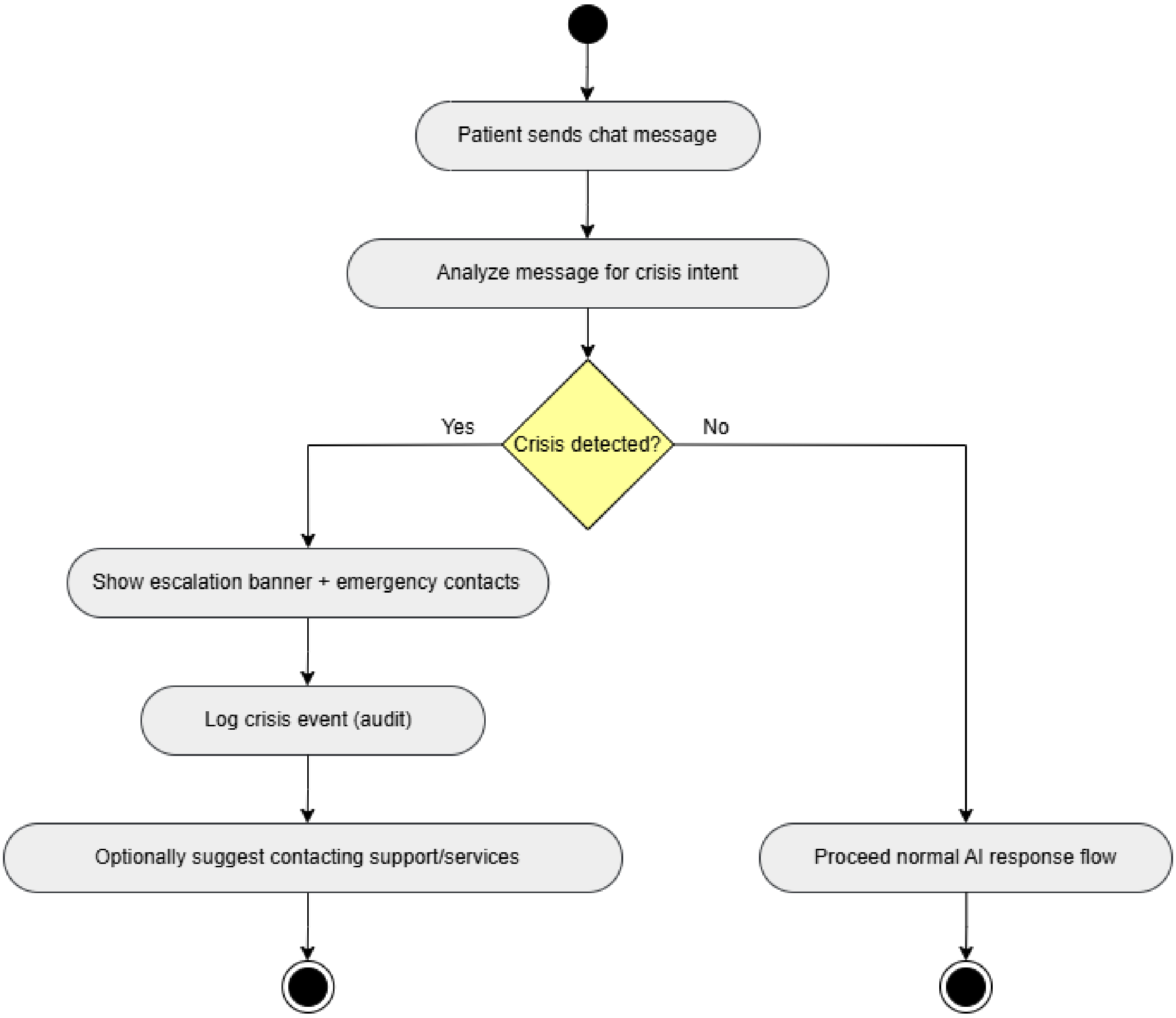


Figure 5.10: Crisis Language Detection Activity Diagram

Figure 5.11 shows the flow of activities relating to updating the appointment requests in CalmConnect AI as a psychiatrist. It starts by the list of outstanding appointment requests being opened by the psychiatrist, who then selects an appointment to look at. The psychiatrist can accept the request where the status of the appointment is changed to approved and they are informed about it. In case of rejection of appointment, the psychiatrist might opt to re-request the appointment by picking a different time slot. At this point, the system verifies slot availability of the recently chosen slot; in the event that slot is free, the appointment information is recorded with a rescheduled status and a notification informing the patient about the change is created. In case the new slot is not available the system will inform the psychiatrist to this extent and the process will conclude. The present diagram indicates that the choice-based workflows, the regulated rescheduling, and the timely notifying patients would guarantee the clarity of communication and the effective management of appointments.

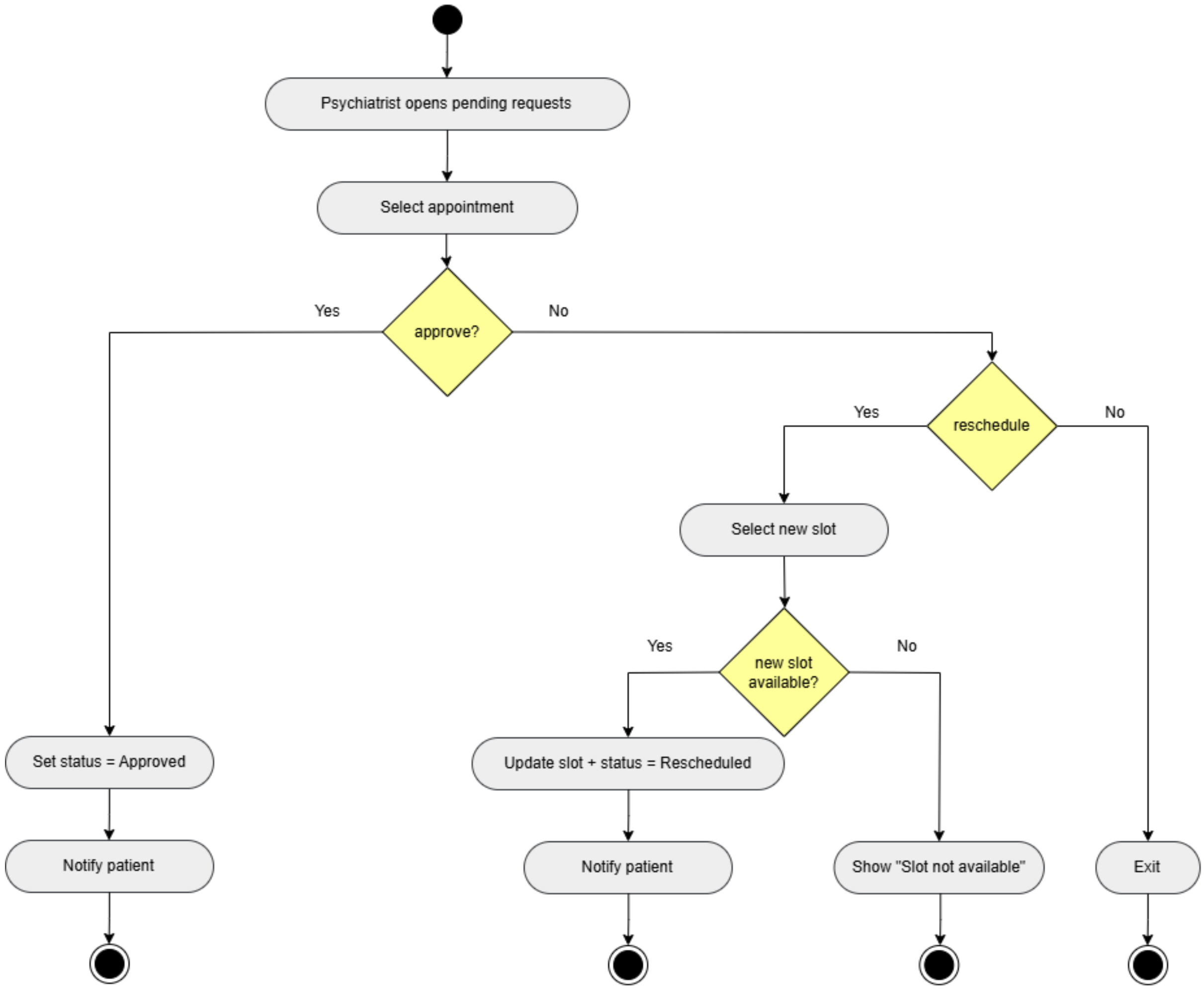


Figure 5.11: Update Appointment Status Activity Diagram

Figure 5.12 shows the steps in the process of developing and recording an appointment summary in CalmConnect AI. This starts with a psychiatrist clicking into the appointment notification and selecting the add a session summary option.

The psychiatrist then keyes in summary notes and this is justified by the system to prevent an empty or invalid record.

Validation can fail and then an error message is displayed and this stops the process. On having a valid summary, it is associated with the relevant appointment and patient record and is stored safely and a success message is shown.

This diagram emphasizes on structured clinical documentation, validation check and appropriate connection of the records of the sessions to continuity of care.

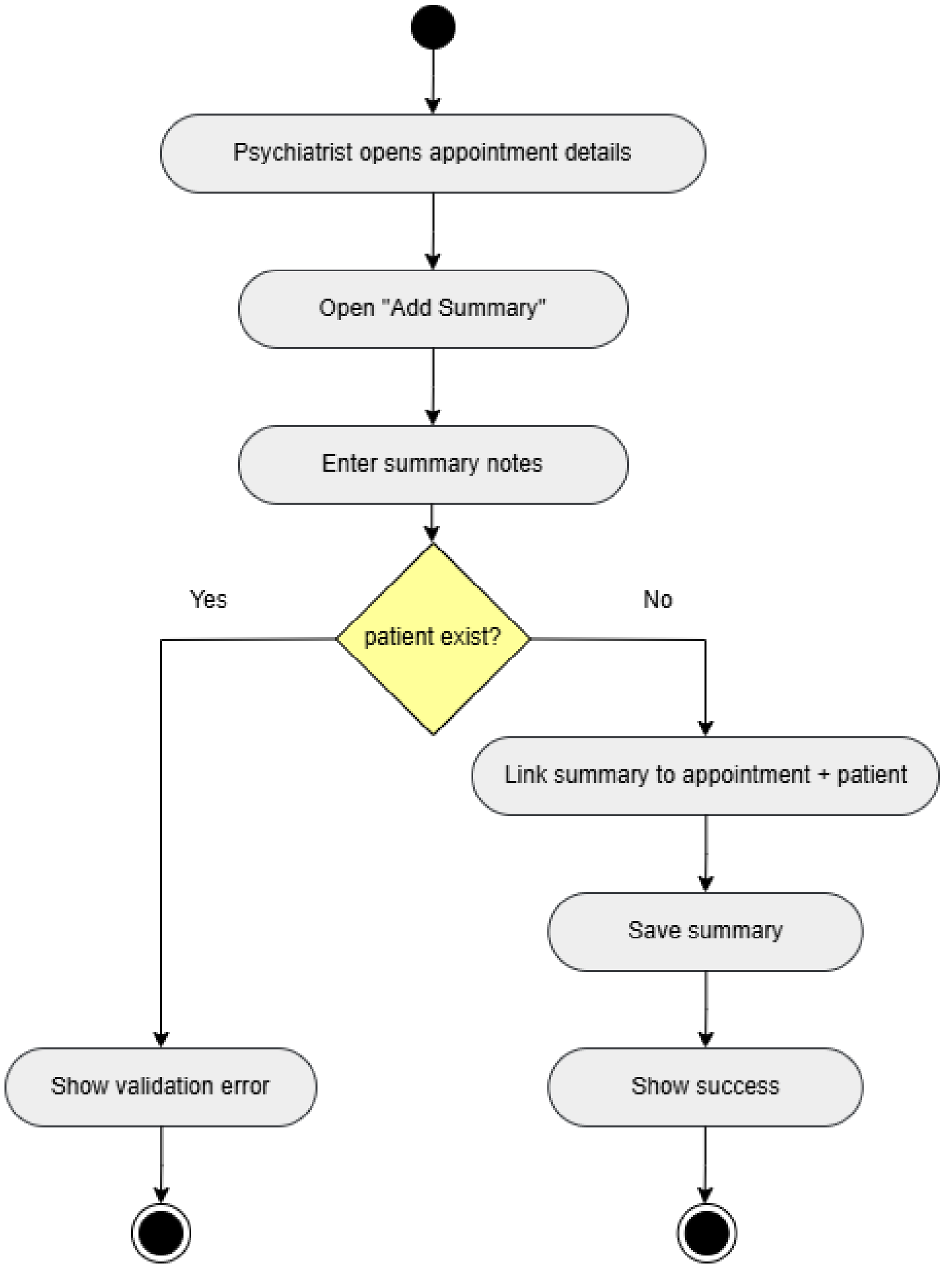


Figure 5.12: Record Summary Activity Diagram

Figure 5.13 indicates the flow of activities in the direct messaging feature of Calm-Connect AI. It begins as far as a user opens up the messaging area, writes a message, picks out a person to send it, and sends the message. The system will look into the recipient account before delivery to ensure that it is not blocked or inactive. In case of the recipient being inaccessible, the delivery error will be presented and the message will not be delivered. When a recipient happens to be active the message is saved, delivered to the recipient and a notification is given. Confirmation is then sent to the sender of the message that the message was sent successfully. This diagram focuses on the controlled communication, delivery validation, and clarity of the user feedback in the messaging module.

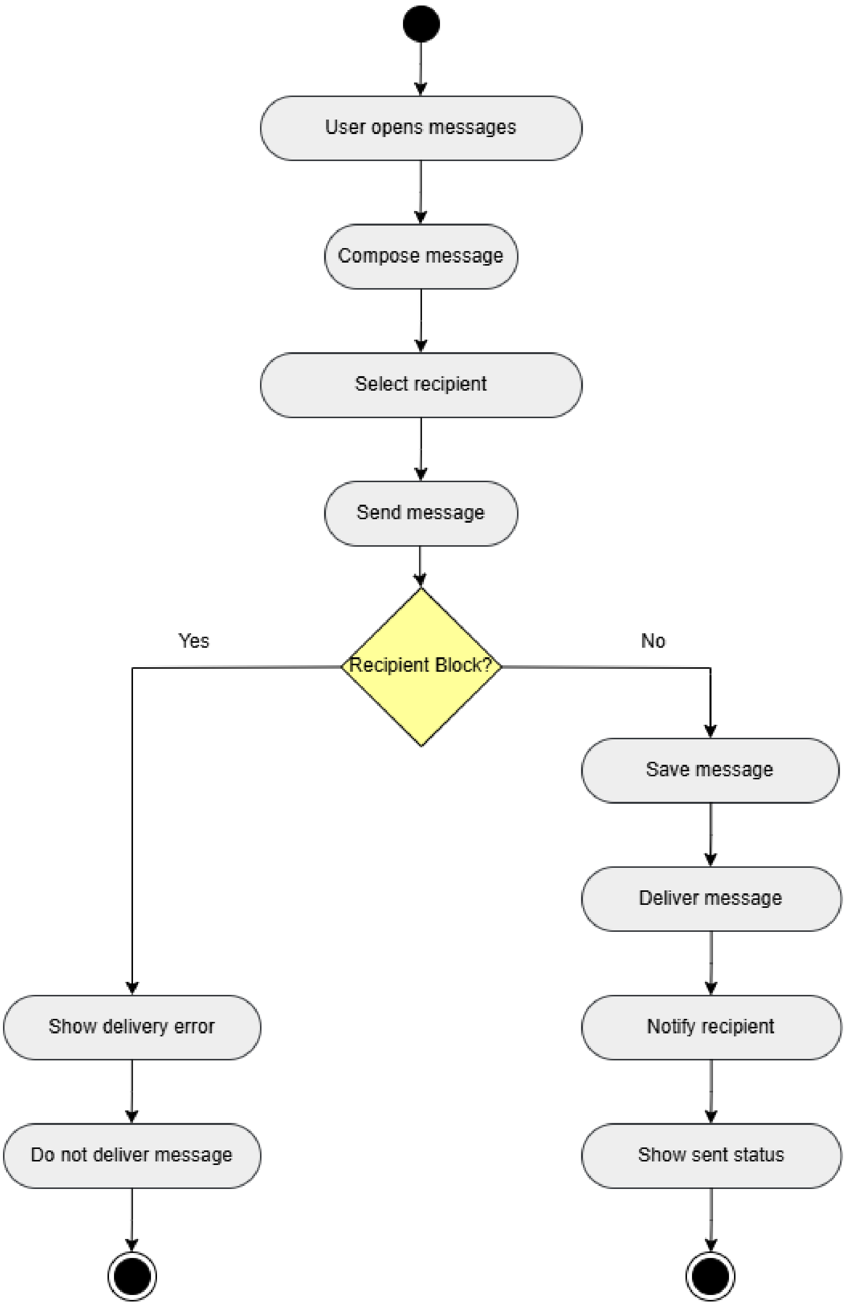


Figure 5.13: Direct Message Activity Diagram

Figure 5.14 displays the activity flow so as to view activity logs on CalmConnect AI in the administrator perspective. It starts with the admin having opened the activity logs page and life-filtered it by either user, day or type of action. The system reads logs with the chosen filters and analyzes by determining the presence of any similar records. In case of records available, it will be presented in a tabular form that is easily readable. In case, no records exist, the system notifies the admin about the same. In this diagram transparency of audits, effective filtering and ensuring secure oversight of systems activities have been highlighted.

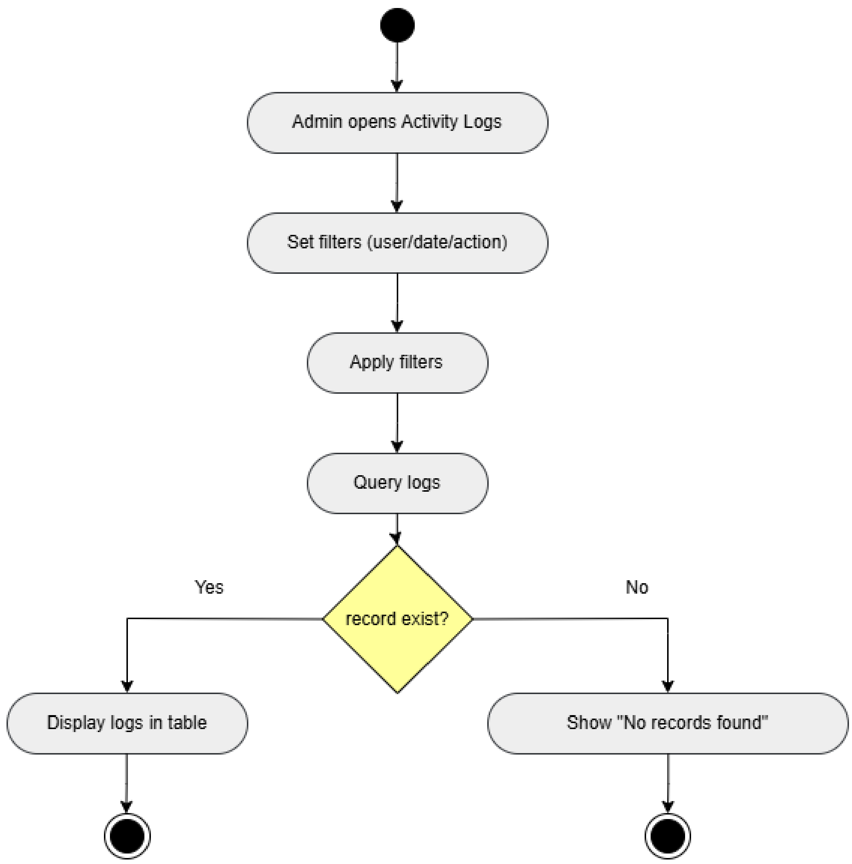


Figure 5.14: Activity Logs Activity Diagram

5.2.2 Communication Diagrams

The communication diagrams dwell on interactions of system objects and messages between them. They depict the interactions that frontend entities, back-end services, and database entities follow in a given case. [20]

These diagrams are used to describe interactions in CalmConnect AI, including the process of user authentication during a log-in, between patients and psychiatrists, updating the status of an appointment, or delivery of a notification. They bring out the chain of messages, explain the part of each part that performs a particular action, and demonstrate dependencies management.

The communication diagrams come in handy particularly when it comes to the way loosely coupled modules combine without jeopardizing the security, efficiency and performance of the system.

User Communication Diagram:
Figure 5.15 illustrates the interaction of a general user to CalmConnect AI core functions. The backend API receives the user requests via the Web UI and processes the completed data with the help of the database and sends an intelligent response to the AI service in case of its necessity. This diagram shows the process of introducing AI features into the normal interactions of users.

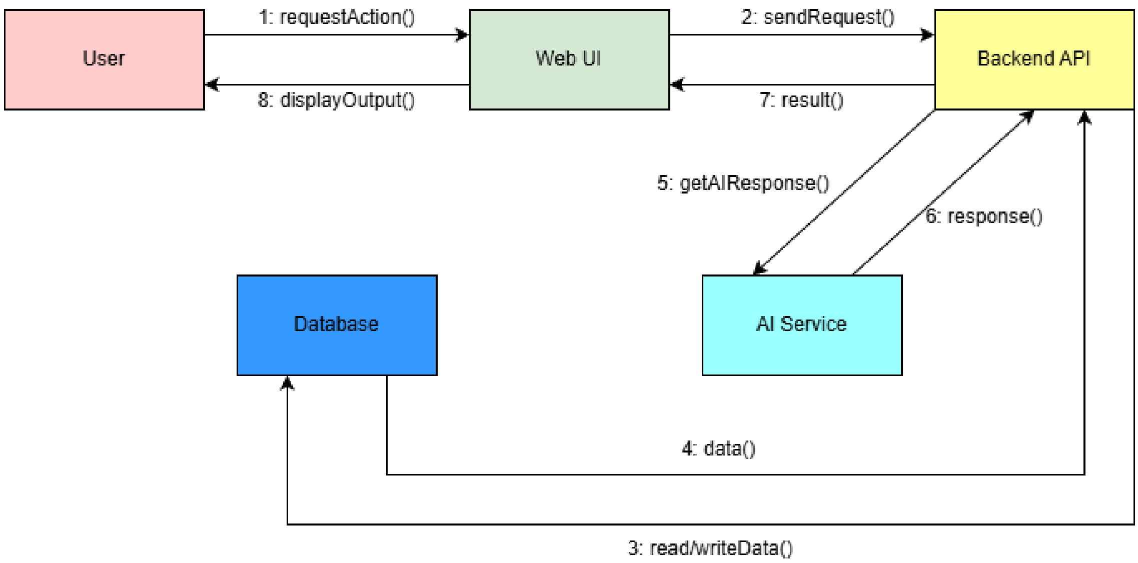


Figure 5.15: Patient Communication Diagram

Admin Communication Diagram:
Figure 5.16 shows the communication flow of the administrative processes in CalmConnect AI. The administrator communicates with the system through the Web UI whereas the backend API is used to process user administration and system administration requests. Every administrative practice is recorded in the database making the data handling centralized and uniform.

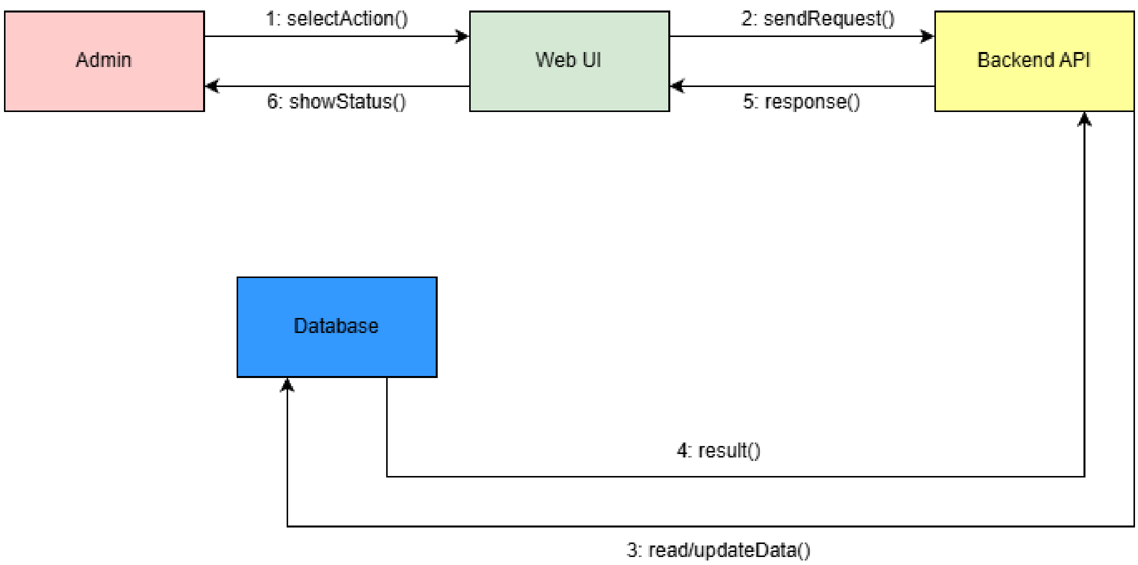


Figure 5.16: Admin Communication Diagram

Psychiatrist Communication Diagram:
Figure 5.17 shows the communication with AI CalmConnect system and the psychiatrist. The psychiatrist does the actions via the Web UI which uses API to communicate with the backend to retrieve and update data related to appointment in the database. This figure

highlights the role of psychiatrist in controlling patient session by controlled interaction system.

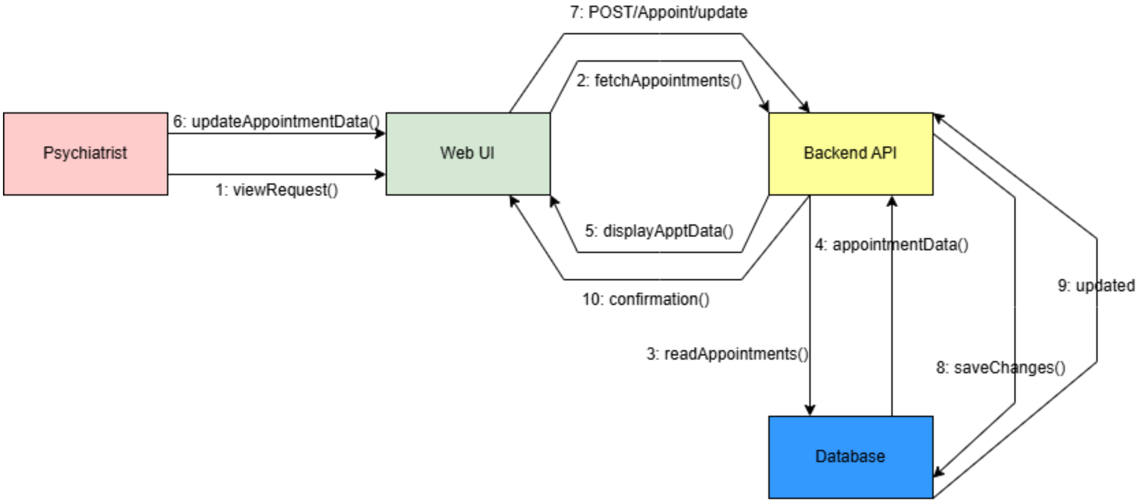


Figure 5.17: Psychiatrist Communication Diagram

5.2.3 Sequence Diagrams

Sequence diagrams: Here, a time ordered diagram that illustrates interactions between actors in the system and internal system components is drawn. They are applied to simulate the most important system circumstances like user log-in, starting an AI talk session, requesting an appointment, and creating a session summary. [26]

Each sequence diagram diagrams depict how the actor communicates with the frontend interface, backend services, database and any external APIs present. Other critical steps like authentication checks, data validation as well as error handling is illustrated explicitly to explain how the system behaves.

Such diagrams will assist in maintaining a track of the execution of a system within the stipulated functional requirements and acceptance criteria. They also aid in defining the operations that could be sensitive to delays or failures so that any possible problems could be handled at an early stage when the design was underway.

The flow of interactions during the registration of the user is shown in Figure 5.18. The user enters registration data by using the web-based or app interface with the request being sent to the authentication service to be verified. The system makes sure it verifies fields required and match email uniqueness with the database. In case of the email being present, a response of error is given back to the user. In case the email is new, it will create a new unverified user and issue a verification token that is stored and has an expiration and a verification email will be sent via email service. The process finishes with confirmation reply where the user is required to confirm his/her email address.

Moreover, the chain imposes a safe onboarding protocol as the accounts are not activated until ownership of the email has been verified. Expiry on token generation deters using it inappropriately, and it minimizes the occurrence of unregistered misuse. This will enhance data integrity and also prevent unauthenticated users to pass through authentication and access to the features of the system.

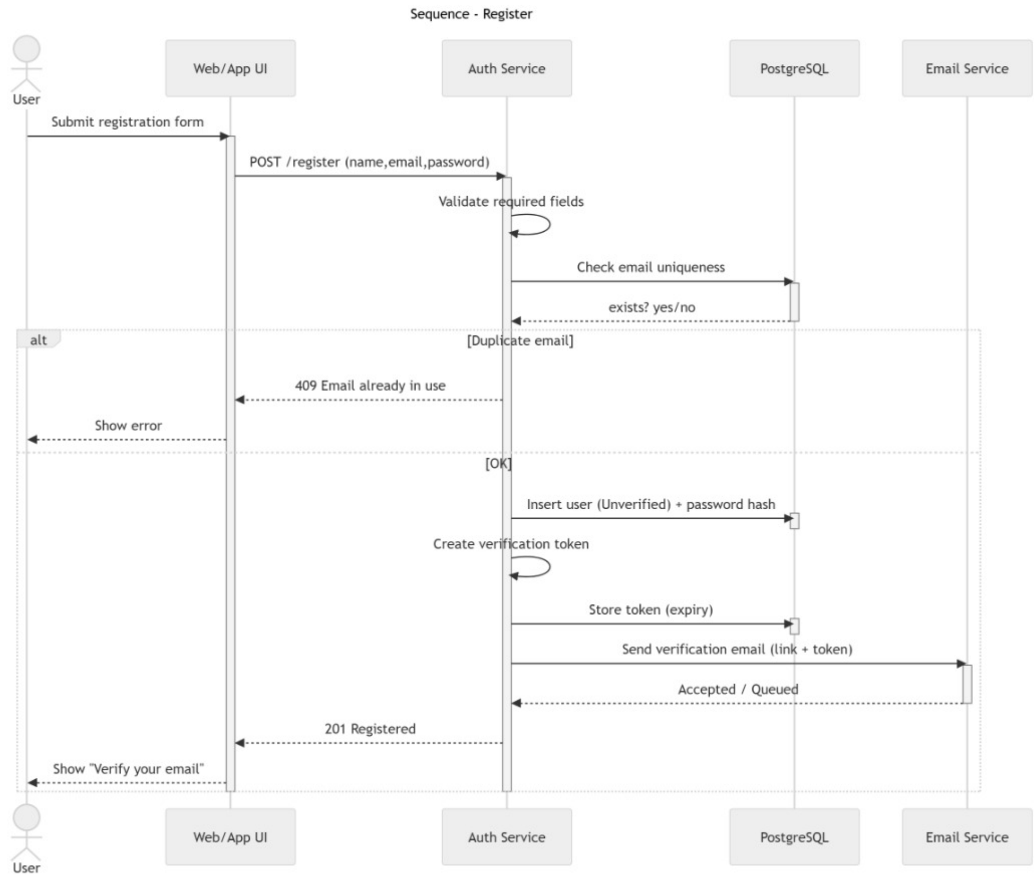


Figure 5.18: Register Sequence Diagram

Figure 5.19 gives the flow chart of the user login process. The user provides the authentication service with login credentials, which is why the authentication service retrieves the user record out of the database. In case the user is not existent or the password is not valid the error response is sent. In case of blockage of account, the access is provided with an adequate message. In the case of valid and active accounts, the system will dig through the password hash and create a secure JWT and role information and send the token back to the UI, which will further redirect the user to the dashboard based on their role.

Moreover, the sequence provides secure handling of the session since the issuance of the authentication tokens is done only after validating credential and account status. Role information is included in the JWT; hence, authorization is achieved efficiently without making possibly repeated database access. Such design enhances the control of access without affecting a smooth security login procedure among the authorized users. It also embraces the use of token expiration and refreshing features to mitigate the threat of hiajacking a session and exposure in case a token is stolen. The token can be stored safely on the client-side to guard against unauthorized users and ensure that user identity is not compromised between uses. All these measures combined provide uniform integrity to sessions; a role-based system-wide access; and maintain usability without the extra authentication load on the legitimate users.

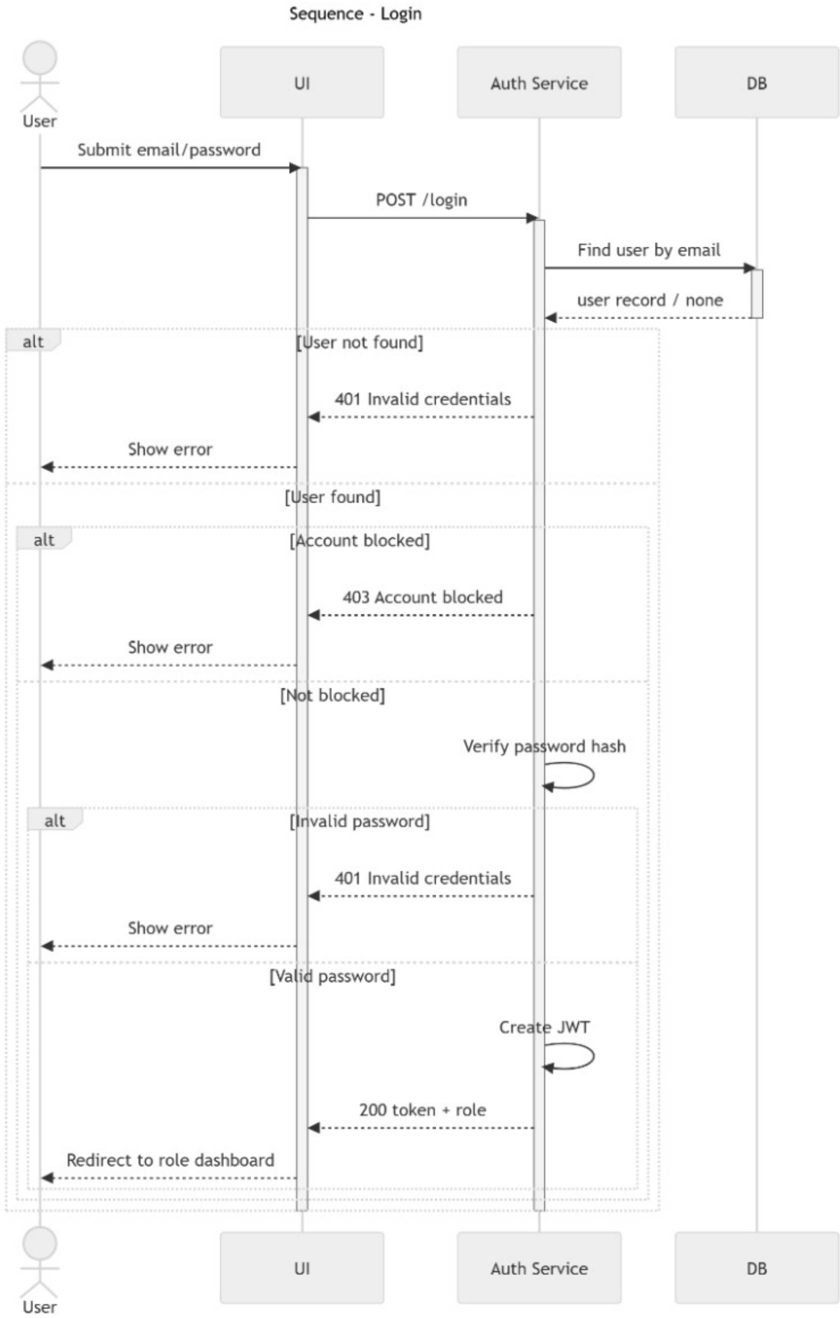


Figure 5.19: Login Sequence Diagram

Figure 5.20 illustrates the flow of sequence of an AI chat interaction. The patient writes a chat message on the UI, and it is relayed to the chat service and stored on the database. The chat service subsequently dispatches a message to the AI API to come up with a response. When the AI service is not available or crashes, the system will record the downtime occurrence and will respond with a service-unavailable response to the user. In response to a successful AI, the response generated is stored and reentered into the UI to be displayed. In order to emphasize message persistence, external AI integration, and fault handling, this sequence is important.

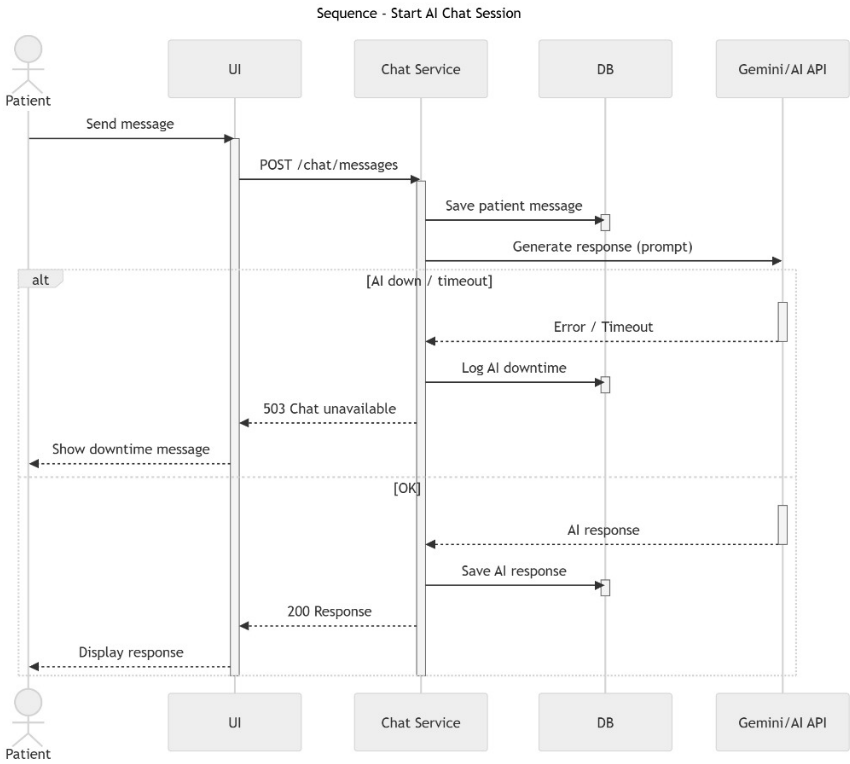


Figure 5.20: AI Chat Sequence Diagram

Figure 5.21 illustrates the flow of interactions with a patient when he/she requests an appointment. The request is placed by the UI and handled by the appointment service which initially verifies the chosen psychiatrist whether it is active or not. On inactivity, the response returned is a response of error. When it is working, the system verifies that slots are available in the database. In the event there are no slots, the user is made to know this. In case of the availability of slots, a pending appointment is established and the notification service is applied to notify the psychiatrist, and a success response is sent to the patient.

Moreover, this chain facilitates a well-managed scheduling based on the fact that only psychiatrists on the active roster and free times are made into appointments. The development of appointments in a pending state enables psychiatrists to go through and attend to requests before confirmation. Notifications that are automated enhance communication and minimize the time lag in the process of appointments approval.

It also implements validation policies avoiding overlapping appointments and booking request within the same time slot. Status tracking allows patients and psychiatrists to track the progress of request to confirmation or rejection. When implemented jointly, these measures promote scheduling integrity and clinician work load management; moreover, they guarantee clear and reliable appointment work flows to all relevant users.

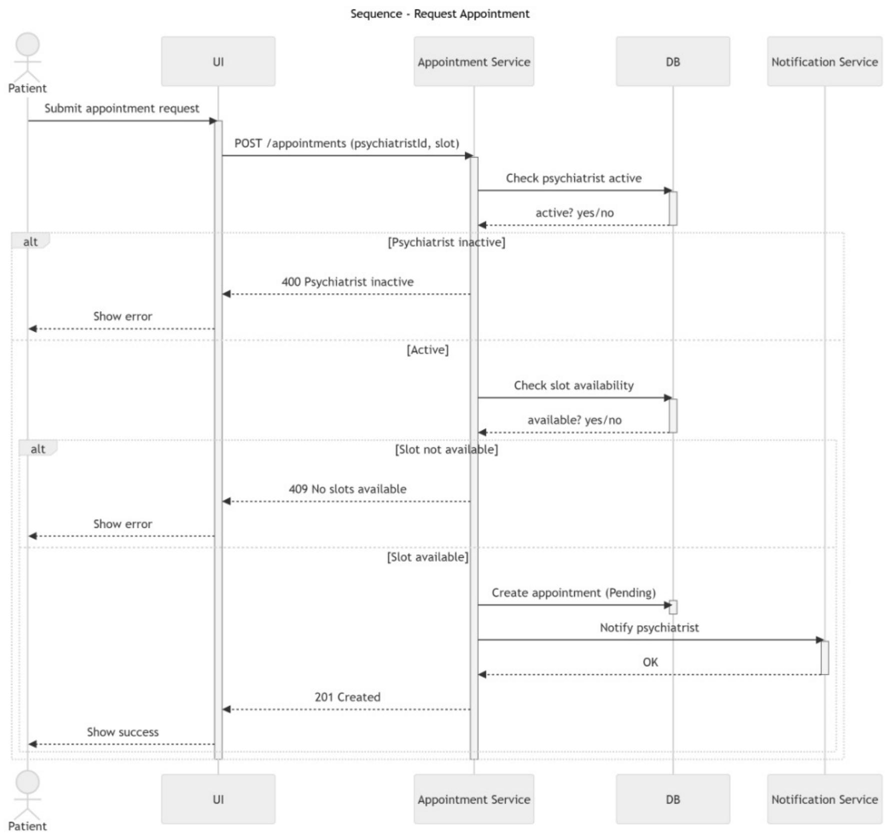


Figure 5.21: Request Appointment Sequence Diagram

Figure 5.22 illustrates the flow of interaction of administrative user management. The UI asked by the admin to the user is retrieved by the user service and the database. When the admin carries out operations like blocking, unblocking, or deleting of a user, the user service also updates the user record. The audit log service logs every administrative action and the end result of the operations is sent back to the UI to be displayed.

Also, such sequence provides accountability and integrity of the system by capturing all administrative actions in the audit logs. Action logging can be used to enforce traceability and facilitate compliance, monitoring and security reviews.

There is also controlled user management operations to avoid unauthorized or accidental modifications to user accounts.

It also aids in role enforcement where the administration is only allowed to access administrative functions. The history of actions can be provided in a logical form of issue investigation and system audits.

All these controls ensure the stability of the platform, the security of user data, and responsible administration throughout the system.

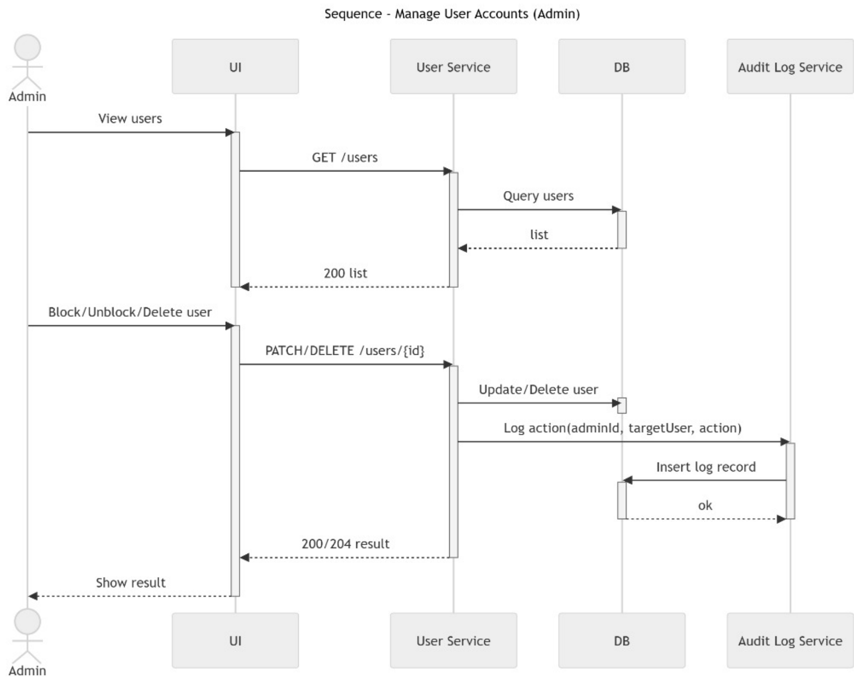


Figure 5.22: User Management Sequence Diagram

Figure 5.23 shows the sequence flow of email verification. On clicking the verification link, the token is forwarded to the authentication service to be verified. The database is used to validate and check the expiry of tokens by the system. In case of the invalidity or expiry of the token, an error is returned. Under the assumption of validity, the user account is considered to be verified, the token is invalidated to avoid resilience, and a success approval is sent to the user.

Besides that, this sequence enhances the security of accounts in the form of enforcing time-limited and single-use verification tokens. Unverification of the token when that is successfully verified blocks replay attacks and inappropriate reuse. There is also the clear response of success and errors that help the user navigate the operation of the verification and prevent confusion on the process of activating the account.

It also makes sure that account access by the user is only achieved once his account status has been confirmed. Expiry processing of tokens prevents any slow/malicious verification requests. Collectively, these measures will enhance confidence in the registration process and have a transparent, secure, and easy-to-use account activation process. It also gives system reliability because the verification attempts are always verified against stored token records. Unsuccessful or time-lapsed verification requests are rejected harmlessly without disclosing sensitive information about the system.

This will ensure that the integrity of the data is maintained, that the verification mechanism is not abused excessively, as well as guarantee that the onboarding experience of a new user is efficient and secure.

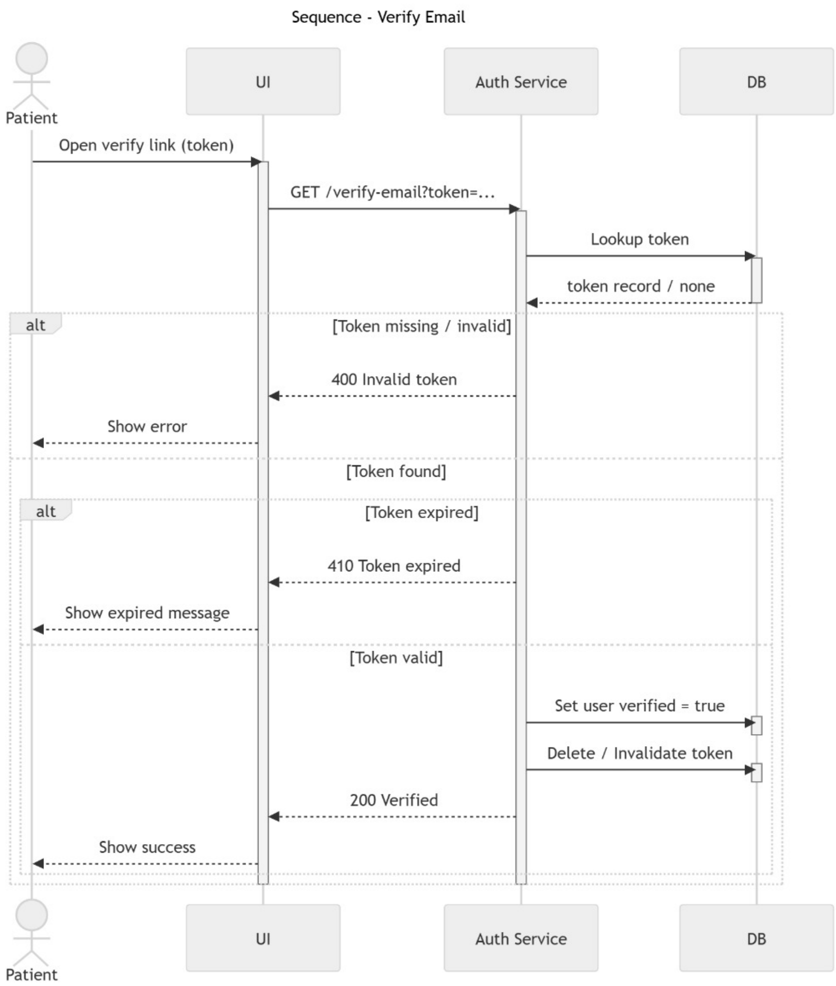


Figure 5.23: Email Verification Sequence Diagram

Figure 5.24 shows the entire steps of password reset. At the request phase, the user provides his/her email address and the authentication service creates a reset token and stores it and mails a reset link through the email service. The reset stage involves the user entering a new password with the token, and this is checked as to be correct and expired. In case of invalid or expired token, an error will be returned. In the event that it is valid, the password hash is revised, the current session and tokens are invalidated and a success notification is sent back. This chain of being provides secure credentials recovery.

Besides that, this chain secures a user account by deactivating previous sessions and tokens in case of a successful password change and minimizes the possibility of unauthorized access. In case a reset link is severed the exposure is minimized by token expiry. These measures combined will make password recovery a safe and regulated procedure. It also provides passwords verification and verification before doing the update, before the update is done, it will not change the password with weak credentials or mismatched credentials. Explicit responses of success and failure also give information to the users on whether the reset has been successful or not. All in all, such a sequence ensures the security of accounts and enables users to regain access effectively and with little wait-time.

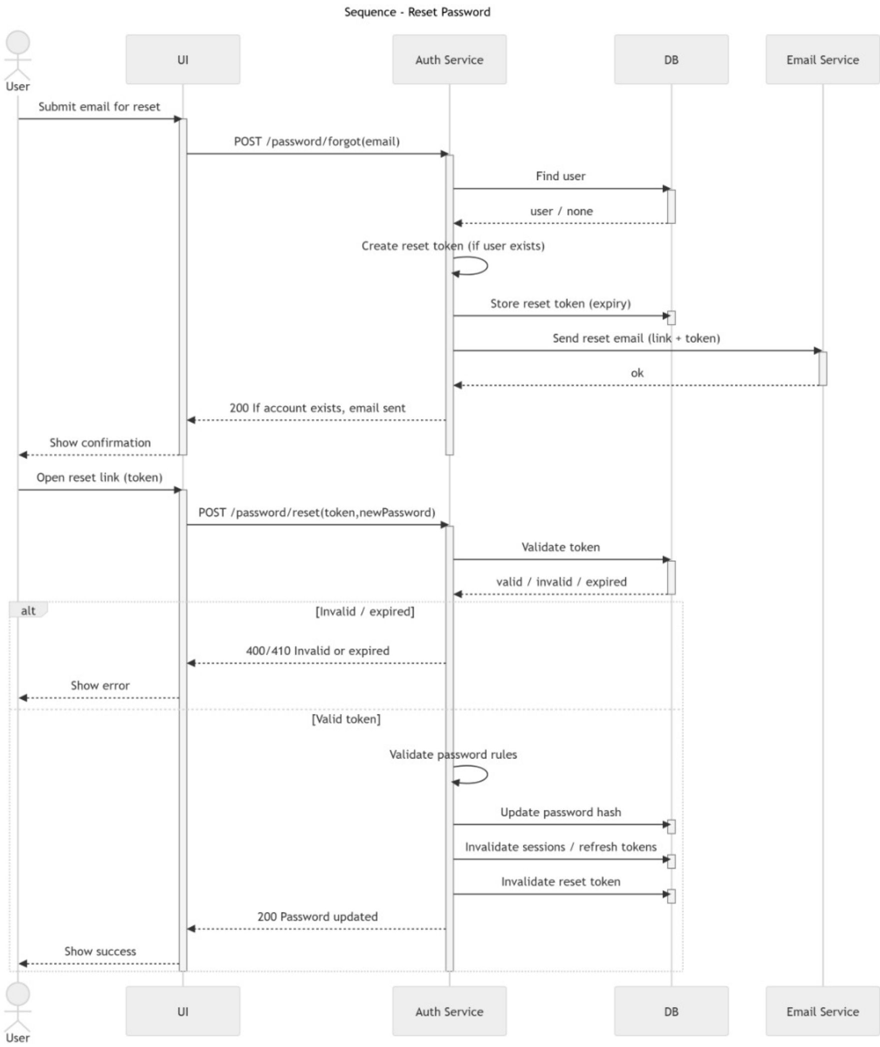


Figure 5.24: Password Reset Sequence Diagram

Figure 5.25 depicts the process flow chart of identifying language use due to crisis in chat interaction. A message is sent by a patient into the chat service where the message is stored and then sent to the safety or crisis detection component where the message is analyzed. In case a crisis is reported, the system documents the incident in case of audit and sends an escalation response, which causes the appearance of emergency instructions and contacts on the UI. In case of absence of a crisis, the conversation follows the regular pattern of interaction. This order of practice is in favor of patient safety and reasonable intervention.

Moreover, the sequence will help to identify risks early because messages are analyzed before the regular response is generated. Crisis events logging makes it possible to monitor and work on safety mechanisms continuously. Immediate direction and emergency contact information will serve to provide time-sensitive assistance and ethical and responsible conduct of the system.

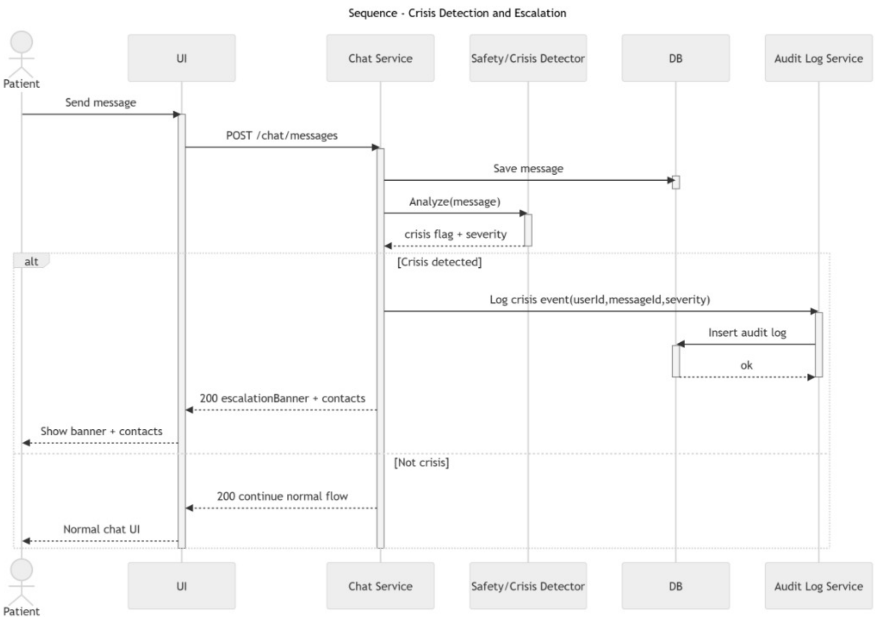


Figure 5.25: Crisis Language Detection Sequence Diagram

Figure 5.26 shows how the interactions occur during an update on the status of an appointment by a psychiatrist. The approval or rescheduling action is submitted by the psychiatrist in the UI and is performed by the appointment service. To get an approval, the appointment status is changed in the database and the patient informed about the change. In the case of rescheduling, the service examines the availability of new slots first, and to show that it is unavailable, an error is sent. In case of availability, the status and slot of appointment is changed and the patient informed. In this order of sequence, emphasis is placed upon controlled decision handling and communication in a timely manner.

Moreover, the sequence guarantees its uniformity in appointment management by authenticating slot availability and implementing change that is made. Uncertainty and lack of coordination between the patients and the psychiatrists are minimized by providing instant information to patients. The well-organized update procedure avoids conflicts in the schedules and keeps the right records of appointments. It also promotes transparency because all parties involved will include changes in the status of the appointment in real-time. Recorded changes enable the changes to be checked again in future and facilitate resolution of disputes in case of necessity.

All of these measures ensure the stability of schedules, effective communication, and proper record-keeping during the process of appointment. It also maintains data integrity by implementing upgrades in a non-atomic manner so that appointment is not left in some irregular positions. The permissions of access control deny unauthorized roles the right to make changes, decreasing the prospects of accidents.

All in all, this flow provides a reliable and auditable process of updating about their appointments that can balance operation accuracy and understandability.

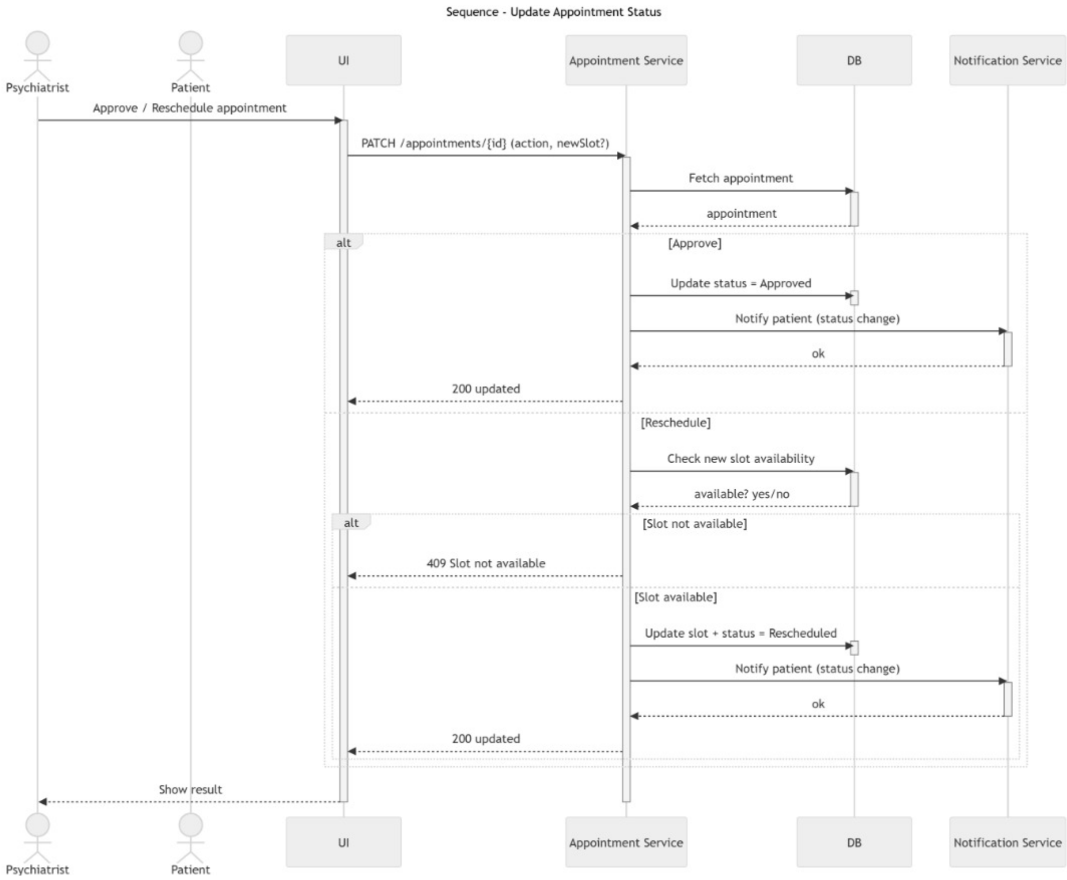


Figure 5.26: Update Appointment Status Sequence Diagram

Figure 5.27 is the order of taking note of the session summary by a psychiatrist. The psychiatrist uploads summary text into the UI, which is sent to the service of session notes to be checked. In case the summary is either empty or invalid the system gives a validation error to the UI. In the case of valid input, the summary gets logged into the database and integrated with the relevant appointment, patient as well as psychiatrist record. A success shot is then sent back and it is a confirmation that the summary has been saved. Such chain facilitates organized records and information security of the clinical records.

Moreover, the continuity of care is aided by this sequence because, in such a way, it is the most effective way to guarantee the safety of the connection between the summaries of the sessions and the given clinical scenario. validation checks are used in the quality and completeness of recorded notes used in storage. Summaries should also be stored in a systematic way that will allow them to be reviewed later, reported on, and used to make clinical decisions. It also provides a high level of confidentiality since only users who have rights to see session summaries are allowed to do so. The audit trails show the time when the summaries are made or updated to back up accountability and compliance. Combined, the measures help maintain clinical accuracy, safety of sensitive information, and serve the long-term care and care of the patient. It also enhances reliability of the system in the sense that once a summaries are saved then it means that the summaries have run successfully. Summaries Versioning provides historical contextual updates. On the whole, this sequence is providing both ongoing and subsequent treatment and analysis-supportive secure, traceable, and clinically meaningful documentation.

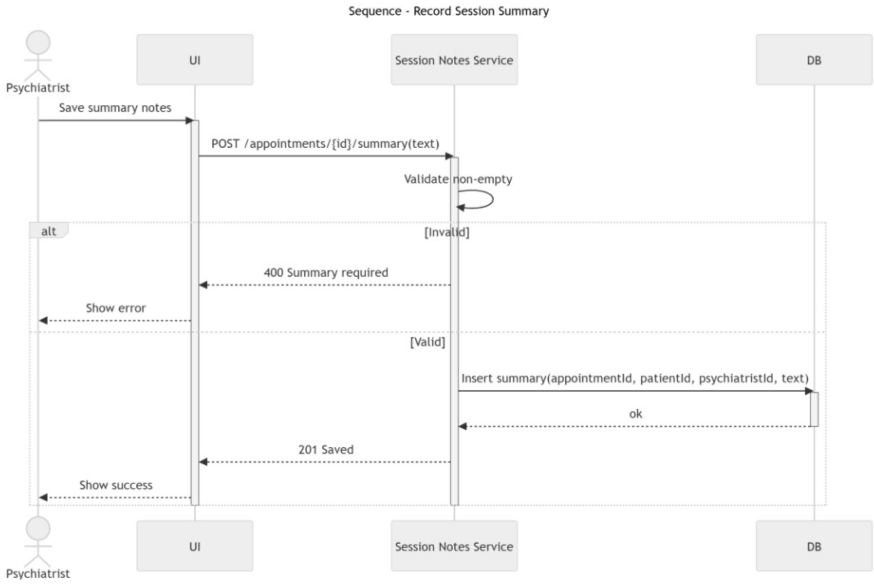


Figure 5.27: Record Summary Sequence Diagram

Figure 5.28 illustrates the process flow of the direct message between the users on the system. A patient sends a message to a psychiatrist using the UI which redirects the request to messaging service. The service verifies the status of the recipient accounts in the database before delivery. in case of a blocked or a non-active recipient, delivery of the message is canceled and a corresponding error response is sent back to the sender. In case the recipient is active, it is stored, delivered successfully and a notification is sent to the recipient. A sender is then sent the confirmation that the message has been sent.

Besides this, this sequence will make sure the communication is controlled in that they do not send messages to inactive or limited accounts. Caching of messages prior to transmission of the message messages helps in maintaining conversation history and aids in reviewing the messages in case of necessity at a later time. Transparency and responsiveness are enhanced by delivery confirmations and notifications, in the messaging feature. It also controls the access to guarantee message exchange between authorized agents. Record Timestamped records are conversation ordered and enable audit or dispute resolution. All these measures offer secure and dependable communication with good tracking that enhances communication between users. It also helps in improving the reliability level by gracefully failing message delivery and trying to resend the message when necessary. Secure storage ensures confidential messages are not accessed by unauthenticated users. On the whole, this sequence ensures message integrity, user communication security and a consistent and reliable messaging experience throughout the platform. It also enhances scaling, where one can use it to handle the large volumes of messages without affecting the performance of the messaging service. Status informers like sent, delivered and failed messages enlighten the users on the success or failure of communication. In all, this design is balanced in terms of security, reliability and usability such that real-time and asynchronous messaging is reliable even as the platform expands.

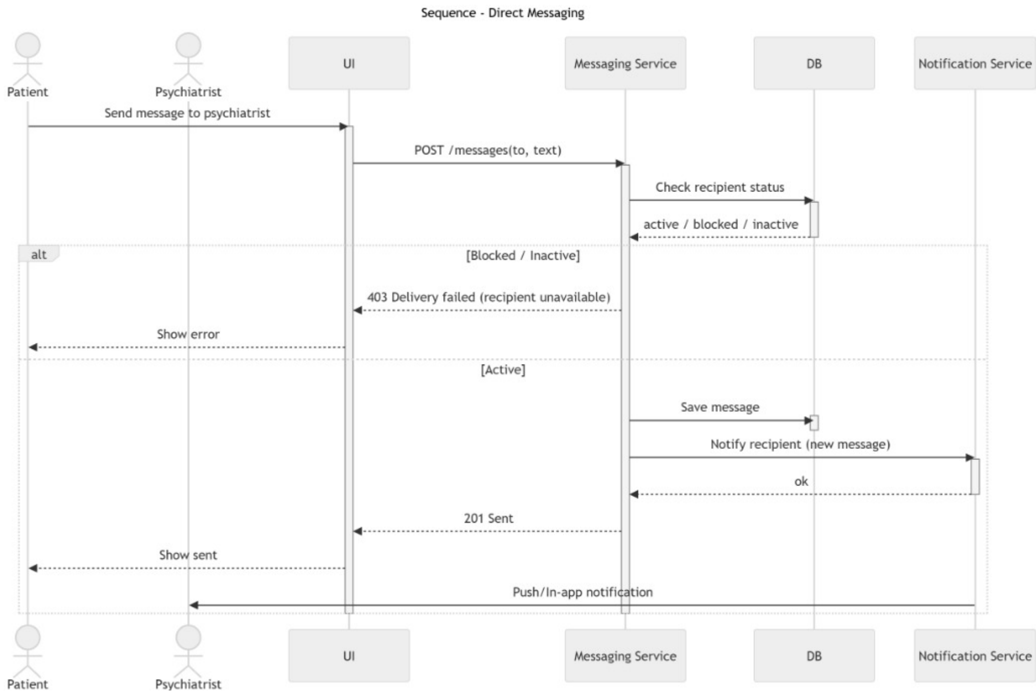


Figure 5.28: Direct Message Sequence Diagram

Figure 5.29 shows the process of interactions during the viewing of system activity logs by an administrator. The audit log service receives a request using the UI where the admin applies filters of the information, e.g. user, date, or action type. The service makes a query call to the database with the given filters, and it provides records according to the query. In case of non-existence of records, blank result set is returned and the UI shows the clear no-records-found message. In case records are present, then the log entries are returned and displayed as a form of a structured table. This sequence puts emphasis on safe access to audit, effective filtering and open monitoring of the systems.

Moreover, this chain of custody is useful in enhancing accountability since the administrators can monitor user and system-activity across a period of time. Filtered log retrieval enhances the performance of an investigation when carrying out an audit or when analyzing an issue. It is better to provide results in a logical order to make them easier to read and keep track of the system action. It also implements access control over sensitive logs to enable only authorized administrators to access them. Unalterable records which appear are time-stamped and guarantee integrity of logs and deter malpractice. Collectively, these measures provide transparency of monitoring, compliance requirements as well as enhance overall system governance. It also enhances reliability because it makes sure that there are logs that have been taken over all the critical components of the system. The historical records are enhanced by secure storage and retention policies that maintain the records and facilitate analysis in the long-term. All in all, this chain offers a strong and auditing logistic system that can be used to facilitate oversight, accountability, and sustained system enhancement.

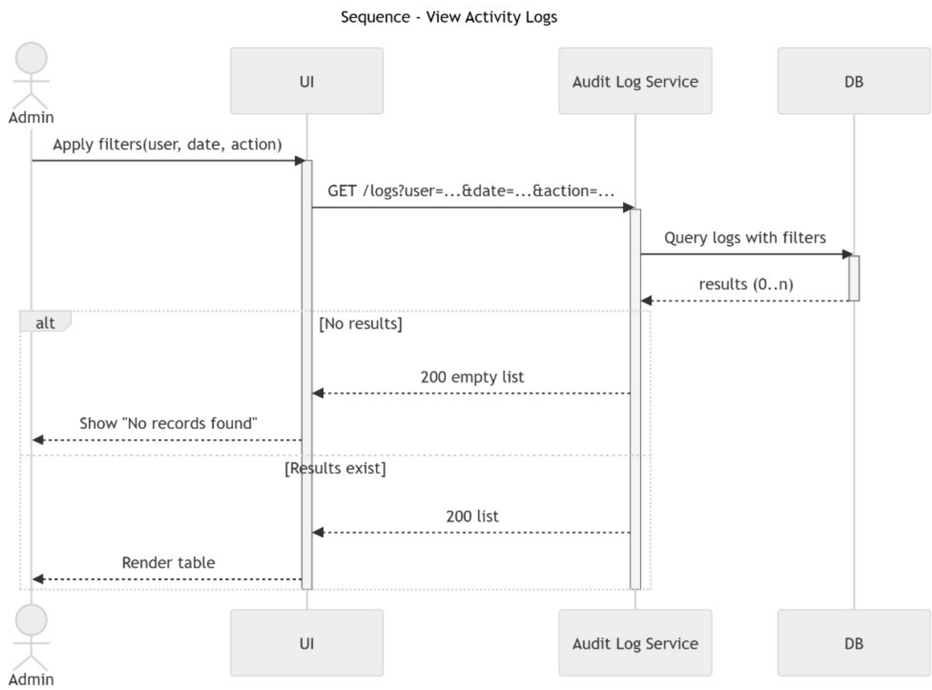


Figure 5.29: Activity Logs Sequence Diagram

5.2.3.1 Login

Figure 5.30 displays the registration screen of the CalmConnect AI site which is intended to offer an easy and safe authentication process. The system name and logo appear on the top and the following points appear namely, input fields where one can put in email address and password with a password visibility control at the convenience of the user. A reminder link with an option of forgot your password can be used to get back into the system provided there is necessity whereas a prominent sign in button can be used to submit the form. There is a Sign up link at the bottom which takes new users to the sign up page. The general design has been minimal and easy to use and is more centered on ease of use, accessibility, and security.

Besides that, the interface is able to handle any errors or provide the user with instructions as the failure to authenticate will show the invalid credential or the fields that are required to be filled in. The visual hierarchy is kept clean, thus, assisting the user in achieving maximum speed regarding the process of completing the login process. The design encourages the first interaction with the platform to be pleasant and straightforward with the necessary level of security. It is also consistent to the overall system design making experience on various modules familiar. Real-time feedback helps in reassuring users to authenticate as well as eliminates the number of reauthentications. Generally, usability and security are well balanced in the interface, which makes the interface reliable and welcoming to the users. It also promotes access with the use of clear labels, legible texts and the placement of controls. The subtle visual effects lead users in terms of direction and they do not overload the interface. All in all, this design will help promote trust, minimize friction throughout the process of logging in, and strengthen the positive initial impression of the site.

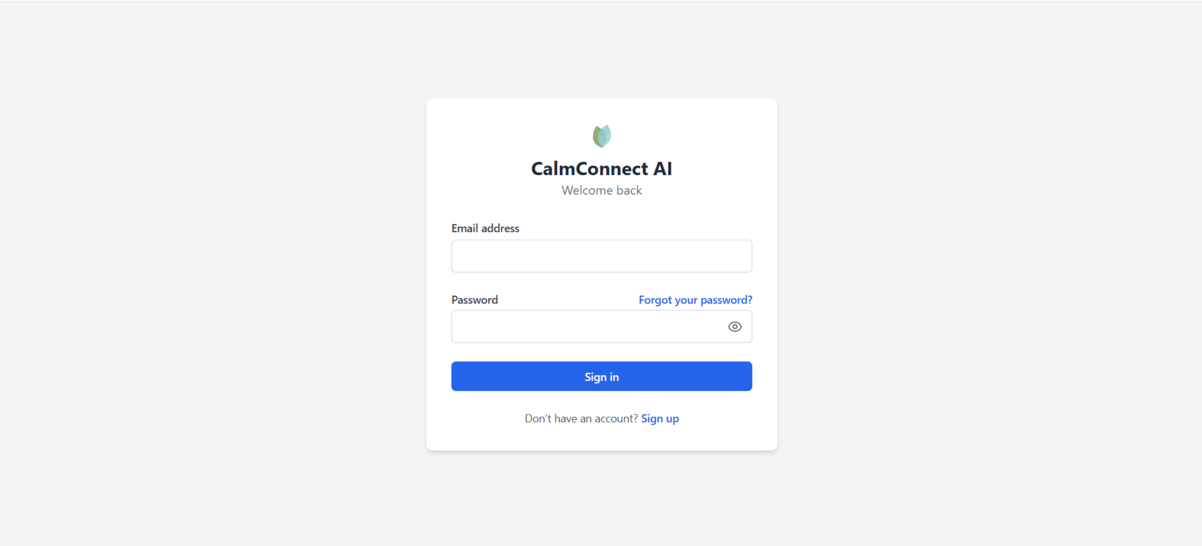


Figure 5.30: Login

5.2.3.2 Logout

Figure 5.31 illustrates the dialog box to be overcome during the logout in the CalmConnect AI platform. The dialog will come as an overlay modal on top of the primary AI chat screen, where the user is required to confirm that he/she wants to quit the system. It has a prominent alert symbol, a brief confirmation message and two action buttons, "Cancel" and "Log Out". The user can have a chance to stay online with the help of the Cancel option and to provide the procedure with the help of the logging out button which authenticates the chosen action and finishes the session safely.

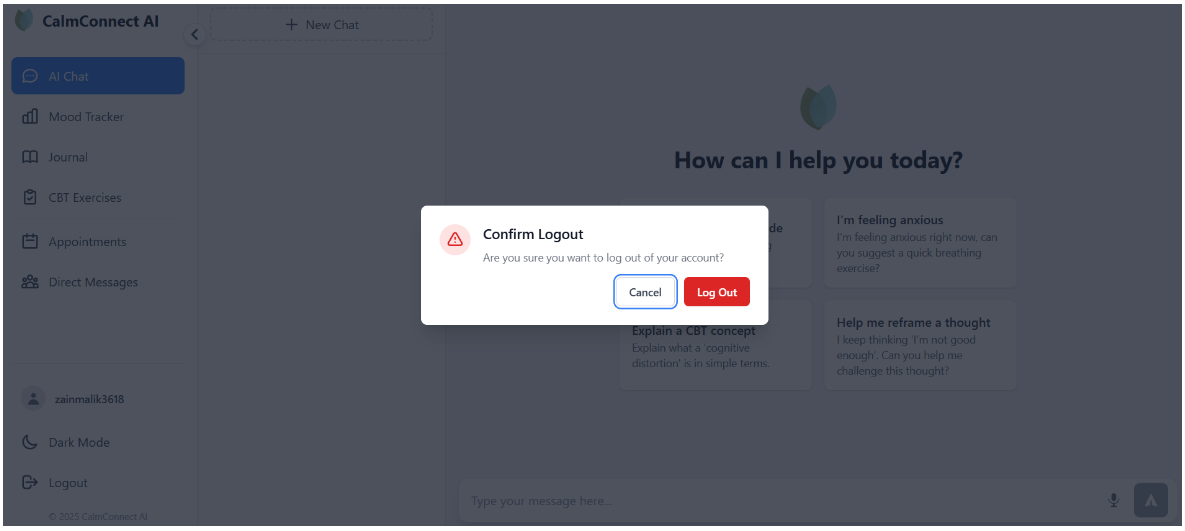


Figure 5.31: Logout

5.2.3.3 Change Password

Figure 5.32 displays the Profile Settings window of CalmConnect AI, where the user can see and update his account details safely. The number of editable profile information, including username and email address, is seen on the screen, and this is followed by the section, where the account password can be changed.

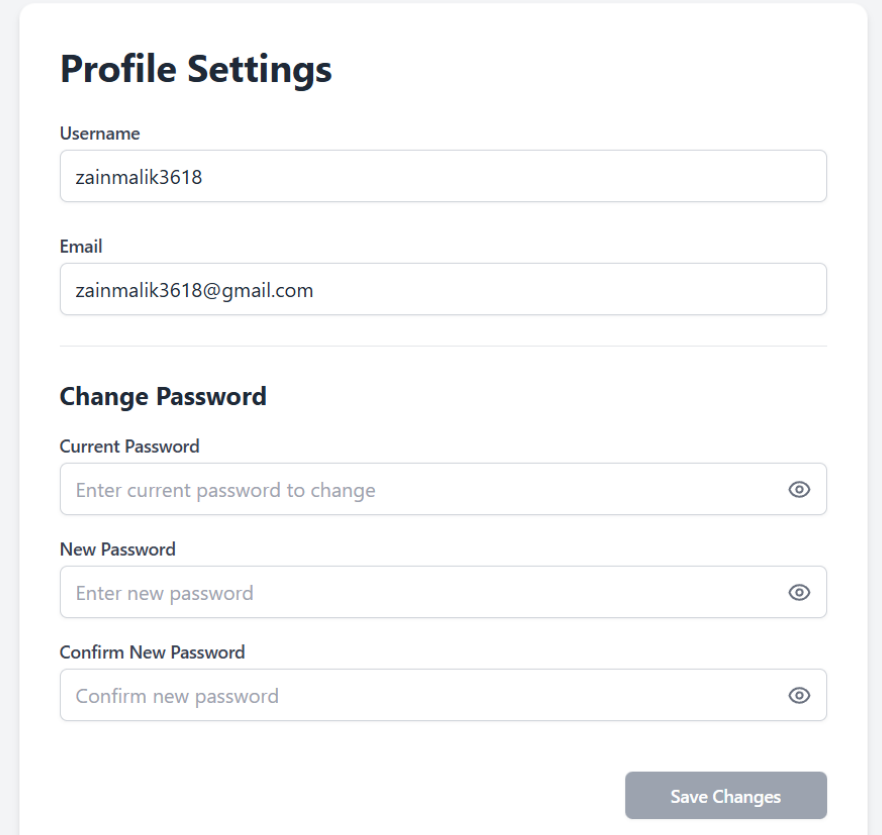


Figure 5.32: Change Password

5.2.3.4 View Database

Figure 5.33 shows the database perspective of the CalmConnect AI system that can be seen in the PostgreSQL management interface.

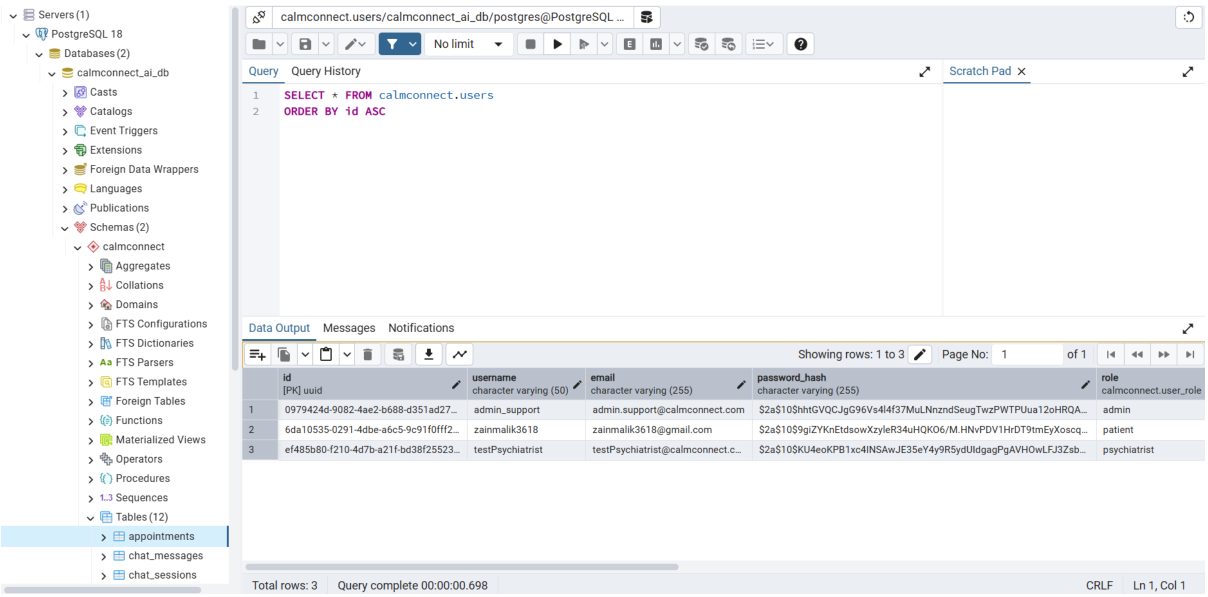


Figure 5.33: Database Schema

Chapter 6

SYSTEM INTERFACE AND PHYSICAL DESIGN

This chapter demonstrates the screens and key physical data structures of CalmConnect AI that are presented to users. This chapter aims at explaining how the system looks inside to the people using it and how the vital information is stored in the database. CalmConnect AI interface design has remained straightforward, safe and comprehensible to give the patients, psychiatrists, and administrators the ability to carry out their duties without creating an unwarranted burden of working. Privacy, clarity and controlled access are also considered to be important in the interface since the system is sensitive to mental health related communication. [27]

Physical architecture Opera of the system is founded on PostgreSQL and accommodates the main functions that have been already mentioned in previous chapters, including, but not limited to authentication, appointments, di- direct messaging, AI chat, and activity logging. The table structures are made such that they support role-based access control, secure record keeping, and scaling the platform in the future.

6.1 System User Interfaces

The CalmConnect AI user interface is structured to have a clean and user-friendly interface. The screens are touch-friendly and can work on desktop and mobile devices. The design is in a basic format, leaving the users to concentrate on core business activities which include logging in, conversing with the AI assistant, making appointments, maintaining moods and analyzing their personal information. All interfaces are labeled, and action buttons visible and easy to navigate across so that any user with various degrees of technical skills can use the system. [28]

6.1.1 Sign-up Interface

Figure 6.1 contains the first entry point of the users who would like to sign up to the CalmConnect AI platform. It is created to gather simplistic user data in a transparent and safe way. The form also has the following fields: username, email address, password and confirm password. Users are automatically created as patients; psychiatrist accounts undergo a different verification after registration.

The interface design is maintained simple and structured to ensure that one can go through the registration process without confusion. All input fields are well marked with validation rules applied to make sure that all data is well entered. The passwords area implements password security rules, including but not limited to length, upper and lower case use, numbers and special characters. Icons that enable password visibility are also offered, which enhances usability without jeopardising privacy.

When the user inputs the invalid or incomplete information, a system shows the appropriate-ate validation messages. Once this is successfully submitted, account creation process will commence and an email verification step will come forth to ensure only verified users can access it.

This interface is significant in the initial interaction with the user with the system. Uncomplicated and facilitated registration process helps to decrease friction and promote usage of the platform. It also helps in making sure that account creation is done in a good security manner.

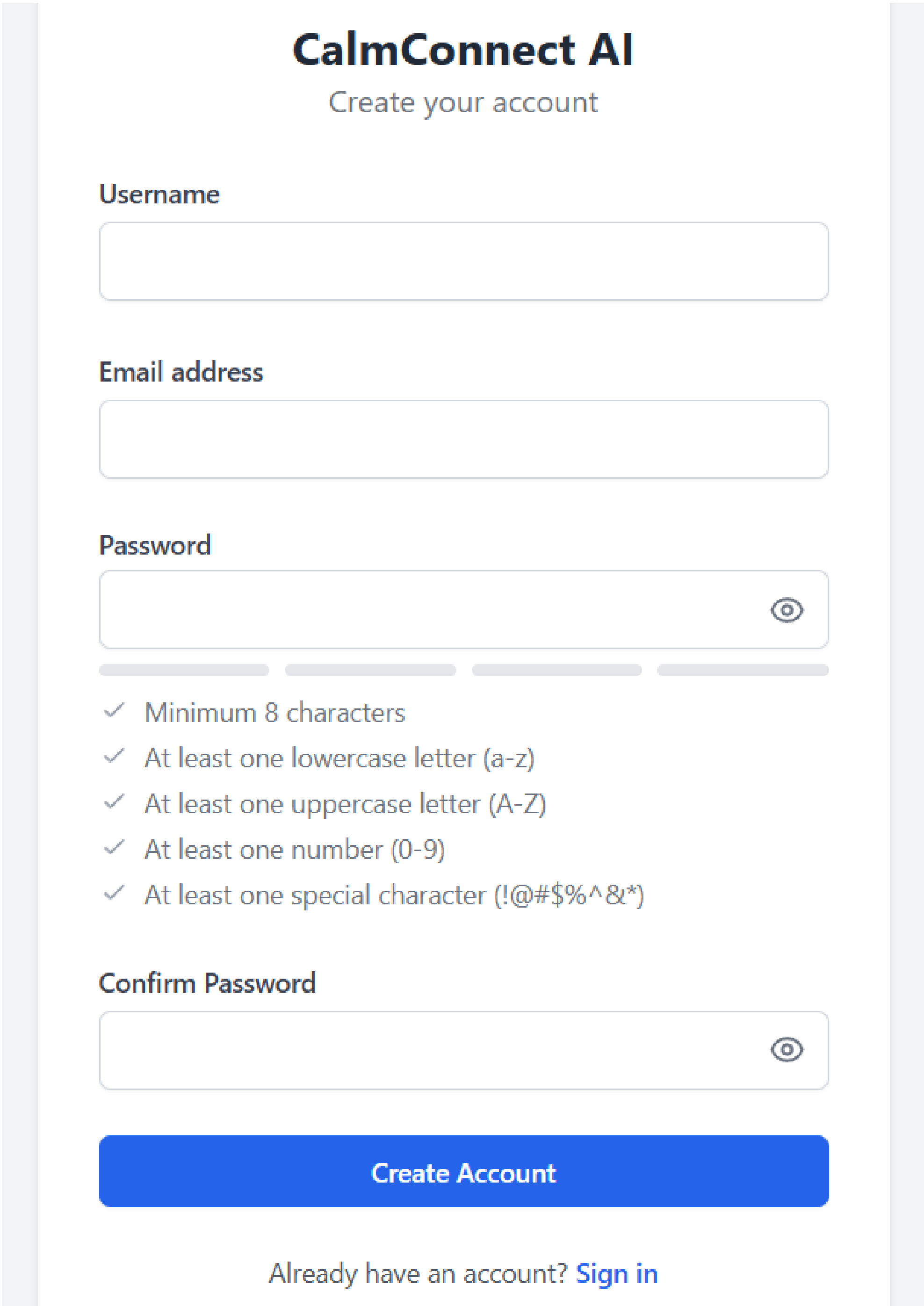


Figure 6.1: Sign-Up Interface

6.1.2 Sign-in Interface

The registered users have secure access to CalmConnect AI plat- form in Figure 6.2. It enables the users to enter their email address and password to login. This interface serves as the central access point into the system and would guarantee that the system allows only authorized users to access the features that are protected.

The sign-in screen is a simple design. It contains well marked entry points where email and password is typed and a sign-in button has been strategically positioned to facilitate easy communication. There is a password visibility switch that can be offered to ensure that users input their details in a manner that is sensitive of their privacy. Moreover, there is a Forgot your pass- word wherein users can log in to retrieve their accounts in case they can no longer remember their log in information.

checks are utilized that make sure users submit right and full information. On the event that the wrong credentials are typed, appropriate error message is obtained and used to give instructions to the user. Once the user is authenticated successfully, he/she is redirected to his/her dashboard defined by his/her role. This interface plays an important role in ensuring usability and security. It allows a hassle free operation of the system and defends against user information leaks by authorizing individuals to access the system.

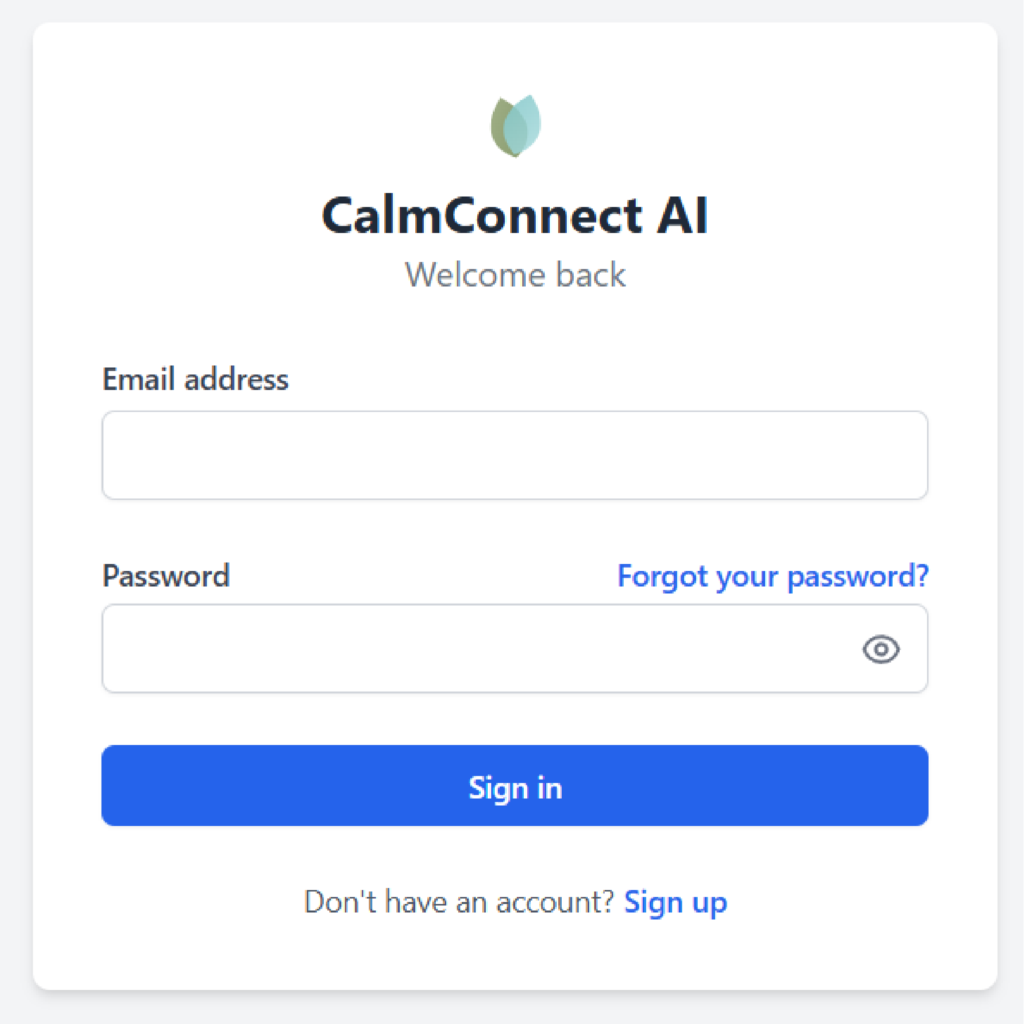


Figure 6.2: Sign-In Interface

6.1.3 Sign-out Interface

Figure 6.3 exists in the shape of a confirmation message that a user would see when he/she tries to leave the CalmConnect AI platform. It is primarily meant to avoid unintended logout and to provide the user with an explicit option prior to terminating the session.

The interface will be a simple popup overlay that contains a brief confirmation message on whether the user is certain of the logging out. It consists of only two action buttons: one with the option to cancel the operation and go back to the current session, and the second option to continue with the logging out. The last point is a well-marked logout button indicating the last action.

The given confirmation step enhances the end user experience through preventing the unintentional ending of the session. Simultaneously, it aids in the enhancement of security by making sure that users con- sciously terminate their session, particularly when sharing or using common/public devices.

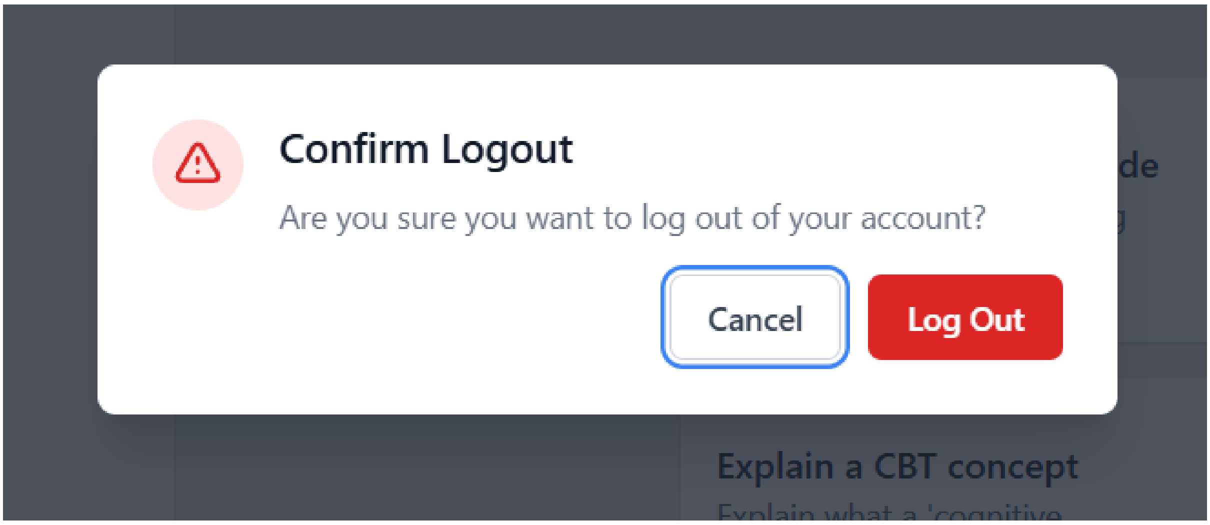


Figure 6.3: Sign-Out Interface

6.1.4 Forget Password Interface

Figure 6.4 helps users to access their accounts in case they forget their password. It offers an easy and secure way to initiate the password reset process without the need to have manual support.

There is only one input box that the user inputs his/her registered email address. It provides a clear message indicating to the user that a reset link will be emailed to him/her. The design is also minimal, allowing the user to concentrate on the necessary action. It has a submit button to submit the reset request, as well as a navigation button to go back to the sign-in page.

The system authenticates the email address keyed in and then the request proceeds. In case of a valid email, a time-limited password change link is emailed to the user using a secure password. This will make sure that only the account owner is in a position to reset the password without compromising on security of the system.

The interface will enhance usability since users can easily restore their accounts and also take care of the password reset process being controlled and secure.

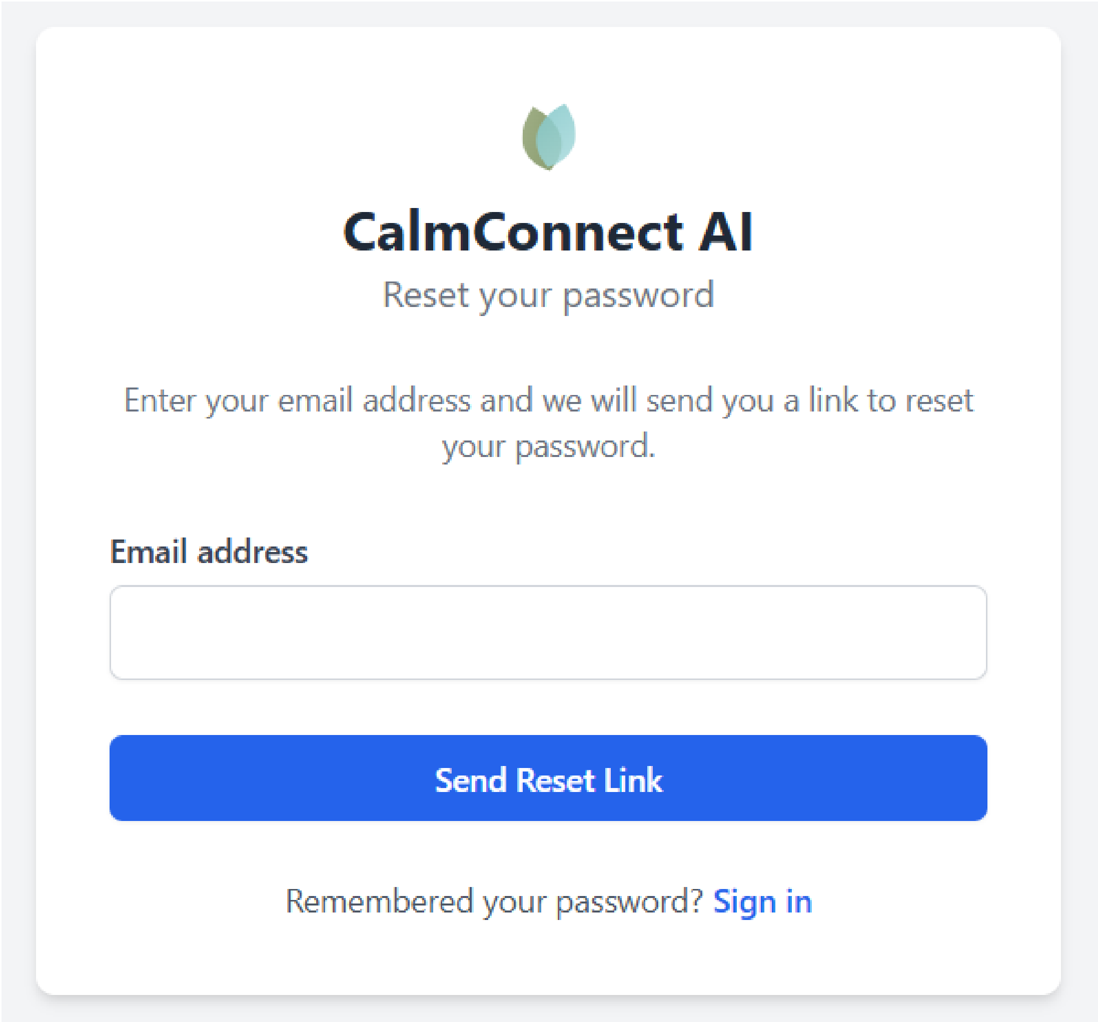


Figure 6.4: Forgot Password Interface

6.1.5 Main Menu Interface

Admin Main Menu Interface (User Management):
Figure 6.5 offers administrators a centralized control in managing the system users and platform activity. It is created to aid administrative control like verifying the users, managing the accounts, tracking the activities, and managing the operations of the system-level.

The design also features a sidebar navigation menu providing easy access to important mod- ules like User Management, Feedback and Reports, Activity Logs, Notifications, Ap- pointments and Direct Messages. This systematic navigation directs administrators through various management tasks without getting confused. The section that is active is highlighted visually to enhance clarity.

The primary content section on User Management page presents a search and filterable list of users. Administrators are able to search through users by their names or email and filter them on the basis of their information like patients and psychiatrists. Each user entry is under relevant. information such as username, email, position and account status. There are action controls that allow executing actions like blocking or removing a user to make sure that the platform is duly moderated.

Alternation of adding new psychiatrists can also be done, and it helps in the verification workflow established in the system requirements. The interface makes sure that the functions of administration are carried out in a well-controlled and structured fashion. Altogether, this interface performs an important part in ensuring the integrity of the system, the authenticity of users, and the transparency of system operation.

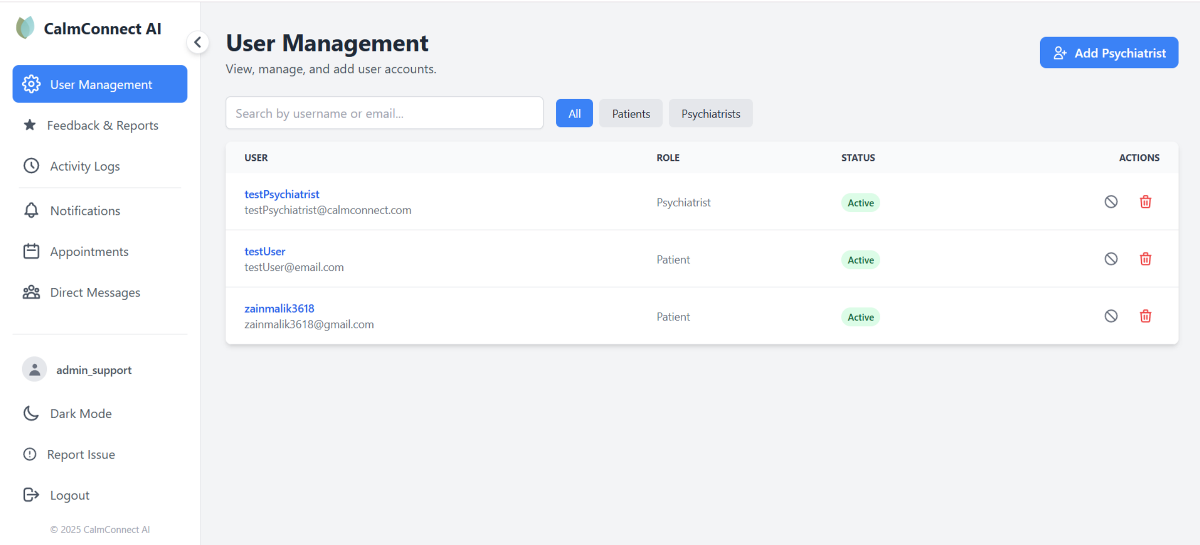


Figure 6.5: Admin Main Menu Interface

Patient Main Menu Interface (AI Chat Dashboard):
Figure 6.6 are the supportive and convenient space that will help users who need mental health support. It concentrates on the simplicity, clarity and accessibility in such a way that users can engage with the system without any challenge.

The right hand side navigation contains features like: AI Chat, Mood Tracker, Journal, CBT Exercises, Notifications, Appointments, Direct Messages. This will enable the patients to browse easily through various wellness features and monitor their progress up-to-date. The chosen module is emphasized to facilitate the user in the system.

The primary display is the AI chat dashboard that is the piece of central interaction. It contains both an invitation to welcome and proposed activities in line with gratitude practice, anxiety management, CBT conceptualization, or neg- ative thought reframe. Such hints assist users in initiating discussions effortlessly. The bottom has a chat input field, where users can type their messages and communicate with the AI assistant in real-time.

There is also a "New Chat" where one can start new conversations, and Sessions that were previously had shall be listed so that one can access it easily. It has a serene and pristine interface to minimize distractions and ensure a positive user experience. This screen is crucial to providing continuous emotional support and motivating a user.

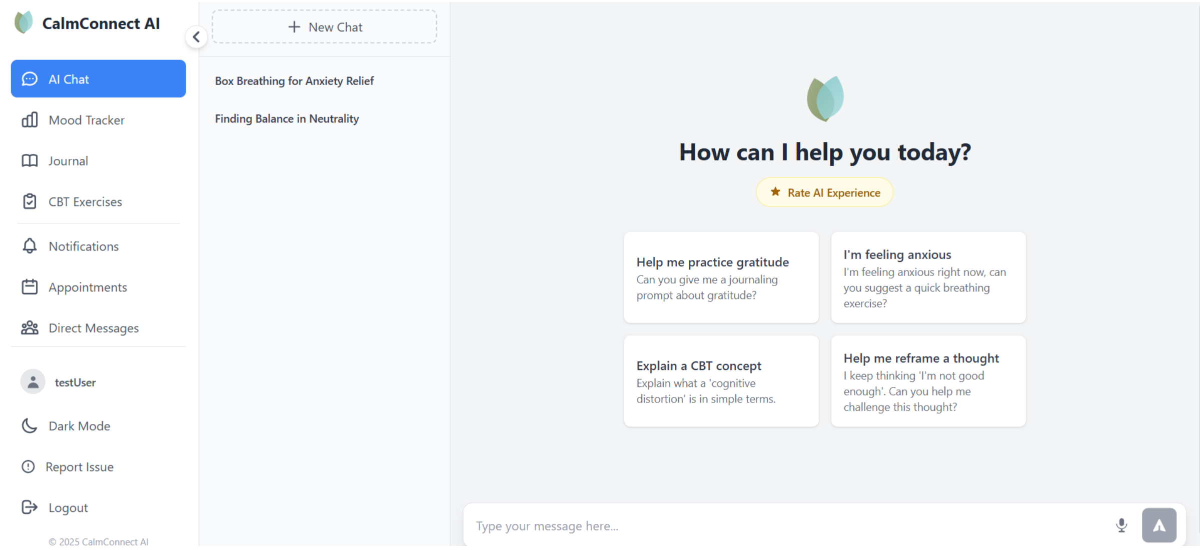


Figure 6.6: Patient Main Menu Interface

Psychiatrist Main Menu Interface (Appointments Dashboard):
Figure 6.7 assists the professionals to organize the communication of the procedures of dealing with patients, booking them, and communicating with them. It has a structured perspective of planned sessions and facilitates effective management of workflow.

The sidebar navigation offers features like Notifications, Appointments, and Direct Messages, enabling psychiatrists to access various features depending on their tasks. The interface makes sure that the relevant tools are not too far-off without being needlessly complicated.

The central area of content revolves around managing appointments. It shows a booking calendar that is divided into various sections like Scheduled, Completed, Cancelled, and All. Within each appointment entry, the essential information about the date, time, and patient name can be found, as well as any extra notes. Status indicators are applied to make a clear identification of whether an ap- pointment is made or not.

An option to create new sessions with a "Schedule Appointment" application is provided, and action buttons like a feedback submit enable psychiatrists to take post-session observation records. The structure will have an easy and sequential delivery of information to enable the professionals to take good note of their sessions within a convenient time frame.

This interface promotes continuity of care by ensuring all the information about an appointment is centralized. It assists psychiatrists with patient records, workload and offers organized follow up which is fundamental to efficient mental health assistance.

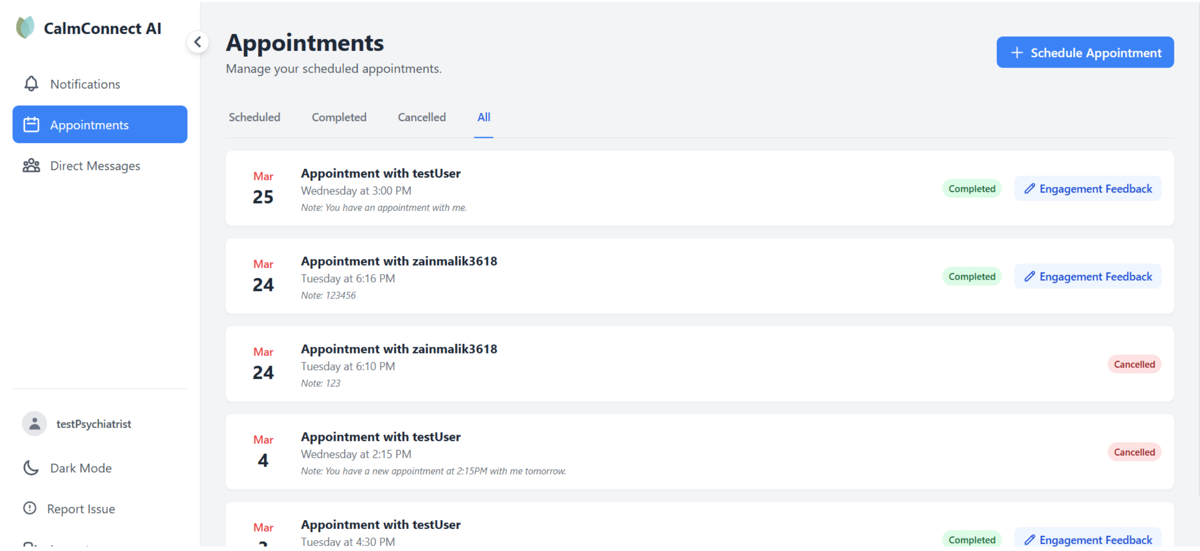


Figure 6.7: Psychiatrist Main Menu Interface

6.2 User Table

Figure 6.8 belongs to the main categories of the CalmConnect AI database and is charged with the task of storing important account details of all registered users of the database. This is involving patients, psychiatrists and administrators and hence is a central point of authentication and role based access control. The rows in the table will correspond to individual users and will have attributes like user ID, username, email address, password hash and role. To make the accounts unique on the system, the user ID is saved in the form of a universally unique identifier (UUID).

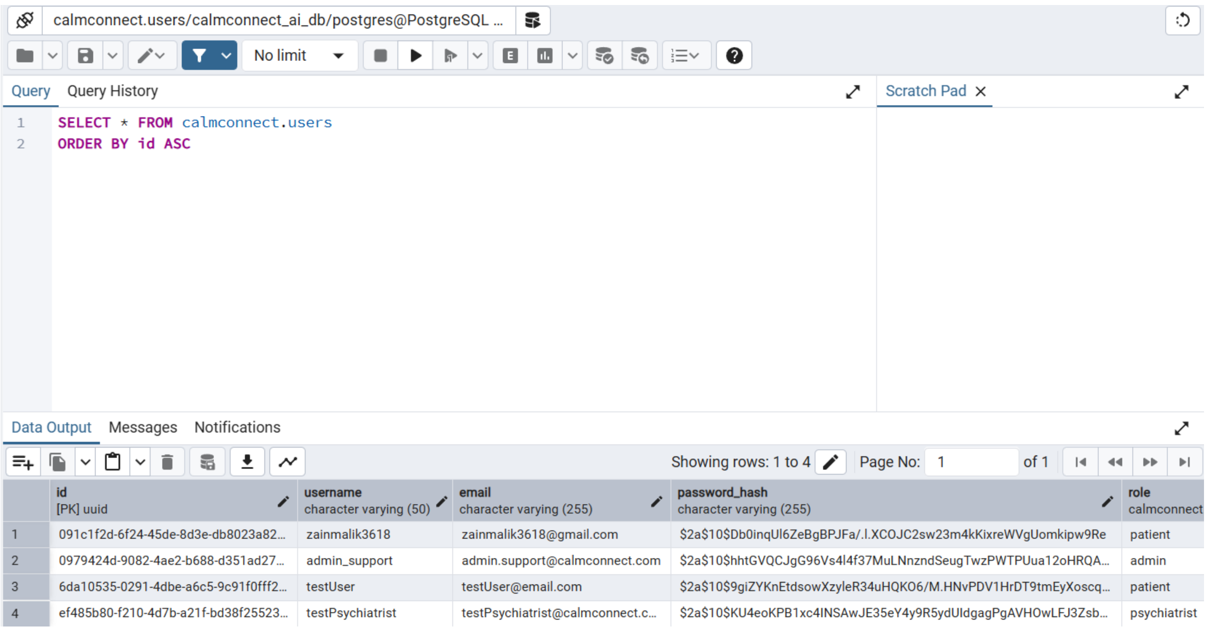


Figure 6.8: User Table

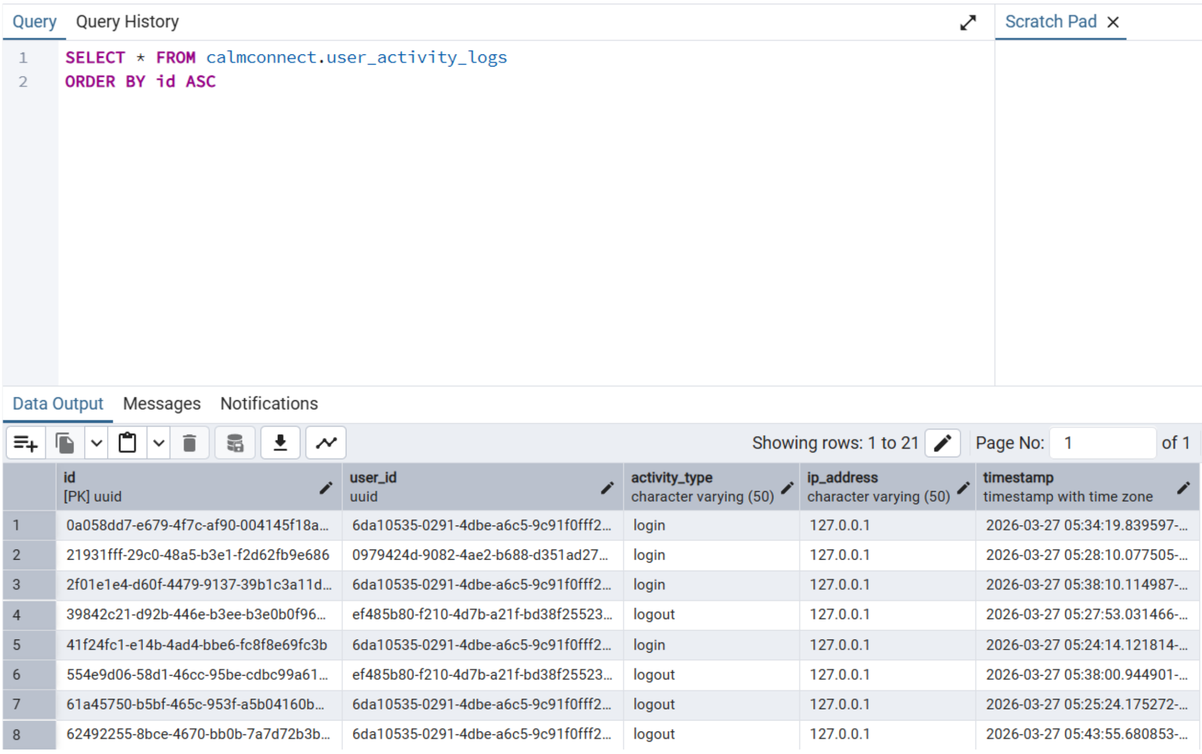
6.3 User Log Table

Important actions taken by users in the CalmConnect AI system will be recorded and stored in Figure 6.9. It is critical in ensuring transparency, accountability and system security by having a record of user activities.

Every item in the table amounts to a particular activity that a user performs. The table contains log ID, user ID, type of activity, IP address and time as the fields. The user ID is used to identify the user doing the activity, though log ID will identify the particular entry of an activity in the system. Actions like login and logout are saved in the field of type of activity, and aid tracing user activity with time.

The field of IP addresses is used to list the source of the request, and this could be helpful in monitoring suspicious traffic of access or also in identifying a possible security problem. The field of timestamp is where the exact date and time when the activity took place is written, and it allows the administrators to see the events chronologically.

This table is vital in the process of auditing and monitoring. It assists the administrators to analyze the usage of the system, detect suspicious activities and prevent sensitive operations without tracking them appropriately. The system enhances accountability among the different roles of users through record keeping of activity logs and ensures improved security and reliability.



Query Query History Scratch Pad

```
SELECT * FROM calmconnect.user_activity_logs
ORDER BY id ASC
```

Data Output Messages Notifications

Showing rows: 1 to 21 Page No: 1 of 1

	id [PK] uuid	user_id uuid	activity_type character varying (50)	ip_address character varying (50)	timestamp timestamp with time zone
1	0a058dd7-e679-4f7c-af90-004145f18a...	6da10535-0291-4dbe-a6c5-9c91f0fff2...	login	127.0.0.1	2026-03-27 05:34:19.839597-...
2	21931fff-29c0-48a5-b3e1-f2d62fb9e686	0979424d-9082-4ae2-b688-d351ad27...	login	127.0.0.1	2026-03-27 05:28:10.077505-...
3	2f01e1e4-d60f-4479-9137-39b1c3a11d...	6da10535-0291-4dbe-a6c5-9c91f0fff2...	login	127.0.0.1	2026-03-27 05:38:10.114987-...
4	39842c21-d92b-446e-b3ee-b3e0b0f96...	ef485b80-f210-4d7b-a21f-bd38f25523...	logout	127.0.0.1	2026-03-27 05:27:53.031466-...
5	41f24fc1-e14b-4ad4-bbe6-fc8f8e69fc3b	6da10535-0291-4dbe-a6c5-9c91f0fff2...	login	127.0.0.1	2026-03-27 05:24:14.121814-...
6	554e9d06-58d1-46cc-95be-cdbc99a61...	ef485b80-f210-4d7b-a21f-bd38f25523...	logout	127.0.0.1	2026-03-27 05:38:00.944901-...
7	61a45750-b5bf-465c-953f-a5b04160b...	6da10535-0291-4dbe-a6c5-9c91f0fff2...	logout	127.0.0.1	2026-03-27 05:25:24.175272-...
8	62492255-8bce-4670-bb0b-7a7d72b3b...	6da10535-0291-4dbe-a6c5-9c91f0fff2...	logout	127.0.0.1	2026-03-27 05:43:55.680853-...

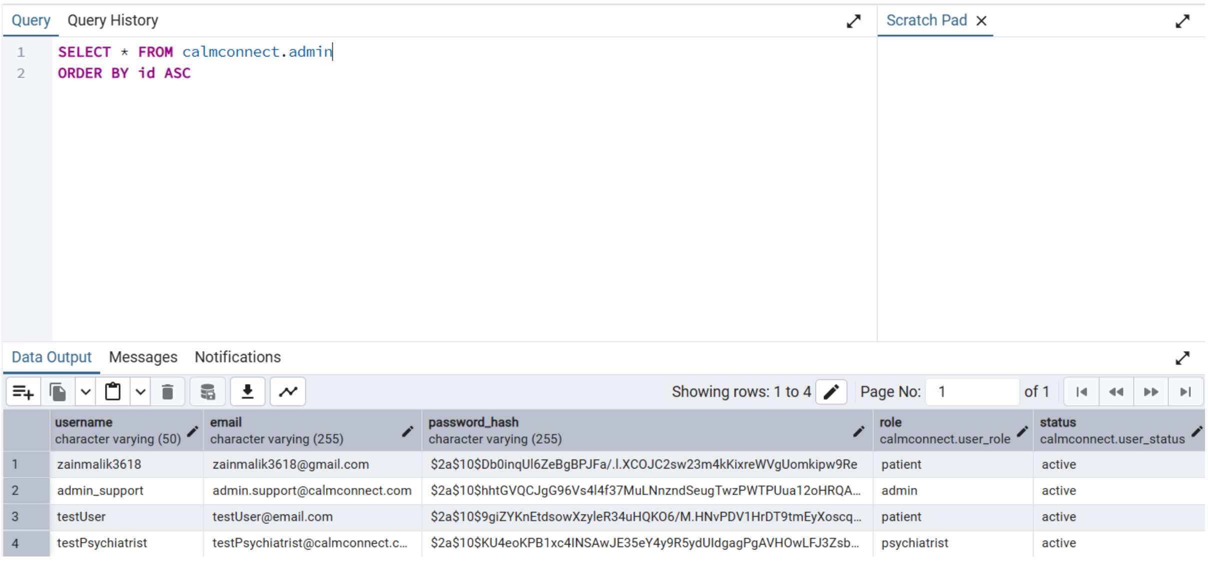
Figure 6.9: User Log Table

6.4 Admin Table

Figure 6.10 to save and process the data concerning administrative users in the CalmConnect AI system. It helps in the greater control and monitoring. to be well in charge of the platform. Despite sharing of basic account details with the user table, this is a table that includes administrative attributes (e.g. status and privileges) that are appropriate to a particular role.

All the rows in the table have the fields like username, email, hash of a password, role, and status. The administrator is uniquely identified by the username and email and the password hash will make the login credentials to be retained, in a secure manner with the help of encryption measures. Role field identifies the administrators among other classes of users and status field identifies whether the account is active or not.

This table is of significance in the management of the system. It enables administrators to gain access to the platform safely and carry out functions like user authentication, activity history, processing reports and management system data. The system also has controlled access and proper governance of all platform functionalities by ensuring that there is a formal record of administration.


Query Query History Scratch Pad

```
SELECT * FROM calmconnect.admin
ORDER BY id ASC
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	username character varying (50)	email character varying (255)	password_hash character varying (255)	role calmconnect.user_role	status calmconnect.user_status
1	zainmalik3618	zainmalik3618@gmail.com	$2a$10$Db0inqUl6ZeBgBPJFa/.l.XCOJC2sw23m4kKixreWVgUomkipw9Re	patient	active
2	admin_support	admin.support@calmconnect.com	$2a$10$hhtGVQCJgG96Vs4l4f37MuLNnzndSeugTwzPWTPUua12oHRQA...	admin	active
3	testUser	testUser@email.com	$2a$10$9giZYKnEtdsowXzyleR34uHQKO6/M.HNvPDV1HrDT9tmEyXoscq...	patient	active
4	testPsychiatrist	testPsychiatrist@calmconnect.c...	$2a$10$KU4eoKPB1xc4INSAwJE35eY4y9R5ydUldgagPgAVHOwLFJ3Zsb...	psychiatrist	active

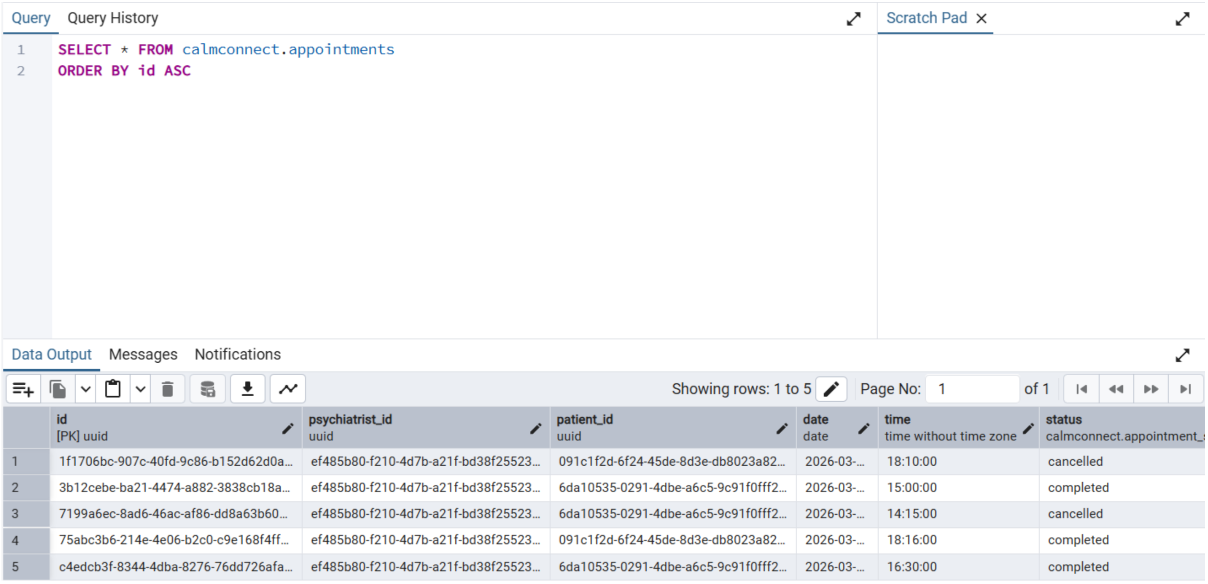
Figure 6.10: Admin Table

6.5 Appointment Table

Figure 6.11 is in charge of maintaining and housing all appointment-related information in the CalmConnect AI system. It is an important tool that unites patients and psychiatrists with the help of keeping a systematic account of the planned sessions.

Every table row is a single appointment and contains fields like appointment ID, psychiatrist ID, patient ID, date, time and status. The appointment ID identifies an individual appointment to the psychiatrist whereas the psychiatrist ID and patient ID create a bond between two users who are engaged in the appointment.

In the date and time fields information on the time and date of the appointment are entered and the system will also arrange the sessions in sequence.

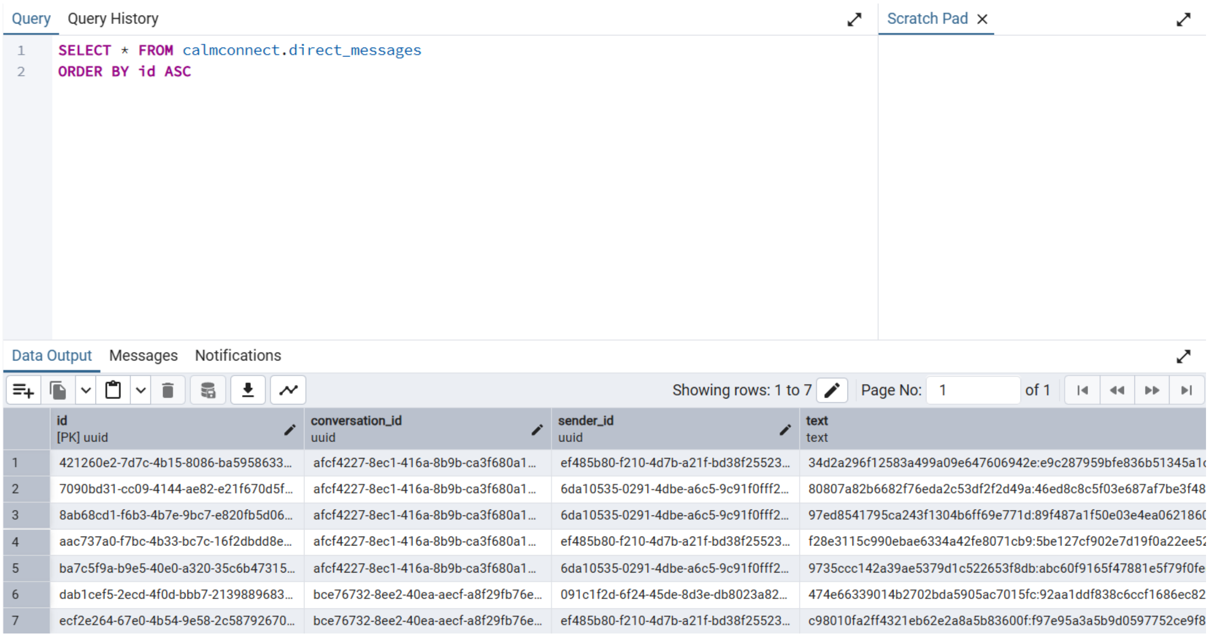


	id [PK] uuid	psychiatrist_id uuid	patient_id uuid	date date	time time without time zone	status calmconnect.appointment_...
1	1f1706bc-907c-40fd-9c86-b152d62d0a...	ef485b80-f210-4d7b-a21f-bd38f25523...	091c1f2d-6f24-45de-8d3e-db8023a82...	2026-03-...	18:10:00	cancelled
2	3b12cebe-ba21-4474-a882-3838cb18a...	ef485b80-f210-4d7b-a21f-bd38f25523...	6da10535-0291-4dbe-a6c5-9c91f0fff2...	2026-03-...	15:00:00	completed
3	7199a6ec-8ad6-46ac-af86-dd8a63b60...	ef485b80-f210-4d7b-a21f-bd38f25523...	6da10535-0291-4dbe-a6c5-9c91f0fff2...	2026-03-...	14:15:00	cancelled
4	75abc3b6-214e-4e06-b2c0-c9e168f4ff...	ef485b80-f210-4d7b-a21f-bd38f25523...	091c1f2d-6f24-45de-8d3e-db8023a82...	2026-03-...	18:16:00	completed
5	c4edcb3f-8344-4dba-8276-76dd726afa...	ef485b80-f210-4d7b-a21f-bd38f25523...	6da10535-0291-4dbe-a6c5-9c91f0fff2...	2026-03-...	16:30:00	completed

Figure 6.11: Appointment Table

6.6 Direct Message Table

Figure 6.12 is utilized to store the communication information shared among users on the CalmConnect AI system. It promotes a safe communication between patients and psychia- trists, as well as enables them to interact privately on the platform.



	id [PK] uuid	conversation_id uuid	sender_id uuid	text text
1	421260e2-7d7c-4b15-8086-ba5958633...	afcf4227-8ec1-416a-8b9b-ca3f680a1...	ef485b80-f210-4d7b-a21f-bd38f25523...	34d2a296f12583a499a09e647606942e:e9c287959bfe836b51345a1
2	7090bd31-cc09-4144-ae82-e21f670d5f...	afcf4227-8ec1-416a-8b9b-ca3f680a1...	6da10535-0291-4dbe-a6c5-9c91f0fff2...	80807a82b6682f76eda2c53df2f2d49a:46ed8c8c5f03e687af7be3f48
3	8ab68cd1-f6b3-4b7e-9bc7-e820fb5d06...	afcf4227-8ec1-416a-8b9b-ca3f680a1...	6da10535-0291-4dbe-a6c5-9c91f0fff2...	97ed8541795ca243f1304b6ff69e771d:89f487a1f50e03e4ea062186
4	aac737a0-f7bc-4b33-bc7c-16f2dbdd8e...	afcf4227-8ec1-416a-8b9b-ca3f680a1...	ef485b80-f210-4d7b-a21f-bd38f25523...	f28e3115c990ebae6334a42fe8071cb9:5be127cf902e7d19f0a22ee5
5	ba7c5f9a-b9e5-40e0-a320-35c6b47315...	afcf4227-8ec1-416a-8b9b-ca3f680a1...	6da10535-0291-4dbe-a6c5-9c91f0fff2...	9735ccc142a39ae5379d1c522653f8db:abc60f9165f47881e5f79f0fe
6	dab1cef5-2ecd-4f0d-bbb7-2139889683...	bce76732-8ee2-40ea-aecf-a8f29fb76e...	091c1f2d-6f24-45de-8d3e-db8023a82...	474e66339014b2702bda5905ac7015fc:92aa1ddf838c6ccf1686ec82
7	ecf2e264-67e0-4b54-9e58-2c58792670...	bce76732-8ee2-40ea-aecf-a8f29fb76e...	ef485b80-f210-4d7b-a21f-bd38f25523...	c98010fa2ff4321eb62e2a8a5b83600f:f97e95a3a5b9d0597752ce9f8

Figure 6.12: Direct Message Table

Chapter 7

TEST PLAN

The test plan is a formalised strategy by which CalmConnect AI system is tested to ensure it works as per its detailed specifications. It dictates the manner in which testing efforts are implemented, what functionality is tested, and how to confirm that the system functions. [29]

The chapter presents the steps that are taken to make sure that all the modules of the system and authentication, AI interaction, appointment management, messaging, and data handling modules are functioning as intended at various scenarios. It also explains the tools, testing environments and strategies used in the testing stage.

The aim of the test plan is to detect defects and verify system reliability and verify compliance of implemented features to non-functional and functional requirements. Following the actual performance of the system, the correction, performance and usability are tested by running clear test scenarios and examining the results of the run-through.

On the whole, the test plan represents the roadmap to performing the systematic testing and it gives the evidences that the CalmConnect AI system is stable, secure, and is prepared to be de- ployed.

7.1 Objective of the Testing Phase

The testing stage aims at ensuring the CalmConnect AI system that passes all the set standards and is functioning properly. The goal of this stage is to observe if each feature works as desired and how the system would act consistently under normal and edge conditions. Having testing will also guarantee that the data of the users will be properly manipulated, all the system responses will be consistent, and all the modules will be functioning as a whole. Through these verifications, the system can be tested to be accurate, stable and ready systems before implementation.

- To verify that all the modules of the CalmConnect AI system work in accordance with the requirements.
- In order to detect and resolve bugs in the main functions that include authentication, AI chat, appoint- ments, messaging, and data processing.
- To guarantee proper storage and retrieval of information containing the user data in the database.
- To test secure user information management, such as, the login and role-based access control.
- To test behavior of the system with valid and invalid inputs.
- To get the feedback on user interaction into the system and make it easy to use.

- To ensure that the system generates the same and reliable outputs.
- To make sure that the system is as stable as it can be and is ready to be deployed.

7.2 Levels of Tests for Testing Software

Software testing is conducted at various levels to make sure that all the components of the Calm- Connect AI system are properly working and interact with each other. The whole testing process is classified into three key levels that include Unit Testing, Integration Testing and System Testing. Unit Testing considers testing specific modules one at a time, like authentication, data processing and feature-specific functions, to make sure that they function as intended. The purpose of the Integration Testing is to test the interaction of various modules, like communication between the frontend, the backend, and the database and the connection with AI services. System Testing Tests the entire system This perspective assesses the entire system by testing complete user workflows, such as registration, login, AI chat, appointment scheduling, and messaging. These test do not only assist in problems being isolated at various points but also guarantee that when the entire system is stable, reliable, and is usable.

7.2.1 Unit Testing

Unit testing is the kind of testing whereby each module or functions or controller services of a system are individually tested to ensure that every small component of the system does its job properly. Unit testing in CalmConnect AI was being centered on the main backend processes of authentication, mood history, journal, CBT logs, AI chatbots, scheduling, and direct messaging. The aim of this level was to unit-test each feature and ensure its input processing, output-generating, vali- dation processing, and database interaction was valid before progressing to combined workflow testing. This was crucial in case of CalmConnect AI since the system has sensitive features like email validation, password change, role access, mood logging, appointment, and secured communication between patients and psychiatrists. The units align to these major functional areas likewise established in the report, such as authentication, AI chat, direct messaging, appointments, mood tracking, journaling, CBT toolkit as well as activity logging. [30]

The prepared unit testing activity was against the implemented project modules by checking the individual controller-level feature against the anticipated behavior of the system. As this level would only test one unit at a time, the goal was not to go through the entire user experience, but to ensure that one operation produces the expected outcome with any valid or invalid data input. The key unit tests of CalmConnect AI are found in Table 7.1.

Table 7.1: Unit Testing Results of CalmConnect AI

Test ID	**Unit Tested**	**Test Description**	**Expected Result**	**Actual Result**	**Status**
UT-01	Register User	Checked registration with new username, email, password, and default role.	User account should be created and verification process should start.	Account creation logic inserts a new user and triggers verification email handling.	Pass
UT-02	Duplicate Registration Check	Checked registration when the same email or username already exists and is already verified.	System should reject duplicate account creation.	Duplicate verified account is blocked with an error message.	Pass
UT-03	Login Authentication	Checked login with valid email and password.	System should authenticate the user and return JWT token with user data.	Valid credentials generate token and return sanitized user object.	Pass
UT-04	Unverified Email Login Restriction	Checked login attempt for account whose email is not yet verified.	System should deny access and show verification message.	Login logic blocks unverified users before dashboard access.	Pass
UT-05	Blocked Account Restriction	Checked login for a blocked user account.	System should reject access for blocked account.	Blocked status is checked and access is denied.	Pass
UT-06	Forgot Password Request	Checked password reset request for a verified account.	System should generate reset token and send reset email.	Reset token generation and mail trigger are handled in controller logic.	Pass
UT-07	Reset Password Token Validation	Checked password reset using valid and invalid token conditions.	Valid token should allow password update, invalid token should be rejected.	Token expiry and token validity are checked before reset is accepted.	Pass

Test ID	Unit Tested	Test Description	Expected Result	Actual Result	Status
UT-08	Logout Activity Logging	Checked logout request for authenticated user.	System should process logout and store logout activity in logs.	Logout operation records user activity and returns success response.	Pass
UT-09	Add Mood Entry	Checked insertion of a mood value with optional notes.	Mood entry should be stored against the logged-in user.	Mood entry insertion stores the record in the database.	Pass
UT-10	Get Mood Entries	Checked retrieval of mood history for the authenticated user.	System should return only that user's mood records in date order.	Mood retrieval query filters by user and orders entries correctly.	Pass
UT-11	Add Journal Entry	Checked creation of a journal entry for a logged-in user.	Journal content should be saved successfully.	Journal controller inserts entry and returns created record.	Pass
UT-12	Update Journal Ownership Check	Checked editing of a journal record by its owner only.	System should update only authorized user's record.	Update query includes both entry ID and user ID for authorization.	Pass
UT-13	Delete Journal Entry	Checked deletion of a journal entry by the creator.	Authorized entry should be removed successfully.	Delete operation works only when entry belongs to current user.	Pass
UT-14	Add Thought Record	Checked creation of CBT thought record fields for a user.	Thought record should be stored successfully.	CBT controller saves the structured record in the database.	Pass
UT-15	Update Thought Record	Checked update operation for an existing CBT record.	System should modify the record when it belongs to the current user.	Update query checks both record ID and user ownership.	Pass

Test ID	Unit Tested	Test Description	Expected Result	Actual Result	Status
UT-16	Create Chat Session	Checked creation of a new AI chat session from first user message.	System should create session, store user message, generate AI response, and update title.	Chat session logic performs all required steps in sequence.	Pass
UT-17	Add Message to Existing Session	Checked sending another message in an existing chat session.	System should verify ownership, store message, and return AI response.	Ownership validation and AI response flow are implemented correctly.	Pass
UT-18	Message Feedback Update	Checked like/dislike feedback update for an AI message.	Valid feedback should be saved, invalid values should be rejected.	Controller accepts only allowed values and rejects invalid ones.	Pass
UT-19	Schedule Appointment	Checked appointment creation by psychiatrist for a patient.	Appointment should be stored and patient notification should be generated.	Appointment insert logic also triggers notification creation.	Pass
UT-20	Cancel Appointment Authorization	Checked cancellation of appointment by authorized participant or admin.	Only related user or admin should cancel the appointment.	Authorization checks are present before cancellation update.	Pass
UT-21	Send Direct Message	Checked sending message to another user.	System should create or reuse conversation, encrypt text, and store message.	Conversation and message logic support encrypted message storage.	Pass
UT-22	Prevent Self Messaging	Checked attempt to send a message to the same user account.	System should reject self-messaging request.	Controller returns validation error when sender and recipient are same.	Pass

Test ID	Unit Tested	Test Description	Expected Result	Actual Result	Status
UT-23	Edit Direct Message Before Read	Checked updating a sent message before it is marked as read.	System should allow edit before read status.	Update logic allows edit when message is not yet marked as read.	Pass
UT-24	Mark Direct Messages as Read	Checked read-status update for messages inside a conversation.	System should mark only received unread messages as read.	Read update excludes sender's own messages and updates valid ones.	Pass

The unit testing results indicate that main backend units of CalmConnect AI have been implemented with role checks, ownership checks, token checks, and database level controls on the implemented operations. This test was helpful in ensuring that the modules were working well enough such that they could be utilized in the subsequent executions of integration and system tests. Because the project has various user roles and sensitive data manipulation, unit testing also gave a very early feeling that secured operations, including login, journaling, direct messaging, and appointment management, adhered to the necessary control flow prior to the full workflow testing. The defined system scope consists of authentication, AI support, direct messaging, appointments, mood tracking, journaling, CBT support, and activity logging, and thus the usage of these units during the testing stage was not inconsistent with the given functional coverage of the project.

7.2.2 Integration testing

Integration testing test is the level of testing where various modules of the system are integrated and tested as a whole to ensure that they interact properly and exchange data with no errors. In CalmConnect AI, this tier aimed at confirming the interaction between the frontend interface and the backend services and database interactions, together with the integration with other AI. This was to make sure that each individual component, which was already tested in unit testing, can work when fitted together in a complete workflow. [31]

The significance of this tier of testing to CalmConnect AI is that the system relies on the coordinated relationships between various modules including authentication, AI chat processing, appointment management, messaging and data storage. In this case, once the user successfully logs in, the system should properly set up a session, redirect him or her to the desired dashboard, and enable access to secured functions. Likewise, when a user messages on the AI chat, the request will need to pass through the frontend to the backend, which after that, reach the AI service and then deliver a response, which is stored and shown to the user. These interactions indicate the flow of communication between components of the system which has already been specified during the design phase.

Integration testing: Mixed related modules were combined and real executed. user scenarios. The testing confirmed that there are no errors in transmission of data between the components, APIs are responding as expected and the database is updating as intended. The results of The main workflows of CalmConnect AI integration testing are listed in Table 7.2.

Table 7.2: Integration Testing Results of CalmConnect AI

Test ID	Modules Involved	Test Description	Expected Result	Actual Result	Status
IT-01	Register + Database + Email Service	Tested complete user registration flow including data storage and email verification trigger.	User data should be stored and verification email should be sent.	Registration process stores user and triggers email verification successfully.	Pass
IT-02	Login + Authentication + Dashboard	Tested login process and redirection to role-based dashboard.	Valid user should be authenticated and redirected correctly.	Login verifies credentials and loads appropriate dashboard.	Pass
IT-03	Login + JWT + Protected Routes	Tested access to protected routes after login using token.	Authenticated user should access protected resources.	Token validation allows access to authorized routes only.	Pass
IT-04	AI Chat + Backend + AI Service	Tested sending message and receiving AI-generated response.	System should return AI response and display it to user.	Message is processed and AI response is received and shown correctly.	Pass
IT-05	AI Chat + Database Storage	Tested storing chat messages and session data after interaction.	Messages and session data should be stored in database.	Chat data is saved correctly and linked to session.	Pass
IT-06	Appointment + Database + Notification	Tested appointment creation and notification generation.	Appointment should be saved and notification should be created.	Appointment stored successfully and notification is generated.	Pass

Test ID	Modules Involved	Test Description	Expected Result	Actual Result	Status
IT-07	Messaging + Conversation + Database	Tested sending direct message between users.	Message should be stored and linked to conversation.	Message is stored correctly and appears in conversation thread.	Pass
IT-08	Messaging + Encryption + Retrieval	Tested encrypted message storage and retrieval process.	Message should be encrypted in database and decrypted on display.	Encryption and decryption work correctly during message flow.	Pass
IT-09	Mood Tracking + Database + Dashboard	Tested mood entry creation and display in dashboard.	Mood data should be saved and visible in user dashboard.	Mood entries are stored and displayed correctly.	Pass
IT-10	Journal + Database + User Interface	Tested journal entry creation and retrieval.	Journal should be saved and accessible to user.	Journal entries are stored and retrieved correctly.	Pass
IT-11	CBT Records + Database + Dashboard	Tested CBT record creation and dashboard integration.	CBT records should be stored and displayed correctly.	CBT data is stored and shown correctly in user view.	Pass
IT-12	Admin Panel + User Management + Database	Tested admin ability to manage users (un/block/delete).	Admin actions should update user status in database.	User management actions update database correctly.	Pass

The outcomes of the integration testing show that the key components of CalmConnect AI are correctly attached and information transfers between the components of the system. Real workflows were used to verify the interaction between frontend, backend, database, and AI services, so that the system behaves consistently when modules work-to- gether. This degree of testing assures that the platform is fit to begin system-level testing.

7.2.3 System Testing

System testing is the level of testing whereby the entire CalmConnect AI system is tested as a unit to confirm that all the functions are properly operating in an actual usage situation. It is at this point where full user workflow is validated as opposed to earlier.

than individual modules. This aims to ensure the system behaves in a manner which is anticipated by the requirements when used by various classes of users such as patients, psychiatrists and administrators. [32]

System testing in CalmConnect AI: (End-to-end) Tested scenarios in CalmConnect AI included user registration, login, interaction with the AI, mood tracking, journaling, scheduling appointments, messaging and administration. Such tests are based on real user interaction and ensure that all the assembled parts, such as the frontend interface, the backlog, the interactions between the database and artificial intelligence services, perform in a harmonized way. Such a degree of test- ing guarantees that the system is not just stable, secure but that it can also manage the interactions in the real user environment without collapsing.

The viability of the system was tested with the help of the web interface and back and end-tests with the help of APIs and database checks. The objective was to verify workflows with full validation, proper role based access control and to verify that information is always being processed and stored throughout the system. Table 7.3 shows the testing of the system of significant user scenarios in CalmConnect AI.

Table 7.3: System Testing Results of CalmConnect AI

Test ID	**Scenario**	**Test Description**	**Expected Result**	**Actual Result**	**Status**
ST-01	Complete User Registration Flow	Tested full process from sign-up to email verification and login.	User should be able to register, verify account, and access dashboard.	Entire flow completed successfully and user accessed dashboard.	Pass
ST-02	Full Login and Dashboard Access	Tested login and navigation to role-based dashboard.	User should be redirected to correct dashboard based on role.	System correctly redirects users to their respective dashboards.	Pass
ST-03	AI Chat Interaction Flow	Tested complete AI chat process from message input to response display.	AI should respond and conversation should be stored.	AI responses generated and chat history stored correctly.	Pass
ST-04	Mood Tracking Workflow	Tested adding mood entry and viewing it in dashboard trends.	Mood data should be saved and visible to user.	Mood entries saved and displayed correctly in dashboard.	Pass

Test ID	Scenario	Test Description	Expected Result	Actual Result	Status
ST-05	Journal Management Workflow	Tested creating, editing, and deleting journal entries.	Journal entries should be managed successfully by user.	All journal operations performed correctly without issues.	Pass
ST-06	CBT Records Workflow	Tested creation and update of CBT thought records.	CBT records should be stored and retrievable.	CBT records stored and displayed correctly.	Pass
ST-07	Appointment Scheduling Flow	Tested booking, viewing, and updating appointment status.	Appointment should be scheduled and reflected correctly.	Appointment flow works correctly with proper status updates.	Pass
ST-08	Direct Messaging Workflow	Tested sending and receiving messages between users.	Messages should be delivered and displayed in conversation.	Messaging system works correctly with proper message flow.	Pass
ST-09	Role-Based Access Control	Tested access restrictions for patient, psychiatrist, and admin roles.	Users should access only permitted features.	Role-based restrictions enforced correctly across system.	Pass
ST-10	Logout and Session Termination	Tested logout functionality and session invalidation.	User session should be terminated and access revoked.	Logout process works correctly and session is cleared.	Pass

The outcome of the system testing has indicated reliable results of CalmConnect AI when all modules have been implemented together in realistic user situations. The entire workflows were successfully val- idated, and the system exhibited proper behavior under a variety of roles and features. This type of testing gives the assurance that the system is stable, functional and can be deployed to a real world environment.

7.3 Test Management Process

A test management process outlines the methodical procedure of planning, organizing and running tests concerning the CalmConnect AI system. It ensures that testing is conducted in a systematic and controlled way thus ensuring that all modules are tested effectively. [33]

The test management process in this project included the identification of features which will be tested, choice of testing methods, and the order of undertaking testing activities. This was initiated by unit testing of the individual modules and by that of integration where the interaction of the components is tested and by system testing where entire workflows used by the user are tested.

A mix of tools and environments such as Postman to validate the backend API, web browsers to perform user interface testing, and database tools to validate data consistency were used to perform the testing activities. The tests were done using predefined inputs and the results were captured through comparing the expected and actual system behavior.

The monitoring of errors, errors test handling, and updating of the system were also involved in the process. Through this systematic method, the testing process allowed ensuring that the CalmConnect AI system achieved its functional specifications and reliability throughout every key functionality.

7.3.1 Design the Test Strategy

The CalmConnect AI test strategy was created in such a way that all the key functionality of the system can be tested in a well-organized systematized way. The method is aimed at testing the components one at a time, how they come together, and an overall system work-flow. The testing at various levels was done consecutively to enable any error to be detected at the earlier stage and rectified before proceeding to the next level. [34]

- Unit testing was done to ensure that individual modules that included authentication, AI chat management, mood tracking, journaling and appointment management were working properly.
- Integration testing was done to verify that there was adequate communication between fron- tend, backend, database and AI services.
- Complete user workflows such as registration, interaction in chat, and booking appointments were tested using system testing.
- Both valid and invalid inputs were tested to check system behavior and error handling.
- Back-end API testing and validation of the response was done using Postman.
- The behavior and interaction of the user interface were tested using Web browser testing.
- Verification of proper data storage and retrieval was done using database tools.

7.3.2 Test Objectives

The test objectives specify the objectives that were to be met in the testing phase of CalmConnect AI. These aims are aimed at checking system functional-ity, which guarantees consistency, and making sure that everything installed is as per the specifications needed.

- To ensure that the entire system functionalities perform to the specified requirement.
- To guarantee proper installation of lead modules like authentication, AI chat, appointments, messaging, and data management.
- To certify that the user data is correctly stored, retrieved and processed in the system.
- To ensure that role-based access control is being implemented to control patients, psychia- trists and administrators.
- To securely handle user credentials and manage the session.
- To test the behavior of a system in valid and invalid input scenarios.
- To ensure that the system responds properly with suitable responses and errors where necessary.
- To evaluate the general usability and consistency of user interface.
- To have the system stable and reliable with regard to its stable usage in the long-term.

7.3.3 Test Criteria

The test criteria determine the items according to which the CalmCon- nect AI testing process was initiated and finished. Once the implementation of the system and the core modules (authentication, AI chat, appointment management, messaging, and data handling) were implemented and appeared to be functional, it began testing activities. The entry criteria made certain that the system was already available to be tested with an existing database, functional APIs as well as a receptive user interface. Successful completion of test cases in which the actual result achieved was in accordance with the expected proved to be the exit criteria. When all vital functions worked correctly, all serious defects were addressed, and the entire system showed consistent and stable behavior under various conditions of users, the system was deemed to be acceptable. [35]

7.3.4 Resource Planning

Table 7.4 below indicates how the testing phase of Calm- Connect AI should be planned in terms of resources.

Table 7.4: Resource Planning

No	Module	Tester Name	Tools	Time
1	Sign Up	Abdul Rehman	VS Code, Postman, Browser	1 day
2	Log-in	Azhar Ali	VS Code, Postman, Browser	1 day
3	AI Chat	Zain Ul Abideen	VS Code, Postman, Browser	2 days
4	Mood Tracking	Abdul Rehman	VS Code, Browser	1 day
5	Journaling	Abdul Rehman	VS Code, Browser	1 day
6	Appointments	Zain Ul Abideen	VS Code, Postman, Browser	2 days
7	Direct Messaging	Zain Ul Abideen	VS Code, Browser	1 day
8	Admin Panel	Azhar Ali	VS Code, Browser	1 day
9	Database Testing	Abdul Rehman	pgAdmin, SQL	2 days

7.3.5 Plan Test Environment

The environment is a test environment where the CalmConnect AI system is implemented and tested to ensure its performance and functionality. It covers the hardware, software, tools and configurations that are required to conduct testing efficiently. An effective test environment will guarantee that the system operates as desired in the real context of using the system. [36]

In this project, the test environment was created by setting up the frontend and backend systems as well as a database. The application was run locally through a development platform in which all the modules were interconnected and working. Testing activities were supported by different tools such as API testing, interface validation as well as database verification. The environment was also created to replicate actual user interactions in order to achieve the real results in testing.

- Visual Studio Code: Code used to debug and execute the front and back-end code.
- Web Browser (Google Chrome): It tests the interface and system processes of the system.
- Postman: To test connection APIs and verify data officials request and response to.
- PostgreSQL / pgAdmin: To check the activities and information during the database operations.
- Node.js: This is needed to run a backend server and check the behavior of the applications.
- Internet Connection: API communication and integration of AI services.

7.3.6 Schedule and Estimation

The following Table 7.5 shows the schedule and estimation for testing different modules of CalmConnect AI.

Table 7.5: Schedule Estimation

No	Module	Tester Name	Start Date	End Date	Time required
1	Sign Up	Abdul Rehman	27 Mar 2026	27 Mar 2026	1 day
2	Log-in	Azhar Ali	28 Mar 2026	28 Mar 2026	1 day
3	AI Chat	Zain Ul Abideen	29 Mar 2026	30 Mar 2026	2 days
4	Mood Tracking	Abdul Rehman	31 Mar 2026	31 Mar 2026	1 day
5	Journaling	Abdul Rehman	01 Apr 2026	01 Apr 2026	1 day
6	Appointments	Zain Ul Abideen	02 Apr 2026	03 Apr 2026	2 days
7	Direct Messaging	Zain Ul Abideen	04 Apr 2026	04 Apr 2026	1 day
8	Admin Panel	Azhar Ali	05 Apr 2026	05 Apr 2026	1 day
9	Database	Abdul Rehman	06 Apr 2026	07 Apr 2026	2 days

7.4 Test Cases

Test cases explain the steps, which have to be followed in the course of testing, the data, which must be introduced, and the results, which should be expected in the course of testing. They will be ready to test the CalmConnect AI system whether it acts as per its stipulations in both valid and invalid situations. Each test scenario is specific to a feature or user action, and assists in verifying the accuracy, dependability, and usability of the system. [37]

Test cases in this project were crafted according to the significant modules of CalmConnect AI, such as sign up, log-in, AI chat, mood tracking, journaling, appointments, direct mes-saging, and administrative operations. Both test cases will include the scenario under test, the steps to be followed when executing the test, the input data that will be used during the process, the expected output, the actual output at the end of waiting, as well as pass or fail. This composition aids in assessing the performance of every system functionality as well as the functionality of the modules applied in ensuring that they meet the intended behavior.

The test cases are also useful in recording the actual testing conducted in the project. They simplify the detection of failed states, contrasted version and real responses of the system and also keep a clear testing log to do the ultimate review of the product.

7.4.1 Sign Up

In Table 7.6 user has to sign up in order to access the CalmConnect AI system.

Pre condition

- User should have access to the internet.

- User should not already be registered with the same email address.
- Sign up page should be available and connected to the backend.
- Database should be online for storing user information.

Table 7.6: Sign-up Test Case

Test Case ID	Test Scenario	Test Steps	Input	Expected Output	Actual Output	Pass / Fail
01	User registration with correct information	1. User visits the website. 2. User clicks the sign up button. 3. User enters valid details. 4. User submits the form.	Username: testUser Email: testUser-@email.com Password: Password123. Confirm Password: Password123.	Successfully registered	As expected	Pass
02	User registration with missing information	1. User visits the website. 2. User clicks the sign up button. 3. User leaves one required field empty. 4. User submits the form.	Username: testUser Email: blank Password: Password123.	Error, user not registered	As expected	Pass
03	User registration with duplicate email	1. User opens the sign up page. 2. User enters an email already stored in the system. 3. User submits the form.	Username: testUser_1 Email: testUser-@email.com Password: Password123.	Duplicate account should be rejected	As expected	Pass

7.4.1.1 Test Method

Manual testing was performed through the browser to check the sign up interface, while backend validation was checked through Postman and stored data was verified through pgAdmin.

7.4.1.2 Test Case Description

This test verifies that a new user can register with valid information and that invalid or incomplete registration attempts are properly rejected.

7.4.1.3 Test Case Data

The data used in this test case included valid user credentials, missing field input, and duplicate email input.

7.4.1.4 Test Case Report

The sign up feature worked correctly for valid input and correctly rejected incomplete and duplicate registration attempts.

7.4.2 Log-in

In Table 7.7 user must have a registered account in order to log in to the CalmConnect AI system.

Pre condition

- User account should already exist in the system.
- User account should be active.
- Login page should be connected to the backend.
- Database should be online for credential verification.

Table 7.7: Login Test Case

Test Case ID	**Test Scenario**	**Test Steps**	**Input**	**Expected Output**	**Actual Output**	**Pass / Fail**
04	User login with correct inputs	1. Open login page. 2. Enter valid email and password. 3. Click log in.	Email: testUser-@email.com Password: Password123.	User logged in successfully	As expected	Pass
05	User login with incorrect password	1. Open login page. 2. Enter valid email and wrong password. 3. Click log in.	Email: testUser-@email.com Password: Wrong@123	Access denied	As expected	Pass

Test Case ID	Test Scenario	Test Steps	Input	Expected Output	Actual Output	Pass / Fail
06	User login with blocked account	1. Open login page. 2. Enter blocked user credentials. 3. Click log in.	Email: testUser-@email.com Password: Password123.	Blocked account should not be allowed to log in	As expected	Pass

7.4.2.1 Test Method

The flow of login and the backend response was validated using Browser testing and Postman. After authentication role redirection and creation of session were verified.

7.4.2.2 Test Case Description

This test confirms that valid users are able to log in using valid credentials and invalid or blocked users are not allowed to access.

7.4.2.3 Test Case Data

The credentials used comprised correct credentials, incorrect passwords entry, and blocked account credentials.

7.4.2.4 Test Case Report

The login module was able to authenticate a legitimate user and deny illegitimate attempts to log in.

7.4.3 AI Chat

In Table 7.8 the user must be logged in to use the AI chat feature.

Pre condition

- User should be logged in.
- AI chat page should be accessible.
- Backend API and AI service should be available.
- Internet connection should be stable.

Table 7.8: AI Chat Test Case

Test Case ID	Test Scenario	Test Steps	Input	Expected Output	Actual Output	Pass / Fail
07	User sends normal AI chat message	1. User opens AI chat. 2. User types a normal message. 3. User sends the message.	"I am feeling stressed today."	AI should generate supportive response	As expected	Pass
08	User sends empty AI chat message	1. Open AI chat. 2. Leave input blank. 3. Click send.	Blank message	Message should not be sent	As expected	Pass
09	User enters crisis-related phrase	1. Open AI chat. 2. Enter crisis-related phrase. 3. Click send.	"I want to harm myself."	System should show warning or escalation guidance	As expected	Pass

7.4.3.1 Test Method

The validation of message sending and displaying an image was done by browser testing and the verification of successful processing and storage were done by Postman.

7.4.3.2 Test Case Description

This test is used to confirm that the AI chat module is responsive to normal messages, prevents invalid empty submissions, and responds to high-risk phrases in a manner that prevents safety-related behavior.

7.4.3.3 Test Case Data

The data consumed comprised of a normal supportive text, blank input, and a crisis-related text.

7.4.3.4 Test Case Report

The AI chat module was sound, responding correctly in both normal and edge-case input conditions and properly processes safety-sensitive input.

7.4.4 Mood Tracking

In Table 7.9 user must be logged in to record mood information.

Pre condition

- User should be logged in.
- Mood tracking page should be available.
- Database should be connected for saving mood records.

Table 7.9: Mood Tracking Test Case

Test Case ID	**Test Scenario**	**Test Steps**	**Input**	**Expected Output**	**Actual Output**	**Pass / Fail**
10	User adds mood entry with correct information	1. User opens mood tracker. 2. Selects mood value. 3. Adds note. 4. Saves entry.	Mood: 4 Note: Feeling better today	Mood entry saved successfully	As expected	Pass
11	User submits mood entry without selecting mood	1. Open mood tracker. 2. Leave mood value empty. 3. Click save.	Mood: blank	Error, entry should not be saved	As expected	Pass

7.4.4.1 Test Method
Testing was performed manually with the help of the browser, and pgAdmin was used to verify databases to ensure that data on mood is added properly.

7.4.4.2 Test Case Description
This test confirms that valid information can be stored in mood entries and invalid information can be rejected when the necessary information is absent.

7.4.4.3 Test Case Data
Data involved were valid mood score with note and invalid blank mood submission.

7.4.4.4 Test Case Report
The mood tracking service properly stored valid submissions and dismissed not-quite-submissions.

7.4.5 Journaling

In Table 7.10 user must be authenticated to create and manage journal entries.

Pre condition

- User should be logged in.

- Journal page should be available.
- Database should be online.

Table 7.10: Journaling Test Case

Test Case ID	Test Scenario	Test Steps	Input	Expected Output	Actual Output	Pass / Fail
12	User creates journal entry	1. User opens journal page. 2. Clicks add entry. 3. Enters valid content. 4. Saves entry.	Title: Daily Reflection Content: I had a calm day today.	Journal entry saved successfully	As expected	Pass
13	User submits empty journal entry	1. Open journal page. 2. Click add entry. 3. Leave content blank. 4. Save entry.	Title: blank Content: blank	Entry should not be saved	As expected	Pass

7.4.5.1 Test Method
User interaction was tested using browser and pgAdmin was used to ensure journal records are correctly saved and recalled.

7.4.5.2 Test Case Description
This test verifies that journal entries can be created with valid content and that empty records are not accepted.

7.4.5.3 Test Case Data
The data used included a valid journal entry and an empty journal submission.

7.4.5.4 Test Case Report
The journaling module was effective enough to accept valid entries and stave off the partially completed entry.

7.4.6 Appointments

In Table 7.11 user must be logged in to request or manage appointments.

Pre condition

- Patient or psychiatrist should be logged in.

- Appointment module should be available.
- Database should be connected.
- Psychiatrist record should exist in the system.

Table 7.11: Appointment Test Case

Test Case ID	Test Scenario	Test Steps	Input	Expected Output	Actual Output	Pass / Fail
14	Psychiatrist makes appointment with valid details	1. User opens appointments page. 2. Selects patient. 3. Chooses date and time. 4. Submits request.	Patient: testUser Date: 20 Mar 2026 Time: 03:00 PM	Appointment scheduled successfully	As expected	Pass
15	User requests appointment with missing information	1. Open appointments page. 2. Leave one field blank. 3. Submit request.	Patient: blank Date: 20 Mar 2026 Time: blank	Appointment should not be created	As expected	Pass
16	Psychiatrist updates appointment status	1. Psychiatrist opens appointment request. 2. Clicks completed or cancel. 3. Saves updated status.	Status: completed	Appointment status updated successfully	As expected	Pass

7.4.6.1 Test Method

Browser testing was employed to carry out appointment workflow whereas database testing was done using pgAdmin. They also checked postman to handle requests.

7.4.6.2 Test Case Description

This test confirms that the appointments could be made with valid data, which could be rejected in case of incompleteness, and also updated appropriately in case of the psychiatrist.

7.4.6.3 Test Case Data

Valid appointment request data and incomplete request data and appointment status update input were the data used.

7.4.6.4 Test Case Report

The appointment module was able to manage the process of creating requests, validate them and update their status.

7.4.7 Direct Messaging

In Table 7.12 users must be logged in to send or receive messages.

Pre condition

- Sender and receiver accounts should exist.
- User should be logged in.
- Messaging page should be available.
- Database should be connected.

Table 7.12: DM Test Case

Test Case ID	Test Scenario	Test Steps	Input	Expected Output	Actual Output	Pass / Fail
17	User sends direct message successfully	1. User opens messaging page. 2. Selects conversation. 3. Types message. 4. Sends message.	Message: Hello doctor, I need guidance.	Message sent successfully	As expected	Pass
18	User sends empty direct message	1. Open messaging page. 2. Leave message box empty. 3. Click send.	Message: blank	Message should not be sent	As expected	Pass

7.4.7.1 Test Method

The messaging interface was tested with browser and Postman plus database validation was done to confirm the processing of the back end and the storage of the message.

7.4.7.2 Test Case Description

This test confirms that users are able to send valid direct messages and blank submissions are rejected.

7.4.7.3 Test Case Data

Data used contained a normal message input and empty message input.

7.4.7.4 Test Case Report

The instant message service was functioning properly with legitimate messages, and did not allow blank messages.

7.4.8 Admin Panel

In Table 7.13 admin must be logged in to manage users and platform records.

Pre condition

- Admin account should exist and be active.
- Admin should be logged in.
- Admin panel should be accessible.
- Database should be connected.

Table 7.13: Admin Panel Test Case

Test Case ID	**Test Scenario**	**Test Steps**	**Input**	**Expected Output**	**Actual Output**	**Pass / Fail**
19	Admin blocks a user account	1. Admin opens user management. 2. Selects a user. 3. Clicks block option.	User: testUser	User account should be blocked	As expected	Pass
20	Admin unblocks a user account	1. Admin opens blocked user record. 2. Clicks unblock option.	User: testUser	User account should be unblocked	As expected	Pass
21	Admin deletes user account	1. Admin opens user list. 2. Selects user. 3. Clicks delete option.	User: unwantedUser	User account should be deleted	As expected	Pass

7.4.8.1 Test Method

Admin panel operations were tested with the help of browser, account status updates and deletions were tested with the help of database.

7.4.8.2 Test Case Description

This test is used to ensure that the administrator is able to do the significant user management tasks such as block, unblocking and destroy accounts.

7.4.8.3 Test Case Data

The data incorporated were current user records of block, unblock and delete features.

7.4.8.4 Test Case Report

The back office was functioning properly and did all the tested user management processes as needed.

7.4.9 Database

In Table 7.14 database testing was carried out to verify whether stored records matched the system actions performed through the user interface and APIs.

Pre condition

- PostgreSQL database should be online.
- pgAdmin should be connected to the project database.
- Test data should already exist in the system through executed workflows.

Table 7.14: Database Test Case

Test Case ID	Test Scenario	Test Steps	Input	Expected Output	Actual Output	Pass / Fail
22	Verify user registration record in database	1. Execute sign up process. 2. Open pgAdmin. 3. Run query on users table.	SELECT * FROM calmconnect.users;	New user record should be visible	As expected	Pass
23	Verify appointment record in database	1. Create appointment. 2. Open pgAdmin. 3. Run query on appointments table.	SELECT * FROM calmconnect. appointments;	Appointment record should be visible	As expected	Pass
24	Verify direct message record in database	1. Send direct message. 2. Open pgAdmin. 3. Run query on direct messages table.	SELECT * FROM calmconnect.direct_messages;	Message record should be visible	As expected	Pass

7.4.9.1 Test Method

The testing of databases was done with the help of pgAdmin and SQL query after performing the functional operations with the help of the application.

7.4.9.2 Test Case Description

This test confirms that information on users, appointments, and messages are properly stored in the database following the working system functions.

7.4.9.3 Test Case Data

The available data consisted of the records created during Sign up, appointment making, and direct messaging.

7.4.9.4 Test Case Report

Database processed all tests in the correct way and was comparable to the test system actions.

7.5 Bugs Report

The bugs report shows the problems that were found in the testing stage of Calm-Connect AI system. It is a documentation of mistakes that are noticed when performing various tests. checks and assists in balancing the anticipated system behaviour against the actual results. The bugs are recorded in detail including the name of the module that it has affected, the type of problem, and how it has affected the functionality of the system. Maintaining a bug report is meant to ensure that all the identified problems are properly analyzed, prioritized and resolved before it goes into deployment. Through this process, the reliability of systems, user experience and the application in various scenarios have been enhanced. [38]

Table 7.15 gives the information about issues observed at the stage of testing AI Calm-connect. It assists in documenting issues of the system, making a comparison between the intended and the actual behavior and also giving a grade of the issues based on the impact of the issue on the functionality of the system. The table below shows the bug report of the system.

Table 7.15: Bug Report

Bug Id	**Module Name**	**Bug Detail**	**OS**	**Browser**	**Expected Results**	**Actual Results**	**Severity**
BG01	Sign Up	System allowed submission when one required field was left empty.	Windows 10/11	Chrome	Registration should fail and validation message should be shown.	Form submitted without proper validation.	High

Bug Id	Module Name	Bug Detail	OS	Browser	Expected Results	Actual Results	Severity
BG02	Sign Up	Duplicate email account was attempted and error message was not clear to the user.	Windows 10/11	Chrome	User should receive clear duplicate email warning.	Generic error response was shown.	Medium
BG03	Log-in	Registered user could not log in because of invalid credential handling during one test cycle.	Windows 10/11	Chrome	User should be logged in with correct credentials.	Login request failed and access was denied.	High
BG04	AI Chat	AI response was delayed or not returned for one request due to service unavailability.	Windows 10/11	Chrome	AI should generate a response after valid message submission.	No response was shown for the submitted message.	High
BG05	Mood Tracking	Mood entry was not saved when note field was left empty in one test run.	Windows 10/11	Chrome	Mood should be saved even if note is optional.	Entry was not stored in database.	Medium
BG06	Journaling	Journal form accepted incomplete content in one invalid submission test.	Windows 10/11	Chrome	Empty journal entry should be rejected.	Incomplete journal entry was accepted.	Medium
BG07	Appointments	Appointment request was submitted with incomplete scheduling information.	Windows 10/11	Chrome	Appointment should not be created without all required fields.	Request was processed with missing values.	High

Bug Id	Module Name	Bug Detail	OS	Browser	Expected Results	Actual Results	Severity
BG08	Direct Messaging	Empty direct message was sent during one validation test.	Windows 10/11	Chrome	Blank message should not be sent.	Message was submitted without text.	Medium
BG09	Admin Panel	Blocked user status was not refreshed immediately on user management page.	Windows 10/11	Chrome	Updated user status should appear immediately after admin action.	Old status remained visible until page reload.	Low
BG10	Logout	User session was not cleared properly in one browser session after logout.	Windows 10/11	Chrome	User should be fully logged out and redirected to login page.	Session remained active until manual refresh.	High

Chapter 8

CONCLUSION

8.1 Conclusion

CalmConnect AI is a web-based mental health aid platform designed to offer emotional support, communication and wellness management in a safe online space. This system offers various user-friendly features and supports with AI, enhancing accessibility and engagement in mental health management.

The platform provides multiple functionalities such as AI chat support, mood tracking, journaling, CBT exercises, appointment scheduling, direct messaging, notifications, feedback management, and administrative monitoring. These modules combine to form a collaborative and comprehensive mental wellness network for patients, psychiatrists and administrators.

The system was built using React (TypeScript for the frontend), Node.js/Express (backend) and PostgreSQL (for database management). By incorporating the Gemini API, our AI chatbot can offer helpful and engaging responses to users' queries.

All important software engineering concepts including requirement analysis were well realized in the development process as well as system design, database management, authentication, role based access control, testing and reporting. The project was also mindful of its users' privacy, security, and usability as well, to make safe usage of the system.

In general, CalmConnect AI was successful in meeting its primary goals, which involved creating a hub for retreating to a tranquil environment where individuals can examine their personal mental practices and foster a more mindful lifestyle. Essentially, this system presents a blueprint for the potential integration of modern web technologies with AI to enhance mental health services, making them more organized and accessible.

8.2 Future Work

While CalmConnect AI offers a wealth of (helpful) capabilities, they can enhance a few other salient factors in future iterations of the system. Future research could involve expanding the scope of functions, enhancing AI functionality, and enhancing user experience.

A potential improvement in the system would be a video consultation system for patients to communicate with psychiatrists via video online. Mobile application support can also be included for better accessibility to users preferring to use smart phones and tablets.

To enhance the AI chatbot's performance, it could be expanded with more sophisticated AI models and emotion detection algorithms, delivering improves, more customized reactions. Further safety features can be put in place to increase the awareness of damaging talk and mental health risks.

Future versions of the system may also include multilingual support, advanced analytics dashboards, wearable device integration, and personalized wellness recommendations based on user behavior and mood history.

More reporting and data visualization features can also be added for administrators and psychiatrists to improve monitoring and decision making. These future enhancements can help make CalmConnect AI more intelligent, scalable, and effective for mental health support service.

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